

1205.5935v1 — formula report

1012 inline formulas (0 shown: the ones that did not render), 479 display equations, 2 tables, 1 unrecovered image regions.

Display equations

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Page | Conf.             | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                               | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Scan image                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| 1205.5935v1_<br>E08323 (313)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 066  | ■ 0.139<br>● N +0 | $\begin{aligned} & a \downharpoonleft A_2 \ \& \ \left.=\frac{1}{2}\right\{ a \right\rfloor \left(\sum_i a^i \wedge F(a_i)\right) \\ & \left.\left(\sum_i a^i\right) \right\rfloor F(a_i) - \frac{1}{2} \sum_i a^i \left(a \right\rfloor F(a_i)) \\ & \left.\left(\sum_i a^i\right) \right\rfloor F(a_i) + \frac{1}{2} \sum_i a^i (F(a) \rfloor a_i) \\ & \left.\left(\sum_i a^i\right) \right\rfloor F(a) + \frac{1}{2} F(a) \\ & = F(a) . \end{aligned}$ | $\begin{aligned} a \downharpoonleft A_2 &= \frac{1}{2} a \rfloor \left( \sum_i a^i \wedge F(a_i) \right) \\ &= \frac{1}{2} \sum_i \left( a \downharpoonleft a^i \right) F(a_i) - \frac{1}{2} \sum_i a^i (a \rfloor F(a_i)) \\ &= \frac{1}{2} \sum_i \left( a \downharpoonleft a^i \right) F(a_i) + \frac{1}{2} \sum_i a^i (F(a) \rfloor a_i) \\ &= \frac{1}{2} F \left( \sum_i \left( a \downharpoonleft a^i \right) a_i \right) + \frac{1}{2} \sum_i (F(a) \rfloor a_i) a^i \\ &= \frac{1}{2} F(a) + \frac{1}{2} F(a) \\ &= F(a) . \end{aligned}$ | $\begin{aligned} a \rfloor A_2 &= \frac{1}{2} a \rfloor \left( \sum_i a^i \wedge F(a_i) \right) \\ &= \frac{1}{2} \sum_i (a \rfloor a^i) F(a_i) - \frac{1}{2} \sum_i a^i (a \rfloor F(a_i)) \\ &= \frac{1}{2} \sum_i (a \rfloor a^i) F(a_i) + \frac{1}{2} \sum_i a^i (F(a) \rfloor a_i) \\ &= \frac{1}{2} F \left( \sum_i (a \rfloor a^i) a_i \right) + \frac{1}{2} \sum_i (F(a) \rfloor a_i) a^i \\ &= \frac{1}{2} F(a) + \frac{1}{2} F(a) \\ &= F(a) . \end{aligned}$ |
| 1205.5935v1_<br>E08239 (232)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 050  | ■ 0.146<br>● N +0 | $\begin{aligned} & \left. P_{\boldsymbol{A}}(B) := B \downharpoonleft \boldsymbol{A} \right\rfloor \boldsymbol{A}^{-1} = (B \downharpoonleft \boldsymbol{A}) \rfloor \boldsymbol{A}^{-1} \\ & R_{\boldsymbol{A}}(B) := B \wedge \boldsymbol{A} \rfloor \boldsymbol{A}^{-1} = B \wedge \boldsymbol{A} \rfloor \boldsymbol{A}^{-1} . \end{aligned}$                                                                                                          | $\begin{aligned} P_{\boldsymbol{A}}(B) &:= B \downharpoonleft \boldsymbol{A} \rfloor \boldsymbol{A}^{-1} = (B \downharpoonleft \boldsymbol{A}) \rfloor \boldsymbol{A}^{-1} \\ R_{\boldsymbol{A}}(B) &:= B \wedge \boldsymbol{A} \rfloor \boldsymbol{A}^{-1} = B \wedge \boldsymbol{A} \rfloor \boldsymbol{A}^{-1} . \end{aligned}$                                                                                                                                                                                                                 | $\begin{aligned} P_{\boldsymbol{A}}(B) &:= B \rfloor \boldsymbol{A} \rfloor \boldsymbol{A}^{-1} = (B \rfloor \boldsymbol{A}) \rfloor \boldsymbol{A}^{-1} \\ R_{\boldsymbol{A}}(B) &:= B \wedge \boldsymbol{A} \rfloor \boldsymbol{A}^{-1} = B \wedge \boldsymbol{A} \rfloor \boldsymbol{A}^{-1} . \end{aligned}$                                                                                                                                                        |
| <div>Conf. MathPix confidence: ■ ≥0.9   ■ 0.5–0.9   ■ &lt;0.5   — no value</div> <div>LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</div> <div>Residual render vs scan (inkdrill): ● C component   ● W weak   ● S stable   ● N noise   ● K clean   ● X crop overran its region   ● not measured</div> <div>[refined: <b>basis</b>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</div> |      |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |



| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Page | Conf.                                                   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    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| 1205.5935v1_<br>EQ0356 (345)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 073  | <div><div>■0.228</div><div><div></div></div></div>      | $\begin{aligned} & \text{\texttt{\textbackslash begin{aligned} a\text{\texttt{\textbackslash rfloor[\text{\texttt{\textbackslash bar{F}}]\text{\texttt{\textbackslash left(\text{\texttt{\textbackslash boldsymbol{B}}_{\text{\texttt{\textbackslash boldsymbol{s}}}\text{\texttt{\textbackslash right)}}}}}}}}}} \& =\text{\texttt{\textbackslash rfloor[\text{\texttt{\textbackslash bar{F}}]\text{\texttt{\textbackslash left(b \text{\texttt{\textbackslash wedge \text{\texttt{\textbackslash boldsymbol{B}}_{\text{\texttt{\textbackslash boldsymbol{s}}-\text{\texttt{\textbackslash mathbf{1}}}\text{\texttt{\textbackslash right)}}}}}}}}}} \& \& =\text{\texttt{\textbackslash rfloor\text{\texttt{\textbackslash left[\text{\texttt{\textbackslash bar{F}}(b \text{\texttt{\textbackslash wedge[\text{\texttt{\textbackslash bar{F}}]\text{\texttt{\textbackslash left(\text{\texttt{\textbackslash boldsymbol{B}}_{\text{\texttt{\textbackslash boldsymbol{s}}-\text{\texttt{\textbackslash mathbf{1}}}\text{\texttt{\textbackslash right)}}}}}}}}}} \& \& \text{\texttt{\textbackslash left}} .=\text{\texttt{\textbackslash rfloor \text{\texttt{\textbackslash bar{F}}(b)}[\text{\texttt{\textbackslash bar{F}}]\text{\texttt{\textbackslash left(\text{\texttt{\textbackslash boldsymbol{B}}_{\text{\texttt{\textbackslash boldsymbol{s}}-\text{\texttt{\textbackslash mathbf{1}}}\text{\texttt{\textbackslash right)}}}}}}}}}}-\text{\texttt{\textbackslash bar{F}}(b \text{\texttt{\textbackslash wedge [\text{\texttt{\textbackslash rfloor[\text{\texttt{\textbackslash bar{F}}]\text{\texttt{\textbackslash left(\text{\texttt{\textbackslash boldsymbol{B}}_{\text{\texttt{\textbackslash boldsymbol{s}}-\text{\texttt{\textbackslash mathbf{1}}}\text{\texttt{\textbackslash right)}}}}}}}}}} \& \& \text{\texttt{\textbackslash left}} .=[\text{\texttt{F(a)\text{\texttt{\textbackslash rfloor b}}] [\text{\texttt{\textbackslash bar{F}}]\text{\texttt{\textbackslash left(\text{\texttt{\textbackslash boldsymbol{B}}_{\text{\texttt{\textbackslash boldsymbol{s}}-\text{\texttt{\textbackslash mathbf{1}}}\text{\texttt{\textbackslash right)}}}}}}}}}}-\text{\texttt{\textbackslash bar{F}}(b \text{\texttt{\textbackslash wedge[\text{\texttt{\textbackslash bar{F}}](\text{\texttt{F(a)\text{\texttt{\textbackslash rfloor \text{\texttt{\textbackslash boldsymbol{B}}_{\text{\texttt{\textbackslash boldsymbol{s}}-\text{\texttt{\textbackslash mathbf{1}}}\text{\texttt{\textbackslash right)}}}}}}}}}} \& \& \text{\texttt{\textbackslash left}} .\text{\texttt{\textbackslash left}} .=[\text{\texttt{\textbackslash bar{F}}][(\text{\texttt{F(a)\text{\texttt{\textbackslash rfloor b}} \text{\texttt{\textbackslash boldsymbol{B}}_{\text{\texttt{\textbackslash boldsymbol{s}}-\text{\texttt{\textbackslash mathbf{1}}}\text{\texttt{\textbackslash right)}}}}}}}}}-[\text{\texttt{\textbackslash bar{F}}]\text{\texttt{\textbackslash left[b \text{\texttt{\textbackslash wedge(\text{\texttt{F(a)\text{\texttt{\textbackslash rfloor \text{\texttt{\textbackslash boldsymbol{B}}_{\text{\texttt{\textbackslash boldsymbol{s}}-\text{\texttt{\textbackslash mathbf{1}}}\text{\texttt{\textbackslash right)}}}}}}}}}} \& \& \text{\texttt{\textbackslash left}} .=[\text{\texttt{\textbackslash bar{F}}][(\text{\texttt{F(a)\text{\texttt{\textbackslash rfloor b}} ) \text{\texttt{\textbackslash boldsymbol{B}}_{\text{\texttt{\textbackslash boldsymbol{s}}-\text{\texttt{\textbackslash mathbf{1}}}\text{\texttt{\textbackslash right)}}}}}}}}}-\text{\texttt{b \text{\texttt{\textbackslash wedge \text{\texttt{\textbackslash left(\text{\texttt{F(a) \text{\texttt{\textbackslash downharpoonleft \text{\texttt{\textbackslash boldsymbol{B}}_{\text{\texttt{\textbackslash boldsymbol{s}}-\text{\texttt{\textbackslash mathbf{1}}}\text{\texttt{\textbackslash right)}}}}}}}}}} \& \& \text{\texttt{\textbackslash left}} .=[\text{\texttt{\textbackslash bar{F}}]}[\text{\texttt{F(a)\text{\texttt{\textbackslash rfloor\text{\texttt{\textbackslash left(b \text{\texttt{\textbackslash wedge \text{\texttt{\textbackslash boldsymbol{B}}_{\text{\texttt{\textbackslash boldsymbol{s}}-\text{\texttt{\textbackslash mathbf{1}}}\text{\texttt{\textbackslash right)}}}}}}}}}} \& \& =[\text{\texttt{\textbackslash bar{F}}]\text{\texttt{\textbackslash left(\text{\texttt{F(a) \text{\texttt{\textbackslash downharpoonleft \text{\texttt{\textbackslash boldsymbol{B}}_{\text{\texttt{\textbackslash boldsymbol{s}}}\text{\texttt{\textbackslash right)}}}}}}}}}} . \text{\texttt{\textbackslash end{aligned}}}}$ | $\begin{aligned} a\downarrow [\bar{F}](\boldsymbol{B}_s) &= a\downarrow [\bar{F}](b \wedge \boldsymbol{B}_{s-1}) \\ &= a\downarrow [\bar{F}(b) \wedge [\bar{F}](\boldsymbol{B}_{s-1})] \\ &= [a\downarrow \bar{F}(b)][\bar{F}](\boldsymbol{B}_{s-1}) - \bar{F}(b) \wedge [a\downarrow [\bar{F}](\boldsymbol{B}_{s-1})] \\ &= [F(a)\downarrow b][\bar{F}](\boldsymbol{B}_{s-1}) - \bar{F}(b) \wedge [\bar{F}](F(a)\downarrow \boldsymbol{B}_{s-1}) \\ &= [\bar{F}][(F(a)\downarrow b)\boldsymbol{B}_{s-1}] - [\bar{F}][b \wedge (F(a)\downarrow \boldsymbol{B}_{s-1})] \\ &= [\bar{F}][(F(a)\downarrow b)\boldsymbol{B}_{s-1} - b \wedge (F(a)\downarrow \boldsymbol{B}_{s-1})] \\ &= [\bar{F}][F(a)\downarrow (b \wedge \boldsymbol{B}_{s-1})] \\ &= [\bar{F}](F(a)\downarrow \boldsymbol{B}_s). \end{aligned}$ | $\begin{aligned} a\downarrow [\bar{F}](\boldsymbol{B}_s) &= a\downarrow [\bar{F}](b \wedge \boldsymbol{B}_{s-1}) \\ &= a\downarrow [\bar{F}(b) \wedge [\bar{F}](\boldsymbol{B}_{s-1})] \\ &= [a\downarrow \bar{F}(b)][\bar{F}](\boldsymbol{B}_{s-1}) - \bar{F}(b) \wedge [a\downarrow [\bar{F}](\boldsymbol{B}_{s-1})] \\ &= [F(a)\downarrow b][\bar{F}](\boldsymbol{B}_{s-1}) - \bar{F}(b) \wedge [\bar{F}](F(a)\downarrow \boldsymbol{B}_{s-1}) \\ &= [\bar{F}][(F(a)\downarrow b)\boldsymbol{B}_{s-1}] - [\bar{F}][b \wedge (F(a)\downarrow \boldsymbol{B}_{s-1})] \\ &= [\bar{F}][(F(a)\downarrow b)\boldsymbol{B}_{s-1} - b \wedge (F(a)\downarrow \boldsymbol{B}_{s-1})] \\ &= [\bar{F}][F(a)\downarrow (b \wedge \boldsymbol{B}_{s-1})] \\ &= [\bar{F}](F(a)\downarrow \boldsymbol{B}_s). \end{aligned}$ |
| 1205.5935v1_<br>EQ0297 (200)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 043  | <div><div>■0.279</div><div><div>●X +1</div></div></div> | $\begin{aligned} & \text{\texttt{\textbackslash begin{aligned} A_{\text{\texttt{2}}} \text{\texttt{\textbackslash times\text{\texttt{\textbackslash left(a_{\text{\texttt{1}}} \text{\texttt{\textbackslash wedge a_{\text{\texttt{2}}} \text{\texttt{\textbackslash wedge \text{\texttt{\textbackslash cdots \text{\texttt{\textbackslash wedge a_{\text{\texttt{r}}}\text{\texttt{\textbackslash right)}}}}}}}}}} \& =\text{\texttt{\textbackslash left(A_{\text{\texttt{2}}} \text{\texttt{\textbackslash times \text{\texttt{\textbackslash boldsymbol{B}}_{\text{\texttt{\textbackslash boldsymbol{r}}-\text{\texttt{\textbackslash mathbf{1}}}\text{\texttt{\textbackslash right)}}}}}}}}}} \text{\texttt{\textbackslash wedge a_{\text{\texttt{r}}}+\text{\texttt{\textbackslash boldsymbol{B}}_{\text{\texttt{\textbackslash boldsymbol{r}}-\text{\texttt{\textbackslash mathbf{1}}}\text{\texttt{\textbackslash right)}}}}}}}} \text{\texttt{\textbackslash wedge \text{\texttt{\textbackslash left(A_{\text{\texttt{2}}} \text{\texttt{\textbackslash times a_{\text{\texttt{r}}}\text{\texttt{\textbackslash right)}}}}}}}}}} \& \& =\text{\texttt{\textbackslash sum_{j=1}^{\text{\texttt{r-1}}}}}\text{\texttt{\textbackslash left[a_{\text{\texttt{1}}} \text{\texttt{\textbackslash wedge a_{\text{\texttt{2}}} \text{\texttt{\textbackslash wedge \text{\texttt{\textbackslash cdots \text{\texttt{\textbackslash wedge\text{\texttt{\textbackslash left(A_{\text{\texttt{2}}\text{\texttt{\textbackslash left\text{\texttt{\textbackslash lfloor a_{\text{\texttt{j}}}\text{\texttt{\textbackslash right)}}}}}}}}}}}} \text{\texttt{\textbackslash wedge \text{\texttt{\textbackslash cdots \text{\texttt{\textbackslash wedge a_{\text{\texttt{r-1}}} \text{\texttt{\textbackslash wedge a_{\text{\texttt{r}}}\text{\texttt{\textbackslash right}}}}}}}}}}+\text{\texttt{\textbackslash right}} . \& \& a_{\text{\texttt{1}}} \text{\texttt{\textbackslash wedge a_{\text{\texttt{2}}} \text{\texttt{\textbackslash wedge \text{\texttt{\textbackslash cdots \text{\texttt{\textbackslash wedge a_{\text{\texttt{r-1}}} \text{\texttt{\textbackslash wedge\text{\texttt{\textbackslash left(A_{\text{\texttt{2}}\text{\texttt{\textbackslash left\text{\texttt{\textbackslash llcorner a_{\text{\texttt{r}}}\text{\texttt{\textbackslash right)}}}}}}}}}}}}\text{\texttt{\textbackslash right}} . \& \& =\text{\texttt{\textbackslash sum_{j=1}^{\text{\texttt{r}}}}}\text{\texttt{\textbackslash left(a_{\text{\texttt{1}}} \text{\texttt{\textbackslash wedge a_{\text{\texttt{2}}} \text{\texttt{\textbackslash wedge \text{\texttt{\textbackslash cdots \text{\texttt{\textbackslash wedge\text{\texttt{\textbackslash left(A_{\text{\texttt{2}}\text{\texttt{\textbackslash left\text{\texttt{\textbackslash llcorner a_{\text{\texttt{j}}}\text{\texttt{\textbackslash right)}}}}}}}}}}}}}} \text{\texttt{\textbackslash wedge \text{\texttt{\textbackslash cdots \text{\texttt{\textbackslash wedge a_{\text{\texttt{r}}}\text{\texttt{\textbackslash right}}}}}}}}}} . \text{\texttt{\textbackslash end{aligned}}}}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $\begin{aligned} A_2 \times (a_1 \wedge a_2 \wedge \cdots \wedge a_r) &= (A_2 \times \boldsymbol{B}_{r-1}) \wedge a_r + \boldsymbol{B}_{r-1} \wedge (A_2 \times a_r) \\ &= \sum_{j=1}^{r-1} [a_1 \wedge a_2 \wedge \cdots \wedge (A_2 \lfloor a_j) \wedge \cdots \wedge a_{r-1} \wedge a_r] + \\ &\quad a_1 \wedge a_2 \wedge \cdots \wedge a_{r-1} \wedge (A_2 \lfloor a_r) \\ &= \sum_{j=1}^r a_1 \wedge a_2 \wedge \cdots \wedge (A_2 \lfloor a_j) \wedge \cdots \wedge a_r \end{aligned}$                                                                                                                                                                                                                                                                                                                   | $\begin{aligned} A_2 \times (a_1 \wedge a_2 \wedge \cdots \wedge a_r) &= (A_2 \times \boldsymbol{B}_{r-1}) \wedge a_r + \boldsymbol{B}_{r-1} \wedge (A_2 \times a_r) \\ &= \sum_{j=1}^{r-1} [a_1 \wedge a_2 \wedge \cdots \wedge (A_2 \lfloor a_j) \wedge \cdots \wedge a_{r-1} \wedge a_r] + \\ &\quad a_1 \wedge a_2 \wedge \cdots \wedge a_{r-1} \wedge (A_2 \lfloor a_r) \\ &= \sum_{j=1}^r a_1 \wedge a_2 \wedge \cdots \wedge (A_2 \lfloor a_j) \wedge \cdots \wedge a_r, \end{aligned}$                                                                                                                                                                                                                                                                                                                  |
| 1205.5935v1_<br>EQ0229 (222)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 048  | <div><div>■0.290</div><div><div>●K +8</div></div></div> | $\text{\texttt{\textbackslash left.\text{\texttt{\textbackslash left.P_{\text{\texttt{u}}}(v)=v \text{\texttt{\textbackslash downharpoonleft u u^{\text{\texttt{-1}}}}}= (v \text{\texttt{\textbackslash rfloor u\text{\texttt{\textbackslash right}}}\text{\texttt{\textbackslash right}\text{\texttt{\textbackslash rfloor u^{\text{\texttt{-1}}}}}}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $P_u(v) = v \downarrow uu^{-1} = (v \downarrow u) \downarrow u^{-1}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $P_u(v) = v \downarrow uu^{-1} = (v \downarrow u) \downarrow u^{-1}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <div><div>Conf.</div><div>MathPix confidence: <div><div>■≥0.9</div><div>■0.5-0.9</div><div>■&lt;0.5</div><div>—no value</div></div></div><div><div>LaTeX source</div><div>MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</div></div><div><div>Residual</div><div>render vs scan (inkdrill): <div><div>●C component</div><div>●W weak</div><div>●S stable</div><div>●N noise</div><div>●K clean</div><div>●X crop overran its region</div><div>●not measured</div></div></div><div><div>[refined: <i>basis</i>]</div><div>beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</div></div></div></div> |      |                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 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| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Page | Conf.                                                        | LaTeX source                                                                                                                                                                                                                                                                                                             | Rendered                                                                                                                                                                                                                                                                                                                    | Scan image                                                                                                                                                                                                                                                                                                                  |
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| 1205.5935v1_E00469 (127)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 089  | <div> <div>■</div>0.290 <div> <div></div> </div> </div>      | $\begin{aligned} & \sum_i a^i a_i \downarrow A_r = \sum_i a^i \wedge (a_i \downarrow A_r) = r A_r \quad \text{for any } A_r. \\ & \sum_i a_i a^i \Big\rfloor A_r = \sum_i a_i \wedge (a^i \Big\rfloor A_r) = r A_r \quad \text{for any } A_r. \end{aligned}$                                                             | $\sum_i a^i a_i \Big\rfloor A_r = \sum_i a^i \wedge (a_i \Big\rfloor A_r) = r A_r \quad \text{for any } A_r.$                                                                                                                                                                                                               | $\sum_i a^i a_i \Big\rfloor A_r = \sum_i a^i \wedge (a_i \Big\rfloor A_r) = r A_r \quad \text{for any } A_r.$                                                                                                                                                                                                               |
| 1205.5935v1_E00312                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 063  | <div> <div>■</div>0.295 <div> <div>●</div>W +0 </div> </div> | $\begin{array}{l} u \in \operatorname{Ker}(F) \text{ iff } F(u)=0 \\ \text{iff } F(u) \Big\rfloor v = 0 \text{ for all } v \in V \\ \text{iff } u \Big\rfloor \overline{F}(v) = 0 \text{ for all } v \in V \\ \text{iff } u \in \operatorname{Range}(\overline{F})^\perp. \end{array}$                                   | $\begin{array}{l} u \in \operatorname{Ker}(F) \text{ iff } F(u)=0 \\ \text{iff } F(u) \Big\rfloor v = 0 \text{ for all } v \in V \\ \text{iff } u \Big\rfloor \overline{F}(v) = 0 \text{ for all } v \in V \\ \text{iff } u \in \operatorname{Range}(\overline{F})^\perp. \end{array}$                                      | $\begin{array}{l} u \in \operatorname{Ker}(F) \text{ iff } F(u)=0 \\ \text{iff } F(u) \Big\rfloor v = 0 \text{ for all } v \in V \\ \text{iff } u \Big\rfloor \overline{F}(v) = 0 \text{ for all } v \in V \\ \text{iff } u \in \operatorname{Range}(\overline{F})^\perp. \end{array}$                                      |
| 1205.5935v1_E00131 (124)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 032  | <div> <div>■</div>0.312 <div> <div>●</div>X +1 </div> </div> | $\begin{aligned} & a \downarrow (A \wedge B) = (a \downarrow A) \wedge B + A^* \wedge (a \downarrow B) \\ & a \wedge (A \Big\rfloor B) = (a \wedge A) \Big\rfloor B - A^* \Big\rfloor (a \Big\rfloor B) \\ & a \wedge (A \Big\rfloor B) = (a \Big\rfloor A) \Big\rfloor B + A^* \Big\rfloor (a \wedge B). \end{aligned}$ | $\begin{aligned} & a \Big\rfloor (A \wedge B) = (a \Big\rfloor A) \wedge B + A^* \wedge (a \Big\rfloor B) \\ & a \wedge (A \Big\rfloor B) = (a \wedge A) \Big\rfloor B - A^* \Big\rfloor (a \Big\rfloor B) \\ & a \wedge (A \Big\rfloor B) = (a \Big\rfloor A) \Big\rfloor B + A^* \Big\rfloor (a \wedge B). \end{aligned}$ | $\begin{aligned} & a \Big\rfloor (A \wedge B) = (a \Big\rfloor A) \wedge B + A^* \wedge (a \Big\rfloor B) \\ & a \wedge (A \Big\rfloor B) = (a \wedge A) \Big\rfloor B - A^* \Big\rfloor (a \Big\rfloor B) \\ & a \wedge (A \Big\rfloor B) = (a \Big\rfloor A) \Big\rfloor B + A^* \Big\rfloor (a \wedge B). \end{aligned}$ |
| 1205.5935v1_E00202 (195)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 043  | <div> <div>■</div>0.315 <div> <div>●</div>N +1 </div> </div> | $\begin{aligned} & A_2 A_r = A_2 \Big\rfloor A_r + \langle A_2 A_r \rangle_r + A_2 \wedge A_r \\ & A_r A_2 = A_2 \Big\rfloor A_r - \langle A_2 A_r \rangle_r + A_2 \wedge A_r. \end{aligned}$                                                                                                                            | $\begin{aligned} & A_2 A_r = A_2 \Big\rfloor A_r + \langle A_2 A_r \rangle_r + A_2 \wedge A_r \\ & A_r A_2 = A_2 \Big\rfloor A_r - \langle A_2 A_r \rangle_r + A_2 \wedge A_r. \end{aligned}$                                                                                                                               | $\begin{aligned} & A_2 A_r = A_2 \Big\rfloor A_r + \langle A_2 A_r \rangle_r + A_2 \wedge A_r \\ & A_r A_2 = A_2 \Big\rfloor A_r - \langle A_2 A_r \rangle_r + A_2 \wedge A_r. \end{aligned}$                                                                                                                               |
| 1205.5935v1_E00209 (201)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 059  | <div> <div>■</div>0.318 <div> <div>●</div>N +0 </div> </div> | $\begin{aligned} & a^I a_N = a^I \Big\rfloor a_N \\ & = (a^{i_1} \wedge \cdots \wedge a^{i_r}) \Big\rfloor a_N \\ & = (a^{i_1} \wedge \cdots \wedge a^{i_{r-1}}) \Big\rfloor (a^{i_r} \Big\rfloor a_N). \end{aligned}$                                                                                                   | $\begin{aligned} & a^I a_N = a^I \Big\rfloor a_N \\ & = (a^{i_1} \wedge \cdots \wedge a^{i_r}) \Big\rfloor a_N \\ & = (a^{i_1} \wedge \cdots \wedge a^{i_{r-1}}) \Big\rfloor (a^{i_r} \Big\rfloor a_N). \end{aligned}$                                                                                                      | $\begin{aligned} & a^I a_N = a^I \Big\rfloor a_N \\ & = (a^{i_1} \wedge \cdots \wedge a^{i_r}) \Big\rfloor a_N \\ & = (a^{i_1} \wedge \cdots \wedge a^{i_{r-1}}) \Big\rfloor (a^{i_r} \Big\rfloor a_N). \end{aligned}$                                                                                                      |
| <p><b>Conf.</b> MathPix confidence: <span style="color: green;">■</span> ≥0.9 <span style="color: orange;">■</span> 0.5–0.9 <span style="color: red;">■</span> &lt;0.5 — no value</p> <p><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</p> <p><b>Residual</b> render vs scan (inkdrill): <span style="color: red;">●</span> C component <span style="color: orange;">●</span> W weak <span style="color: blue;">●</span> S stable <span style="color: grey;">●</span> N noise <span style="color: green;">●</span> K clean <span style="color: blue;">●</span> X crop overran its region <span style="color: grey;">●</span> not measured</p> <p><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</p> |      |                                                              |                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                             |

| Identifier                                                                                                                                                                                                                                                                                                           | Page | Conf.             | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Scan image |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| 1205.5935v1_<br>EQ0301 (293)                                                                                                                                                                                                                                                                                         | 062  | ■ 0.323<br>● N +0 | $\begin{aligned} u\rfloor F(v) &= u \downharpoonleft \left( \sum_i (a^i \circ v) a_i \right) \\ &= \sum_i (a^i \circ v) (u \downharpoonleft a_i) \\ &= \left( \sum_i (u \downharpoonleft a_i) a^i \right) \circ v \\ &= u \circ v. \end{aligned}$                                                                                                                                                                                                                                             | $\begin{aligned} u \downharpoonleft F(v) &= u \downharpoonleft \left( \sum_i (a^i \circ v) a_i \right) \\ &= \sum_i (a^i \circ v) (u \downharpoonleft a_i) \\ &= \left( \sum_i (u \downharpoonleft a_i) a^i \right) \circ v \\ &= u \circ v. \end{aligned}$                                                                                                                                                                                                                                   |            |
| 1205.5935v1_<br>EQ0080 (78)                                                                                                                                                                                                                                                                                          | 022  | ■ 0.328<br>●      | $\begin{aligned} a \rfloor (a_1 a_2 \cdots a_r) &= \frac{1}{2} (a a_1 a_2 \cdots a_r - (-1)^r a_1 a_2 \cdots a_r a) \\ &= \sum_{j=1}^r (-1)^{j-1} (a \downharpoonleft a_j) a_1 a_2 \cdots \check{a}_j \cdots a_r. \end{aligned}$                                                                                                                                                                                                                                                              | $\begin{aligned} a \downharpoonleft (a_1 a_2 \cdots a_r) &= \frac{1}{2} (a a_1 a_2 \cdots a_r - (-1)^r a_1 a_2 \cdots a_r a) \\ &= \sum_{j=1}^r (-1)^{j-1} (a \downharpoonleft a_j) a_1 a_2 \cdots \check{a}_j \cdots a_r. \end{aligned}$                                                                                                                                                                                                                                                     |            |
| 1205.5935v1_<br>EQ0115 (109)                                                                                                                                                                                                                                                                                         | 029  | ■ 0.344<br>● N +0 | $\begin{aligned} \boldsymbol{A}_{\boldsymbol{r}} \boldsymbol{B}_{\boldsymbol{s}} &= \boldsymbol{A}_{\boldsymbol{r}-1} a \downharpoonleft \boldsymbol{B}_{\boldsymbol{s}} = \boldsymbol{A}_{\boldsymbol{r}-1} \downharpoonleft (a \downharpoonleft \boldsymbol{B}_{\boldsymbol{s}}) \\ &= (\boldsymbol{A}_{\boldsymbol{r}-1} \wedge a) \downharpoonleft \boldsymbol{B}_{\boldsymbol{s}} \\ &= \boldsymbol{A}_{\boldsymbol{r}} \downharpoonleft \boldsymbol{B}_{\boldsymbol{s}}, \end{aligned}$ | $\begin{aligned} \boldsymbol{A}_{\boldsymbol{r}} \boldsymbol{B}_{\boldsymbol{s}} &= \boldsymbol{A}_{\boldsymbol{r}-1} a \downharpoonleft \boldsymbol{B}_{\boldsymbol{s}} = \boldsymbol{A}_{\boldsymbol{r}-1} \downharpoonleft (a \downharpoonleft \boldsymbol{B}_{\boldsymbol{s}}) \\ &= (\boldsymbol{A}_{\boldsymbol{r}-1} \wedge a) \downharpoonleft \boldsymbol{B}_{\boldsymbol{s}} \\ &= \boldsymbol{A}_{\boldsymbol{r}} \downharpoonleft \boldsymbol{B}_{\boldsymbol{s}}, \end{aligned}$ |            |
| <div><div>Conf.</div><div>MathPix confidence: <div><div>■ ≥0.9</div><div>■ 0.5-0.9</div><div>■ &lt;0.5</div><div>— no value</div></div></div></div>                                                                                                                                                                  |      |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |
| <div><div>LaTeX source</div><div>MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</div></div>                           |      |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |
| <div><div>Residual</div><div>render vs scan (inkdrill): <div><div>● C component</div><div>● W weak</div><div>● S stable</div><div>● N noise</div><div>● K clean</div><div>● X crop overran its region</div><div>● not measured</div></div></div></div>                                                               |      |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |
| <div><div>[refined: <i>basis</i>]</div><div>beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</div></div> |      |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Page | Conf.                                                                      | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Scan image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| 1285.5935v1_<br>E08360 (348)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 074  | <div><span>■</span> 0.345</div> <div> <div></div> <div>● X +8</div> </div> | $\begin{aligned} & \backslash \text{begin}{aligned} F\left(a_{\{1\}} a_{\{2\}} \cdots a_{\{r\}}\right) \& \\ & \left.=F\left[a_{\{1\}}\right] \backslash \text{right} \rfloor \left(a_{\{2\}} \cdots a_{\{r\}}\right) \backslash \text{right} \rfloor+F\left[a_{\{1\}} \wedge\left(a_{\{2\}} \cdots a_{\{r\}}\right) \backslash \text{right} \rfloor \right] \backslash \& \\ & \left.=F\left[\sum_{j=2}^r(-1)^{j-2} a_{\{1\}} \backslash \text{right} \rfloor a_{\{j\}} a_{\{2\}} \cdots \check{a}_{\{j\}} \cdots a_{\{r\}}\right] \backslash \& \\ & \left.=F\left(a_{\{1\}}\right) \backslash \text{right} \rfloor\left\{F\left(a_{\{2\}}\right) \cdots F\left(a_{\{r\}}\right)\right\}+F\left(a_{\{1\}}\right) \wedge\left\{F\left(a_{\{2\}}\right) \cdots F\left(a_{\{r\}}\right)\right\} \end{aligned}$ | $\begin{aligned} F\left(a_1 a_2 \cdots a_r\right) &=F\left[a_1\right] \left(a_2 \cdots a_r\right)+F\left[a_1 \wedge\left(a_2 \cdots a_r\right)\right] \\ &=F\left[\sum_{j=2}^r(-1)^{j-2} a_1\right] a_j a_2 \cdots \check{a}_j \cdots a_r+F\left(a_1\right) \wedge F\left[a_2 \cdots a_r\right] \\ &=\sum_{j=2}^r(-1)^{j-2} a_1\left[a_j F\left(a_2\right) \cdots \check{F}\left(a_j\right) \cdots F\left(a_r\right)+F\left(a_1\right) \wedge\left\{F\left(a_2\right) \cdots F\left(a_r\right)\right\}\right] \\ &=\sum_{j=2}^r(-1)^{j-2} F\left(a_1\right)\left[F\left(a_j\right) F\left(a_2\right) \cdots \check{F}\left(a_j\right) \cdots F\left(a_r\right)+F\left(a_1\right) \wedge\left\{F\left(a_2\right) \cdots F\left(a_r\right)\right\}\right] \\ &=F\left(a_1\right)\left\{F\left(a_2\right) \cdots F\left(a_r\right)\right\}+F\left(a_1\right) \wedge\left\{F\left(a_2\right) \cdots F\left(a_r\right)\right\} \\ &=F\left(a_1\right) F\left(a_2\right) \cdots F\left(a_r\right) \end{aligned}$ | $\begin{aligned} F\left(a_1 a_2 \cdots a_r\right) &=F\left[a_1\right] \left(a_2 \cdots a_r\right)+F\left[a_1 \wedge\left(a_2 \cdots a_r\right)\right] \\ &=F\left[\sum_{j=2}^r(-1)^{j-2} a_1\right] a_j a_2 \cdots \check{a}_j \cdots a_r+F\left(a_1\right) \wedge F\left[a_2 \cdots a_r\right] \\ &=\sum_{j=2}^r(-1)^{j-2} a_1\left[a_j F\left(a_2\right) \cdots \check{F}\left(a_j\right) \cdots F\left(a_r\right)+F\left(a_1\right) \wedge\left\{F\left(a_2\right) \cdots F\left(a_r\right)\right\}\right] \\ &=\sum_{j=2}^r(-1)^{j-2} F\left(a_1\right)\left[F\left(a_j\right) F\left(a_2\right) \cdots \check{F}\left(a_j\right) \cdots F\left(a_r\right)+F\left(a_1\right) \wedge\left\{F\left(a_2\right) \cdots F\left(a_r\right)\right\}\right] \\ &=F\left(a_1\right)\left\{F\left(a_2\right) \cdots F\left(a_r\right)\right\}+F\left(a_1\right) \wedge\left\{F\left(a_2\right) \cdots F\left(a_r\right)\right\} \\ &=F\left(a_1\right) F\left(a_2\right) \cdots F\left(a_r\right) . \end{aligned} \tag{348}$ |
| 1285.5935v1_<br>E08112 (106)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 028  | <div><span>■</span> 0.348</div> <div> <div></div> <div>● K +0</div> </div> | $\left.a_{\{1\}} \downarrow \boldsymbol{B}_{\{s\}}=\left(a_{\{1\}} \backslash \text{right} \rfloor b_{\{1\}}\right) b_{\{2\}} \wedge \cdots \wedge b_{\{s\}}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $a_1 \downarrow \boldsymbol{B}_s=\left(a_1 \rfloor b_1\right) b_2 \wedge \cdots \wedge b_s .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $a_1 \rfloor \boldsymbol{B}_s=\left(a_1 \rfloor b_1\right) b_2 \wedge \cdots \wedge b_s .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 1285.5935v1_<br>E08465 (121)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 089  | <div><span>■</span> 0.349</div> <div> <div></div> <div>●</div> </div>      | $\begin{aligned} & \& a_{\{I\}}=a_{\{i_{\{1\}}\}} \wedge a_{\{i_{\{2\}}\}} \wedge \cdots \wedge a_{\{i_{\{r\}}\}} \backslash \& \\ & \& a^{\{I\}}=a^{\{i_{\{1\}}\}} \wedge a^{\{i_{\{2\}}\}} \wedge \cdots \wedge a^{\{i_{\{r\}}\}} . \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | $\begin{aligned} a_I &=a_{i_1} \wedge a_{i_2} \wedge \cdots \wedge a_{i_r} \\ a^I &=a^{i_1} \wedge a^{i_2} \wedge \cdots \wedge a^{i_r} . \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $\begin{aligned} a_I &=a_{i_1} \wedge a_{i_2} \wedge \cdots \wedge a_{i_r} \\ a^I &=a^{i_1} \wedge a^{i_2} \wedge \cdots \wedge a^{i_r} . \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <p><b>Conf.</b> MathPix confidence: <span>■</span> <math>\geq 0.9</math> <span>■</span> <math>0.5-0.9</math> <span>■</span> <math>&lt; 0.5</math> — no value</p> <p><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</p> <p><b>Residual</b> render vs scan (inkdrill): <span>●</span> C component <span>●</span> W weak <span>●</span> S stable <span>●</span> N noise <span>●</span> K clean <span>●</span> X crop overran its region <span>●</span> not measured</p> <p><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</p> |      |                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Page | Conf.             | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Scan image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| 1205.5935v1_<br>EQ0283 (275)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 058  | ■ 0.350<br>● N +0 | $\begin{aligned} & \begin{aligned} & \sum_i a^i a_i \downarrow A_r = \sum_i a^i \left[ (a_i \downarrow A_r) + \sum_i a^i \wedge (a_i \downarrow A_r) \right. \\ & \quad \left. = \sum_i (a^i \wedge a_i) \right] A_r + \sum_i a^i \wedge (a_i \downarrow A_r) \\ & = \left( \sum_i a^i \wedge a_i \right) \Bigg] A_r + \sum_i a^i \wedge (a_i \downarrow A_r) \\ & = \sum_i a^i \wedge (a_i \downarrow A_r) \end{aligned} \end{aligned}$                                                                                                                                                                                                                                                                                    | $\begin{aligned} \sum_i a^i a_i \downarrow A_r &= \sum_i a^i \left[ (a_i \downarrow A_r) + \sum_i a^i \wedge (a_i \downarrow A_r) \right. \\ & \quad \left. = \sum_i (a^i \wedge a_i) \right] A_r + \sum_i a^i \wedge (a_i \downarrow A_r) \\ & = \left( \sum_i a^i \wedge a_i \right) \Bigg] A_r + \sum_i a^i \wedge (a_i \downarrow A_r) \\ & = \sum_i a^i \wedge (a_i \downarrow A_r) \end{aligned}$                                                                                                                                                                                                                                                                                                                    | $\begin{aligned} \sum_i a^i a_i \downarrow A_r &= \sum_i a^i \left[ (a_i \downarrow A_r) + \sum_i a^i \wedge (a_i \downarrow A_r) \right. \\ & \quad \left. = \sum_i (a^i \wedge a_i) \right] A_r + \sum_i a^i \wedge (a_i \downarrow A_r) \\ & = \left( \sum_i a^i \wedge a_i \right) \Bigg] A_r + \sum_i a^i \wedge (a_i \downarrow A_r) \\ & = \sum_i a^i \wedge (a_i \downarrow A_r) \end{aligned}$                                                                                                                                                                                                                                                                                                                    |
| 1205.5935v1_<br>EQ0297 (289)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 060  | ■ 0.390<br>● N +0 | $\begin{aligned} & P_{\boldsymbol{A}_r} (B_s) = B_s \downarrow \boldsymbol{A}_r \boldsymbol{A}_r^{-1} \\ & = B_s \downarrow (a_1 \wedge \cdots \wedge a_r) \boldsymbol{A}_r^{-1} \\ & = \sum (-1)^{\sum_{j=1}^s (i_j - j)} (B_s \downarrow a_{i_1} \wedge \cdots \wedge a_{i_s}) (a_{i_{s+1}} \wedge \cdots \wedge a_{i_r}) \boldsymbol{A}_r^{-1} \\ & = (-1)^{s(s-1)/2} \sum (-1)^{\sum_{j=1}^s (i_j - 1)} (B_s \downarrow a_{i_1} \wedge \cdots \wedge a_{i_s}) (a_{i_{s+1}} \wedge \cdots \wedge a_{i_r}) \boldsymbol{A}_r^{-1} \\ & = \sum (-1)^{\sum_{j=1}^s (i_j - 1)} (B_s^\dagger \downarrow a_{i_1} \wedge \cdots \wedge a_{i_s}) (a_{i_{s+1}} \wedge \cdots \wedge a_{i_r}) \boldsymbol{A}_r^{-1}, \end{aligned}$ | $\begin{aligned} P_{\boldsymbol{A}_r} (B_s) &= B_s \downarrow \boldsymbol{A}_r \boldsymbol{A}_r^{-1} \\ & = B_s \downarrow (a_1 \wedge \cdots \wedge a_r) \boldsymbol{A}_r^{-1} \\ & = \sum (-1)^{\sum_{j=1}^s (i_j - j)} (B_s \downarrow a_{i_1} \wedge \cdots \wedge a_{i_s}) (a_{i_{s+1}} \wedge \cdots \wedge a_{i_r}) \boldsymbol{A}_r^{-1} \\ & = (-1)^{s(s-1)/2} \sum (-1)^{\sum_{j=1}^s (i_j - 1)} (B_s \downarrow a_{i_1} \wedge \cdots \wedge a_{i_s}) (a_{i_{s+1}} \wedge \cdots \wedge a_{i_r}) \boldsymbol{A}_r^{-1} \\ & = \sum (-1)^{\sum_{j=1}^s (i_j - 1)} (B_s^\dagger \downarrow a_{i_1} \wedge \cdots \wedge a_{i_s}) (a_{i_{s+1}} \wedge \cdots \wedge a_{i_r}) \boldsymbol{A}_r^{-1}, \end{aligned}$ | $\begin{aligned} P_{\boldsymbol{A}_r} (B_s) &= B_s \downarrow \boldsymbol{A}_r \boldsymbol{A}_r^{-1} \\ & = B_s \downarrow (a_1 \wedge \cdots \wedge a_r) \boldsymbol{A}_r^{-1} \\ & = \sum (-1)^{\sum_{j=1}^s (i_j - j)} (B_s \downarrow a_{i_1} \wedge \cdots \wedge a_{i_s}) (a_{i_{s+1}} \wedge \cdots \wedge a_{i_r}) \boldsymbol{A}_r^{-1} \\ & = (-1)^{s(s-1)/2} \sum (-1)^{\sum_{j=1}^s (i_j - 1)} (B_s \downarrow a_{i_1} \wedge \cdots \wedge a_{i_s}) (a_{i_{s+1}} \wedge \cdots \wedge a_{i_r}) \boldsymbol{A}_r^{-1} \\ & = \sum (-1)^{\sum_{j=1}^s (i_j - 1)} (B_s^\dagger \downarrow a_{i_1} \wedge \cdots \wedge a_{i_s}) (a_{i_{s+1}} \wedge \cdots \wedge a_{i_r}) \boldsymbol{A}_r^{-1}, \end{aligned}$ |
| 1205.5935v1_<br>EQ0104 (99)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 026  | ■ 0.392<br>● N +0 | $\begin{aligned} & C(a_{i_1}, \dots, a_{i_r}) = \sum_{j=1}^r (-1)^{\epsilon_j} \boldsymbol{B}_{r-1} \downarrow (a_{i_1} \wedge \cdots \wedge \check{a}_{i_j} \wedge \cdots \wedge a_{i_r}) a \downarrow a_{i_j} \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $C(a_{i_1}, \dots, a_{i_r}) = \sum_{j=1}^r (-1)^{\epsilon_j} \boldsymbol{B}_{r-1} \downarrow (a_{i_1} \wedge \cdots \wedge \check{a}_{i_j} \wedge \cdots \wedge a_{i_r}) a \downarrow a_{i_j}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | $C(a_{i_1}, \dots, a_{i_r}) = \sum_{j=1}^r (-1)^{\epsilon_j} \boldsymbol{B}_{r-1} \downarrow (a_{i_1} \wedge \cdots \wedge \check{a}_{i_j} \wedge \cdots \wedge a_{i_r}) a \downarrow a_{i_j}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 1205.5935v1_<br>EQ0117 (111)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 030  | ■ 0.404<br>● N +0 | $\begin{aligned} & A_r^2 B_s = A_r (A_r B_s) = A_r (A_r \downarrow B_s). \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $A_r^2 B_s = A_r (A_r B_s) = A_r (A_r \downarrow B_s).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $\boldsymbol{A}_r^2 \boldsymbol{B}_s = \boldsymbol{A}_r (\boldsymbol{A}_r \boldsymbol{B}_s) = \boldsymbol{A}_r (\boldsymbol{A}_r \downarrow \boldsymbol{B}_s).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <div><div>Conf. MathPix confidence: <span>■ ≥0.9</span> <span>■ 0.5–0.9</span> <span>■ &lt;0.5</span> <span>— no value</span></div><div>LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</div><div>Residual render vs scan (inkdrill): <span>● C component</span> <span>● W weak</span> <span>● S stable</span> <span>● N noise</span> <span>● K clean</span> <span>● X crop overran its region</span> <span>● not measured</span></div><div>[refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</div></div> |      |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Page | Conf.                                                                                                 | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                  | Rendered                                                                                                                                                                                                                                                                                                               | Scan image                                                                                                                                                                                                                                                                                                             |
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| 1205.5935v1_<br>E00377 (363)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 077  | <div> <div>■ 0.415</div> <ul style="list-style-type: none"> <li></li> </ul> </div>                    | $\begin{aligned} \operatorname{det}(P_{\boldsymbol{A}_r}) &= \delta_{rn} \\ \operatorname{det}(R_{\boldsymbol{A}_r}) &= \delta_{r0} . \end{aligned}$                                                                                                                                                                                                                                                          | $\det(P_{\boldsymbol{A}_r}) = \delta_{rn}$ $\det(R_{\boldsymbol{A}_r}) = \delta_{r0}.$                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                        |
| 1205.5935v1_<br>E00068 (67)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 020  | <div> <div>■ 0.419</div> <ul style="list-style-type: none"> <li>• <math>X +1</math></li> </ul> </div> | $\begin{aligned} A_r \lfloor a &= \langle A_r a \rangle_{r-1} \\ A_r \wedge a &= \langle A_r a \rangle_{r+1} \end{aligned}$                                                                                                                                                                                                                                                                                   | $A_r \lfloor a = \langle A_r a \rangle_{r-1}$ $A_r \wedge a = \langle A_r a \rangle_{r+1}$                                                                                                                                                                                                                             | $A_r \lfloor a = \langle A_r a \rangle_{r-1}$ $A_r \wedge a = \langle A_r a \rangle_{r+1} .$                                                                                                                                                                                                                           |
| 1205.5935v1_<br>E00238 (231)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 050  | <div> <div>■ 0.442</div> <ul style="list-style-type: none"> <li>• <math>W +0</math></li> </ul> </div> | $\begin{aligned} P_{\boldsymbol{A}_r}(b_1) \wedge \cdots \wedge P_{\boldsymbol{A}_r}(b_s) &= (b_1 \wedge \cdots \wedge b_s) \rfloor \boldsymbol{A}_r \boldsymbol{A}_r^{-1} \\ R_{\boldsymbol{A}_r}(b_1) \wedge \cdots \wedge R_{\boldsymbol{A}_r}(b_s) &= (b_1 \wedge \cdots \wedge b_s) \wedge \boldsymbol{A}_r \boldsymbol{A}_r^{-1} . \end{aligned}$                                                       | $P_{\boldsymbol{A}_r}(b_1) \wedge \cdots \wedge P_{\boldsymbol{A}_r}(b_s) = (b_1 \wedge \cdots \wedge b_s) \rfloor \boldsymbol{A}_r \boldsymbol{A}_r^{-1}$ $R_{\boldsymbol{A}_r}(b_1) \wedge \cdots \wedge R_{\boldsymbol{A}_r}(b_s) = (b_1 \wedge \cdots \wedge b_s) \wedge \boldsymbol{A}_r \boldsymbol{A}_r^{-1} .$ | $P_{\boldsymbol{A}_r}(b_1) \wedge \cdots \wedge P_{\boldsymbol{A}_r}(b_s) = (b_1 \wedge \cdots \wedge b_s) \rfloor \boldsymbol{A}_r \boldsymbol{A}_r^{-1}$ $R_{\boldsymbol{A}_r}(b_1) \wedge \cdots \wedge R_{\boldsymbol{A}_r}(b_s) = (b_1 \wedge \cdots \wedge b_s) \wedge \boldsymbol{A}_r \boldsymbol{A}_r^{-1} .$ |
| 1205.5935v1_<br>E00101 (96)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 026  | <div> <div>■ 0.448</div> <ul style="list-style-type: none"> <li>• <math>N +0</math></li> </ul> </div> | $\begin{aligned} &\& \left. = a \rfloor \left\lfloor \left( \boldsymbol{B}_{r-1} \rfloor (a_1 \wedge a_2 \wedge \cdots \wedge a_s) \right) \right. \right. \\ &\& \left. = \sum (-1)^{\sum_{j=1}^{r-1} (i_j - j)} \boldsymbol{B}_{r-1} \rfloor (a_{i_1} \wedge a_{i_2} \wedge \cdots \wedge a_{i_{r-1}}) \times \right. \\ &\& \left. a \rfloor [a_{i_r} \wedge \cdots \wedge a_{i_s}] \right. \end{aligned}$ | $= a \rfloor [\boldsymbol{B}_{r-1}] (a_1 \wedge a_2 \wedge \cdots \wedge a_s)]$ $= \sum (-1)^{\sum_{j=1}^{r-1} (i_j - j)} \boldsymbol{B}_{r-1} \rfloor (a_{i_1} \wedge a_{i_2} \wedge \cdots \wedge a_{i_{r-1}}) \times$ $a \rfloor [a_{i_r} \wedge \cdots \wedge a_{i_s}]$                                            | $= a \rfloor [\boldsymbol{B}_{r-1} \rfloor (a_1 \wedge a_2 \wedge \cdots \wedge a_s)]$ $= \sum (-1)^{\sum_{j=1}^{r-1} (i_j - j)} \boldsymbol{B}_{r-1} \rfloor (a_{i_1} \wedge a_{i_2} \wedge \cdots \wedge a_{i_{r-1}}) \times$ $a \rfloor [a_{i_r} \wedge \cdots \wedge a_{i_s}]$                                     |
| 1205.5935v1_<br>E00234 (227)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 049  | <div> <div>■ 0.450</div> <ul style="list-style-type: none"> <li>• <math>N +0</math></li> </ul> </div> | $\begin{aligned} a \downharpoonleft A_r^{\perp} (A_r^{\perp})^{-1} &= (a \wedge A_r)^{\perp} (A_r^{\perp})^{-1} \\ &= a \wedge A_r I^{-1} \boldsymbol{I} \boldsymbol{A}_r^{-1} \\ &= a \wedge A_r \boldsymbol{A}_r^{-1} . \end{aligned}$                                                                                                                                                                      | $a \rfloor A_r^{\perp} (A_r^{\perp})^{-1} = (a \wedge A_r)^{\perp} (A_r^{\perp})^{-1}$ $= a \wedge A_r I^{-1} \boldsymbol{I} \boldsymbol{A}_r^{-1}$ $= a \wedge A_r \boldsymbol{A}_r^{-1} .$                                                                                                                           | $a \rfloor \boldsymbol{A}_r^{\perp} (\boldsymbol{A}_r^{\perp})^{-1} = (a \wedge \boldsymbol{A}_r)^{\perp} (\boldsymbol{A}_r^{\perp})^{-1}$ $= a \wedge \boldsymbol{A}_r \boldsymbol{I}^{-1} \boldsymbol{I} \boldsymbol{A}_r^{-1}$ $= a \wedge \boldsymbol{A}_r \boldsymbol{A}_r^{-1} .$                                |
| 1205.5935v1_<br>E00231 (224)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 048  | <div> <div>■ 0.474</div> <ul style="list-style-type: none"> <li>• <math>X +1</math></li> </ul> </div> | $\begin{aligned} P_{\boldsymbol{A}_r}(a) &= a \rfloor \boldsymbol{A}_r \boldsymbol{A}_r^{-1} = (a \downharpoonleft \boldsymbol{A}_r) \rfloor \boldsymbol{A}_r^{-1} \\ R_{\boldsymbol{A}_r}(a) &= a \wedge \boldsymbol{A}_r \boldsymbol{A}_r^{-1} = a \wedge \boldsymbol{A}_r \rfloor \boldsymbol{A}_r^{-1} \end{aligned}$                                                                                     | $P_{\boldsymbol{A}_r}(a) = a \rfloor \boldsymbol{A}_r \boldsymbol{A}_r^{-1} = (a \downharpoonleft \boldsymbol{A}_r) \rfloor \boldsymbol{A}_r^{-1}$ $R_{\boldsymbol{A}_r}(a) = a \wedge \boldsymbol{A}_r \boldsymbol{A}_r^{-1} = a \wedge \boldsymbol{A}_r \rfloor \boldsymbol{A}_r^{-1}$                               | $P_{\boldsymbol{A}_r}(a) = a \rfloor \boldsymbol{A}_r \boldsymbol{A}_r^{-1} = (a \rfloor \boldsymbol{A}_r) \rfloor \boldsymbol{A}_r^{-1}$ $R_{\boldsymbol{A}_r}(a) = a \wedge \boldsymbol{A}_r \boldsymbol{A}_r^{-1} = a \wedge \boldsymbol{A}_r \rfloor \boldsymbol{A}_r^{-1} .$                                      |
| <p><b>Conf.</b> MathPix confidence: <span style="color: green;">■</span> <math>\geq 0.9</math> <span style="color: orange;">■</span> <math>0.5-0.9</math> <span style="color: red;">■</span> <math>&lt; 0.5</math> — no value</p> <p><b>LaTeX source</b> MathPix's reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</p> <p><b>Residual</b> render vs scan (inkdrill): <span style="color: red;">●</span> C component <span style="color: orange;">●</span> W weak <span style="color: blue;">●</span> S stable <span style="color: green;">●</span> N noise <span style="color: green;">●</span> K clean <span style="color: blue;">●</span> X crop overran its region <span style="color: grey;">●</span> not measured</p> <p><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</p> |      |                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                        |



| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Page | Conf.                                                            | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                     | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                         | Scan image                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| 1205.5935v1_E00236 (229)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 049  | <div><div>0.523</div><div></div></div>                           | $\begin{aligned} &P_{\boldsymbol{A}_r \wedge \boldsymbol{B}_s}(a) = a \downarrow \boldsymbol{A}_r \boldsymbol{A}_r^{-1} + (-1)^{r+r(s-1)+rs}(a \downarrow \boldsymbol{B}_s) \boldsymbol{A}_r \boldsymbol{A}_r^{-1} \boldsymbol{B}_s^{-1} \\ &= a \downarrow \boldsymbol{A}_r \boldsymbol{A}_r^{-1} + a \downarrow \boldsymbol{B}_s \boldsymbol{B}_s^{-1} \\ &= P_{\boldsymbol{A}_r}(a) + P_{\boldsymbol{B}_s}(a). \end{aligned}$ | $\begin{aligned} P_{\boldsymbol{A}_r \wedge \boldsymbol{B}_s}(a) &= a \downarrow \boldsymbol{A}_r \boldsymbol{A}_r^{-1} + (-1)^{r+r(s-1)+rs}(a \downarrow \boldsymbol{B}_s) \boldsymbol{A}_r \boldsymbol{A}_r^{-1} \boldsymbol{B}_s^{-1} \\ &= a \downarrow \boldsymbol{A}_r \boldsymbol{A}_r^{-1} + a \downarrow \boldsymbol{B}_s \boldsymbol{B}_s^{-1} \\ &= P_{\boldsymbol{A}_r}(a) + P_{\boldsymbol{B}_s}(a). \end{aligned}$ | $\begin{aligned} P_{\boldsymbol{A}_r \wedge \boldsymbol{B}_s}(a) &= a \downarrow \boldsymbol{A}_r \boldsymbol{A}_r^{-1} + (-1)^{r+r(s-1)+rs}(a \downarrow \boldsymbol{B}_s) \boldsymbol{A}_r \boldsymbol{A}_r^{-1} \boldsymbol{B}_s^{-1} \\ &= a \downarrow \boldsymbol{A}_r \boldsymbol{A}_r^{-1} + a \downarrow \boldsymbol{B}_s \boldsymbol{B}_s^{-1} \\ &= P_{\boldsymbol{A}_r}(a) + P_{\boldsymbol{B}_s}(a). \end{aligned}$ |
| 1205.5935v1_E00416 (20)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 084  | <div><div>0.529</div><div><div></div><div>N +0</div></div></div> | $\begin{aligned} &a \wedge (AB) = (a \wedge A)B - A^*(a \downarrow B) \\ &= (a \downarrow A)B + A^*(a \wedge B) \end{aligned}$                                                                                                                                                                                                                                                                                                   | $\begin{aligned} a \wedge (AB) &= (a \wedge A)B - A^*(a \downarrow B) \\ &= (a \downarrow A)B + A^*(a \wedge B) \end{aligned}$                                                                                                                                                                                                                                                                                                   | $\begin{aligned} a \wedge (AB) &= (a \wedge A)B - A^*(a \downarrow B) \\ &= (a \downarrow A)B + A^*(a \wedge B) \end{aligned}$                                                                                                                                                                                                                                                                                                   |
| 1205.5935v1_E00282 (274)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 058  | <div><div>0.531</div><div><div></div><div>X +1</div></div></div> | $\sum_i a^i a_i \downarrow A_r = \sum_i a^i \wedge (a_i \downarrow A_r) = r A_r$                                                                                                                                                                                                                                                                                                                                                 | $\sum_i a^i a_i \downarrow A_r = \sum_i a^i \wedge (a_i \downarrow A_r) = r A_r$                                                                                                                                                                                                                                                                                                                                                 | $\sum_i a^i a_i \downarrow A_r = \sum_i a^i \wedge (a_i \downarrow A_r) = r A_r,$                                                                                                                                                                                                                                                                                                                                                |
| 1205.5935v1_E00200 (193)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 042  | <div><div>0.531</div><div><div></div><div>N +0</div></div></div> | $\begin{aligned} &a \times A = a \downarrow \langle A \rangle_+ + a \wedge \langle A \rangle_- \\ &A \times a = \langle A \rangle_+ \downarrow a + \langle A \rangle_- \wedge a. \end{aligned}$                                                                                                                                                                                                                                  | $\begin{aligned} a \times A &= a \downarrow \langle A \rangle_+ + a \wedge \langle A \rangle_- \\ A \times a &= \langle A \rangle_+ \downarrow a + \langle A \rangle_- \wedge a. \end{aligned}$                                                                                                                                                                                                                                  | $\begin{aligned} a \times A &= a \downarrow \langle A \rangle_+ + a \wedge \langle A \rangle_- \\ A \times a &= \langle A \rangle_+ \downarrow a + \langle A \rangle_- \wedge a. \end{aligned}$                                                                                                                                                                                                                                  |
| 1205.5935v1_E00078 (76)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 022  | <div><div>0.531</div><div><div></div><div>N +0</div></div></div> | $\begin{aligned} &a \downarrow (a_1 a_2 \cdots a_r) = \frac{1}{2}(a a_1 a_2 \cdots a_r - (-1)^r a_1 a_2 \cdots a_r a) \\ &a \wedge (a_1 a_2 \cdots a_r) = \frac{1}{2}(a a_1 a_2 \cdots a_r + (-1)^r a_1 a_2 \cdots a_r a) \end{aligned}$                                                                                                                                                                                         | $\begin{aligned} a \downarrow (a_1 a_2 \cdots a_r) &= \frac{1}{2}(a a_1 a_2 \cdots a_r - (-1)^r a_1 a_2 \cdots a_r a) \\ a \wedge (a_1 a_2 \cdots a_r) &= \frac{1}{2}(a a_1 a_2 \cdots a_r + (-1)^r a_1 a_2 \cdots a_r a) \end{aligned}$                                                                                                                                                                                         | $\begin{aligned} a \downarrow (a_1 a_2 \cdots a_r) &= \frac{1}{2}(a a_1 a_2 \cdots a_r - (-1)^r a_1 a_2 \cdots a_r a) \\ a \wedge (a_1 a_2 \cdots a_r) &= \frac{1}{2}(a a_1 a_2 \cdots a_r + (-1)^r a_1 a_2 \cdots a_r a) \end{aligned}$                                                                                                                                                                                         |
| 1205.5935v1_E00037 (37)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 015  | <div><div>0.538</div><div><div></div><div>N +0</div></div></div> | $\begin{aligned} &a \downarrow \boldsymbol{A}_r = a \downarrow (a_1 \wedge a_2 \wedge \cdots \wedge a_r) \\ &= \sum_{j=1}^r (-1)^{j-1} (a \downarrow a_j) a_1 \wedge \cdots \wedge \check{a}_j \wedge \cdots \wedge a_r, \end{aligned}$                                                                                                                                                                                          | $\begin{aligned} a \downarrow \boldsymbol{A}_r &= a \downarrow (a_1 \wedge a_2 \wedge \cdots \wedge a_r) \\ &= \sum_{j=1}^r (-1)^{j-1} (a \downarrow a_j) a_1 \wedge \cdots \wedge \check{a}_j \wedge \cdots \wedge a_r, \end{aligned}$                                                                                                                                                                                          | $\begin{aligned} a \downarrow \boldsymbol{A}_r &= a \downarrow (a_1 \wedge a_2 \wedge \cdots \wedge a_r) \\ &= \sum_{j=1}^r (-1)^{j-1} (a \downarrow a_j) a_1 \wedge \cdots \wedge \check{a}_j \wedge \cdots \wedge a_r, \end{aligned}$                                                                                                                                                                                          |
| 1205.5935v1_E00100                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 025  | <div><div>0.539</div><div><div></div><div>N +0</div></div></div> | $\begin{aligned} &\boldsymbol{B}_r \downarrow (a_1 \wedge a_2 \wedge \cdots \wedge a_s) = (a \wedge \boldsymbol{B}_{r-1}) \downarrow (a_1 \wedge a_2 \wedge \cdots \wedge a_s) \end{aligned}$                                                                                                                                                                                                                                    | $\boldsymbol{B}_r \downarrow (a_1 \wedge a_2 \wedge \cdots \wedge a_s) = (a \wedge \boldsymbol{B}_{r-1}) \downarrow (a_1 \wedge a_2 \wedge \cdots \wedge a_s)$                                                                                                                                                                                                                                                                   | $\boldsymbol{B}_r \downarrow (a_1 \wedge a_2 \wedge \cdots \wedge a_s) = (a \wedge \boldsymbol{B}_{r-1}) \downarrow (a_1 \wedge a_2 \wedge \cdots \wedge a_s)$                                                                                                                                                                                                                                                                   |
| <div><div>Conf.</div><div>MathPix confidence:</div><div><div><div></div><div>≥0.9</div></div><div><div></div><div>0.5-0.9</div></div><div><div></div><div>&lt;0.5</div></div><div><div></div><div>— no value</div></div></div></div> <div><div>LaTeX source</div><div>MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</div></div> <div><div>Residual</div><div>render vs scan (inkdrill):<div><div></div><div>C component</div><div></div><div>W weak</div><div></div><div>S stable</div><div></div><div>N noise</div><div></div><div>K clean</div><div></div><div>X crop overran its region</div><div></div><div>not measured</div></div></div></div> <div><div>[refined: <i>basis</i>]</div><div>beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</div></div> |      |                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                  |

| Identifier                   | Page | Conf.                                                                  | LaTeX source                                                                                                                                                                                                                                                   | Rendered                                                                                                                                                                                                                                                                                                    | Scan image                                                                                                                                                                                                                                                                                                  |
|------------------------------|------|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1205.5935v1_<br>E00415 (18)  | 084  | <div> <div>0.541</div> <div> <div></div> <div>N +0</div> </div> </div> | $\begin{aligned} a \downharpoonleft (A \wedge B) &= (a \downharpoonleft A) \wedge B \\ &= (a \wedge A) \downharpoonleft B \end{aligned}$                                                                                                                       | $a \downharpoonleft (AB) = (a \downharpoonleft A)B + A^*(a \downharpoonleft B) = (a \wedge A)B - A^*(a \wedge B)$                                                                                                                                                                                           | $a \downharpoonleft (AB) = (a \downharpoonleft A)B + A^*(a \downharpoonleft B) = (a \wedge A)B - A^*(a \wedge B)$                                                                                                                                                                                           |
| 1205.5935v1_<br>E00271 (264) | 056  | <div> <div>0.544</div> <div> <div></div> <div></div> </div> </div>     | $\begin{aligned} \sum_i a_i \wedge a^i &= \sum_i a_i \wedge \left( \sum_j a^i \downharpoonleft a^j a_j \right) \\ &= \sum_{i,j} (a_i \wedge a_j) (a^i \downharpoonleft a^j) \\ &= 0 \end{aligned}$                                                             | $\sum_i a_i \wedge a^i = \sum_i a_i \wedge \left( \sum_j a^i \downharpoonleft a^j a_j \right) = \sum_{i,j} (a_i \wedge a_j) (a^i \downharpoonleft a^j) = 0$                                                                                                                                                 | $\sum_i a_i \wedge a^i = \sum_i a_i \wedge \left( \sum_j a^i \downharpoonleft a^j a_j \right) = \sum_{i,j} (a_i \wedge a_j) (a^i \downharpoonleft a^j) = 0$                                                                                                                                                 |
| 1205.5935v1_<br>E00103 (98)  | 026  | <div> <div>0.554</div> <div> <div></div> <div>N +0</div> </div> </div> | $\left. \boldsymbol{B} \right\rfloor (a_1 \wedge a_2 \wedge \dots \wedge a_s) = \sum C(a_{i_1}, \dots, a_{i_r}) a_{i_{r+1}} \wedge \dots \wedge a_{i_s}$                                                                                                       | $\boldsymbol{B}_r \downharpoonleft (a_1 \wedge a_2 \wedge \dots \wedge a_s) = \sum C(a_{i_1}, \dots, a_{i_r}) a_{i_{r+1}} \wedge \dots \wedge a_{i_s}$                                                                                                                                                      | $\boldsymbol{B}_r \downharpoonleft (a_1 \wedge a_2 \wedge \dots \wedge a_s) = \sum C(a_{i_1}, \dots, a_{i_r}) a_{i_{r+1}} \wedge \dots \wedge a_{i_s}$                                                                                                                                                      |
| 1205.5935v1_<br>E00166 (159) | 036  | <div> <div>0.555</div> <div> <div></div> <div>W +0</div> </div> </div> | $\begin{aligned} A * (B \downharpoonleft C) &= (A \downharpoonleft B) * C \\ A * (B \downharpoonleft C) &= (A \downharpoonleft B) * C \\ A * (B \downharpoonleft C) &= (A \downharpoonleft B) * C \end{aligned}$                                               | $\begin{aligned} A * (BC) &= (B^\dagger A) * C \\ A * (B \downharpoonleft C) &= (B^\dagger \downharpoonleft A) * C \\ A * (B \downharpoonleft C) &= (B^\dagger \wedge A) * C \\ A * (B \wedge C) &= (B^\dagger \downharpoonleft A) * C \end{aligned}$                                                       | $\begin{aligned} A * (BC) &= (B^\dagger A) * C \\ A * (B \downharpoonleft C) &= (B^\dagger \downharpoonleft A) * C \\ A * (B \downharpoonleft C) &= (B^\dagger \wedge A) * C \\ A * (B \wedge C) &= (B^\dagger \downharpoonleft A) * C \end{aligned}$                                                       |
| 1205.5935v1_<br>E00046 (46)  | 017  | <div> <div>0.555</div> <div> <div></div> <div>K +0</div> </div> </div> | $\frac{1}{2} (a_1 a_2 \dots a_{r-1}) = \frac{1}{2} (-1)^{r-1} (a_1 a_2 \dots a_{r-1} a) + \sum_{j=1}^{r-1} (-1)^{j-1} (a \downharpoonleft a_j) a_1 a_2 \dots \check{a}_j \dots a_{r-1}.$                                                                       | $\frac{1}{2} (a_1 a_2 \dots a_{r-1}) = \frac{1}{2} (-1)^{r-1} (a_1 a_2 \dots a_{r-1} a) + \sum_{j=1}^{r-1} (-1)^{j-1} (a \downharpoonleft a_j) a_1 a_2 \dots \check{a}_j \dots a_{r-1}.$                                                                                                                    | $\frac{1}{2} (a_1 a_2 \dots a_{r-1}) = \frac{1}{2} (-1)^{r-1} (a_1 a_2 \dots a_{r-1} a) + \sum_{j=1}^{r-1} (-1)^{j-1} (a \downharpoonleft a_j) a_1 a_2 \dots \check{a}_j \dots a_{r-1}.$                                                                                                                    |
| 1205.5935v1_<br>E00232 (225) | 049  | <div> <div>0.557</div> <div> <div></div> <div>N +0</div> </div> </div> | $\left. \boldsymbol{P} \right\rfloor (a \wedge \boldsymbol{A}_r) = (a \downharpoonleft \boldsymbol{A}_r) \wedge \boldsymbol{A}_r = \langle a \downharpoonleft \boldsymbol{A}_r \rangle_{r+1} = \langle a \downharpoonleft \boldsymbol{A}_r \rangle_{r+1} = 0.$ | $P_{\boldsymbol{A}_r}(a) \wedge \boldsymbol{A}_r = (a \downharpoonleft \boldsymbol{A}_r \boldsymbol{A}_r^{-1}) \wedge \boldsymbol{A}_r = \langle a \downharpoonleft \boldsymbol{A}_r \boldsymbol{A}_r^{-1} \boldsymbol{A}_r \rangle_{r+1} = \langle a \downharpoonleft \boldsymbol{A}_r \rangle_{r+1} = 0.$ | $P_{\boldsymbol{A}_r}(a) \wedge \boldsymbol{A}_r = (a \downharpoonleft \boldsymbol{A}_r \boldsymbol{A}_r^{-1}) \wedge \boldsymbol{A}_r = \langle a \downharpoonleft \boldsymbol{A}_r \boldsymbol{A}_r^{-1} \boldsymbol{A}_r \rangle_{r+1} = \langle a \downharpoonleft \boldsymbol{A}_r \rangle_{r+1} = 0.$ |

**Conf.** MathPix confidence:

≥0.9

0.5–0.9

<0.5

— no value

**LaTeX source** MathPix’s reading, unchanged;
**Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

**Residual** render vs scan (inkdrill):

C component

W weak

S stable

N noise

K clean

X crop overran its region

not measured

**[refined: *basis*]** beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.



| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Page | Conf.                                         | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                    | Rendered                                                                                                                                                                                                                                                                                                                                              | Scan image                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1285.5935v1_<br>E09467 (124)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 089  | <div> <div>0.593</div> <div>K +0</div> </div> | $\begin{aligned} A &= \sum_I A^I a_I \quad \text{where} \quad A^I = A * a^I \\ &= \sum_I A_I a^I \quad \text{where} \quad A_I = A * a_I. \end{aligned}$                                                                                                                                                                                                                                                         | $A = \sum_I A^I a_I \quad \text{where} \quad A^I = A * a^I$ $= \sum_I A_I a^I \quad \text{where} \quad A_I = A * a_I.$                                                                                                                                                                                                                                | $A = \sum_I A^I a_I \quad \text{where} \quad A^I = A * a^I$ $= \sum_I A_I a^I \quad \text{where} \quad A_I = A * a_I.$                                                                                                                                                                                                                                      |
| 1285.5935v1_<br>E09188 (181)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 040  | <div> <div>0.597</div> <div>N +0</div> </div> | $\begin{aligned} &\left. (A \wedge B)^{\perp} = A \right\rfloor B^{\perp} \\ &\left. (A \right\rfloor B)^{\perp} = A \wedge B^{\perp}. \end{aligned}$                                                                                                                                                                                                                                                           | $(A \wedge B)^{\perp} = A \right\rfloor B^{\perp}$ $(A \right\rfloor B)^{\perp} = A \wedge B^{\perp}.$                                                                                                                                                                                                                                                | $(A \wedge B)^{\perp} = A \right\rfloor B^{\perp}$ $(A \right\rfloor B)^{\perp} = A \wedge B^{\perp}.$                                                                                                                                                                                                                                                      |
| 1285.5935v1_<br>E09270 (263)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 056  | <div> <div>0.600</div> <div></div> </div>     | $\begin{aligned} \sum_i a_i a^i &= \sum_i a_i \downarrow a^i + \sum_i a_i \wedge a^i \\ &= n + \sum_i a_i \wedge a^i \end{aligned}$                                                                                                                                                                                                                                                                             | $\sum_i a_i a^i = \sum_i a_i \downarrow a^i + \sum_i a_i \wedge a^i$ $= n + \sum_i a_i \wedge a^i$                                                                                                                                                                                                                                                    | $\sum_i a_i a^i = \sum_i a_i \right\rfloor a^i + \sum_i a_i \wedge a^i$ $= n + \sum_i a_i \wedge a^i.$                                                                                                                                                                                                                                                      |
| 1285.5935v1_<br>E09084 (82)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 023  | <div> <div>0.605</div> <div>W +0</div> </div> | $\begin{aligned} &\left. a \downarrow (A_r \wedge B_s) = (a \downarrow A_r) \wedge B_s + (-1)^r A_r \wedge (a \right\rfloor B_s) \right. \\ &a \wedge (A_r \left\rfloor B_s) = (a \wedge A_r) \left\rfloor B_s - (-1)^r A_r \left\rfloor (a \right\rfloor B_s) \right. \\ &a \wedge (A_r \right\rfloor B_s) = (a \right\rfloor A_r) \right\rfloor B_s + (-1)^r A_r \right\rfloor (a \wedge B_s). \end{aligned}$ | $a \downarrow (A_r \wedge B_s) = (a \downarrow A_r) \wedge B_s + (-1)^r A_r \wedge (a \right\rfloor B_s)$ $a \wedge (A_r \left\rfloor B_s) = (a \wedge A_r) \left\rfloor B_s - (-1)^r A_r \left\rfloor (a \right\rfloor B_s)$ $a \wedge (A_r \right\rfloor B_s) = (a \right\rfloor A_r) \right\rfloor B_s + (-1)^r A_r \right\rfloor (a \wedge B_s).$ | $a \right\rfloor (A_r \wedge B_s) = (a \right\rfloor A_r) \wedge B_s + (-1)^r A_r \wedge (a \right\rfloor B_s)$ $a \wedge (A_r \left\rfloor B_s) = (a \wedge A_r) \left\rfloor B_s - (-1)^r A_r \left\rfloor (a \right\rfloor B_s)$ $a \wedge (A_r \right\rfloor B_s) = (a \right\rfloor A_r) \right\rfloor B_s + (-1)^r A_r \right\rfloor (a \wedge B_s).$ |
| 1285.5935v1_<br>E09039 (39)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 016  | <div> <div>0.608</div> <div>X +1</div> </div> | $\begin{aligned} &\boldsymbol{A}_{\boldsymbol{r}} = a_1 \wedge a_2 \wedge \cdots \wedge a_r \\ &= a_1 \wedge b_2 \wedge \cdots \wedge b_s \wedge a_{s+1} \wedge \cdots \wedge a_r. \end{aligned}$                                                                                                                                                                                                               | $\boldsymbol{A}_{\boldsymbol{r}} = a_1 \wedge a_2 \wedge \cdots \wedge a_r$ $= a_1 \wedge b_2 \wedge \cdots \wedge b_s \wedge a_{s+1} \wedge \cdots \wedge a_r.$                                                                                                                                                                                      | $\boldsymbol{A}_{\boldsymbol{r}} = a_1 \wedge a_2 \wedge \cdots \wedge a_r$ $= a_1 \wedge b_2 \wedge \cdots \wedge b_s \wedge a_{s+1} \wedge \cdots \wedge a_r.$                                                                                                                                                                                            |
| 1285.5935v1_<br>E09389 (375)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 079  | <div> <div>0.608</div> <div>N +0</div> </div> | $l^2 :=  L ^2 = L^{\dagger} L = -L^2 = m^2 r^4 \dot{\hat{r}}^2.$                                                                                                                                                                                                                                                                                                                                                | $l^2 :=  L ^2 = L^{\dagger} L = -L^2 = m^2 r^4 \dot{\hat{r}}^2.$                                                                                                                                                                                                                                                                                      | $l^2 :=  L ^2 = L^{\dagger} L = -L^2 = m^2 r^4 \dot{\hat{r}}^2.$                                                                                                                                                                                                                                                                                            |
| 1285.5935v1_<br>E09022 (22)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 006  | <div> <div>0.615</div> <div>X +1</div> </div> | $\begin{aligned} v^{\prime} &= -(n \cdot v) n^{-1} - (n \wedge v) n^{-1} \\ &= -(n \cdot v + n \wedge v) n^{-1} \\ &= -n v n^{-1}. \end{aligned}$                                                                                                                                                                                                                                                               | $v' = -(n \cdot v) n^{-1} - (n \wedge v) n^{-1}$ $= -(n \cdot v + n \wedge v) n^{-1}$ $= -n v n^{-1}.$                                                                                                                                                                                                                                                | $v' = -(n \cdot v) n^{-1} - (n \wedge v) n^{-1}$ $= -(n \cdot v + n \wedge v) n^{-1}$ $= -n v n^{-1}.$                                                                                                                                                                                                                                                      |
| 1285.5935v1_E09094                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 024  | <div> <div>0.624</div> <div>U +0</div> </div> | $\left. \left. \left. A_r \right\rfloor (B_s \right\rfloor C_t) = A_r \right\rfloor \langle B_s C_t \rangle_{t-s} \right.$                                                                                                                                                                                                                                                                                      | $A_r \right\rfloor (B_s \right\rfloor C_t) = A_r \right\rfloor \langle B_s C_t \rangle_{t-s}$                                                                                                                                                                                                                                                         | $A_r \right\rfloor (B_s \right\rfloor C_t) = A_r \right\rfloor \langle B_s C_t \rangle_{t-s}$                                                                                                                                                                                                                                                               |
| 1285.5935v1_<br>E09045 (45)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 017  | <div> <div>0.629</div> <div>N +0</div> </div> | $\frac{1}{2} [aa_1 a_2 \cdots a_r - (-1)^r a_1 a_2 \cdots a_r a] = \sum_{j=1}^r (-1)^{j-1} (a \downarrow a_j) a_1 a_2 \cdots \check{a}_j \cdots a_r.$                                                                                                                                                                                                                                                           | $\frac{1}{2} [aa_1 a_2 \cdots a_r - (-1)^r a_1 a_2 \cdots a_r a] = \sum_{j=1}^r (-1)^{j-1} (a \downarrow a_j) a_1 a_2 \cdots \check{a}_j \cdots a_r.$                                                                                                                                                                                                 | $\frac{1}{2} [aa_1 a_2 \cdots a_r - (-1)^r a_1 a_2 \cdots a_r a] = \sum_{j=1}^r (-1)^{j-1} (a \right\rfloor a_j) a_1 a_2 \cdots \check{a}_j \cdots a_r.$                                                                                                                                                                                                    |
| <p><b>Conf.</b> MathPix confidence: <span style="color: green;">■</span> <math>\geq 0.9</math> <span style="color: orange;">■</span> <math>0.5\text{--}0.9</math> <span style="color: red;">■</span> <math>&lt; 0.5</math> <span style="color: gray;">—</span> no value</p> <p><b>LaTeX source</b> MathPix's reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</p> <p><b>Residual</b> render vs scan (inkdrill): <span style="color: red;">●</span> C component <span style="color: orange;">●</span> W weak <span style="color: blue;">●</span> S stable <span style="color: gray;">●</span> N noise <span style="color: green;">●</span> K clean <span style="color: blue;">●</span> X crop overran its region <span style="color: gray;">●</span> not measured</p> <p><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</p> |      |                                               |                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                             |



| Identifier                | Page | Conf.           | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Scan image                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------|------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1205.5935v1_ E00167 (160) | 036  | ■0.641<br>●W +0 | $\begin{aligned} A*(B\ C) &= \left\langle A^{\dagger} B\ C \right\rangle \\ &= \left( A^{\dagger} B \right)^{\dagger} * C \\ &= \left( B^{\dagger} A \right) * C. \end{aligned}$                                                                                                                                                                                                                                                                                                                                         | $\begin{aligned} A*(BC) &= \left\langle A^{\dagger} BC \right\rangle \\ &= \left( A^{\dagger} B \right)^{\dagger} * C \\ &= \left( B^{\dagger} A \right) * C. \end{aligned}$                                                                                                                                                                                                                                                                                                  | $\begin{aligned} A*(BC) &= \left\langle A^{\dagger} BC \right\rangle \\ &= \left( A^{\dagger} B \right)^{\dagger} * C \\ &= \left( B^{\dagger} A \right) * C. \end{aligned}$                                                                                                                                                                                                                                                                                                  |
| 1205.5935v1_ E00418 (25)  | 084  | ■0.642<br>●     | $\begin{aligned} a \downharpoonleft \left( a_1 \wedge a_2 \wedge \cdots \wedge a_r \right) &= \sum_{j=1}^r (-1)^{j-1} (a \downharpoonleft a_j) a_1 \wedge a_2 \wedge \cdots \wedge \check{a}_j \wedge \cdots \wedge a_r \\ a_1 \wedge (a_2 \wedge \cdots \wedge a_r) &= a_1 \wedge a_2 \wedge \cdots \wedge a_r \end{aligned}$                                                                                                                                                                                           | $\begin{aligned} a \downharpoonleft (a_1 \wedge a_2 \wedge \cdots \wedge a_r) &= \sum_{j=1}^r (-1)^{j-1} (a \downharpoonleft a_j) a_1 \wedge a_2 \wedge \cdots \wedge \check{a}_j \wedge \cdots \wedge a_r \\ a_1 \wedge (a_2 \wedge \cdots \wedge a_r) &= a_1 \wedge a_2 \wedge \cdots \wedge a_r \end{aligned}$                                                                                                                                                             | $\begin{aligned} a \downharpoonleft (a_1 \wedge a_2 \wedge \cdots \wedge a_r) &= \sum_{j=1}^r (-1)^{j-1} (a \downharpoonleft a_j) a_1 \wedge a_2 \wedge \cdots \wedge \check{a}_j \wedge \cdots \wedge a_r \\ a_1 \wedge (a_2 \wedge \cdots \wedge a_r) &= a_1 \wedge a_2 \wedge \cdots \wedge a_r \end{aligned}$                                                                                                                                                             |
| 1205.5935v1_ E00242 (235) | 051  | ■0.643<br>●N +0 | $\begin{aligned} &P_{\boldsymbol{A}_r}(\boldsymbol{I}) = \boldsymbol{I} \delta_{rn} \\ &R_{\boldsymbol{A}_r}(\boldsymbol{I}) = \boldsymbol{I} \delta_{r0}. \end{aligned}$                                                                                                                                                                                                                                                                                                                                                | $\begin{aligned} P_{\boldsymbol{A}_r}(\boldsymbol{I}) &= \boldsymbol{I} \delta_{rn} \\ R_{\boldsymbol{A}_r}(\boldsymbol{I}) &= \boldsymbol{I} \delta_{r0}. \end{aligned}$                                                                                                                                                                                                                                                                                                     | $\begin{aligned} P_{\boldsymbol{A}_r}(\boldsymbol{I}) &= \boldsymbol{I} \delta_{rn} \\ R_{\boldsymbol{A}_r}(\boldsymbol{I}) &= \boldsymbol{I} \delta_{r0}. \end{aligned}$                                                                                                                                                                                                                                                                                                     |
| 1205.5935v1_ E00249 (242) | 052  | ■0.648<br>●N +0 | $\begin{aligned} a B_s a &= \left( a \downharpoonleft B_s \right) \lfloor a + (a \downharpoonleft B_s) \wedge a + (a \wedge B_s) \lfloor a + a \wedge B_s \wedge a. \end{aligned}$                                                                                                                                                                                                                                                                                                                                       | $a B_s a = (a \downharpoonleft B_s) \lfloor a + (a \downharpoonleft B_s) \wedge a + (a \wedge B_s) \lfloor a + a \wedge B_s \wedge a.$                                                                                                                                                                                                                                                                                                                                        | $a B_s a = (a \downharpoonleft B_s) \lfloor a + (a \downharpoonleft B_s) \wedge a + (a \wedge B_s) \lfloor a + a \wedge B_s \wedge a.$                                                                                                                                                                                                                                                                                                                                        |
| 1205.5935v1_ E00102 (97)  | 026  | ■0.664<br>●W +0 | $\begin{aligned} \left. \left( \boldsymbol{B}_{\boldsymbol{r}} \downharpoonleft \left( a_1 \wedge a_2 \wedge \cdots \wedge a_s \right) \right) \right. &= \sum_{j=1}^s (-1)^{j-1} \left( \boldsymbol{B}_{\boldsymbol{r}} \downharpoonleft a_j \right) a_1 \wedge a_2 \wedge \cdots \wedge \check{a}_j \wedge \cdots \wedge a_s \\ &= \sum_{j=1}^s (-1)^{j-1} \left( \boldsymbol{B}_{\boldsymbol{r}} \downharpoonleft a_j \right) a_1 \wedge a_2 \wedge \cdots \wedge \check{a}_j \wedge \cdots \wedge a_s \end{aligned}$ | $\begin{aligned} \boldsymbol{B}_{\boldsymbol{r}} \downharpoonleft (a_1 \wedge a_2 \wedge \cdots \wedge a_s) &= \sum_{j=1}^s (-1)^{j-1} \left( \boldsymbol{B}_{\boldsymbol{r}} \downharpoonleft a_j \right) a_1 \wedge a_2 \wedge \cdots \wedge \check{a}_j \wedge \cdots \wedge a_s \\ &= \sum_{j=1}^s (-1)^{j-1} \left( \boldsymbol{B}_{\boldsymbol{r}} \downharpoonleft a_j \right) a_1 \wedge a_2 \wedge \cdots \wedge \check{a}_j \wedge \cdots \wedge a_s \end{aligned}$ | $\begin{aligned} \boldsymbol{B}_{\boldsymbol{r}} \downharpoonleft (a_1 \wedge a_2 \wedge \cdots \wedge a_s) &= \sum_{j=1}^s (-1)^{j-1} \left( \boldsymbol{B}_{\boldsymbol{r}} \downharpoonleft a_j \right) a_1 \wedge a_2 \wedge \cdots \wedge \check{a}_j \wedge \cdots \wedge a_s \\ &= \sum_{j=1}^s (-1)^{j-1} \left( \boldsymbol{B}_{\boldsymbol{r}} \downharpoonleft a_j \right) a_1 \wedge a_2 \wedge \cdots \wedge \check{a}_j \wedge \cdots \wedge a_s \end{aligned}$ |
| 1205.5935v1_ E00417 (23)  | 084  | ■0.669<br>●N +0 | $\begin{aligned} a \downharpoonleft (A \wedge B) &= (a \downharpoonleft A) \wedge B + A^* \wedge (a \downharpoonleft B) \\ a \wedge (A \downharpoonleft B) &= (a \wedge A) \downharpoonleft B - A^* \downharpoonleft (a \downharpoonleft B) \\ a \wedge (A \downharpoonleft B) &= (a \downharpoonleft A) \downharpoonleft B + A^* \downharpoonleft (a \wedge B) \end{aligned}$                                                                                                                                           | $\begin{aligned} a \downharpoonleft (A \wedge B) &= (a \downharpoonleft A) \wedge B + A^* \wedge (a \downharpoonleft B) \\ a \wedge (A \downharpoonleft B) &= (a \wedge A) \downharpoonleft B - A^* \downharpoonleft (a \downharpoonleft B) \\ a \wedge (A \downharpoonleft B) &= (a \downharpoonleft A) \downharpoonleft B + A^* \downharpoonleft (a \wedge B) \end{aligned}$                                                                                                | $\begin{aligned} a \downharpoonleft (A \wedge B) &= (a \downharpoonleft A) \wedge B + A^* \wedge (a \downharpoonleft B) \\ a \wedge (A \downharpoonleft B) &= (a \wedge A) \downharpoonleft B - A^* \downharpoonleft (a \downharpoonleft B) \\ a \wedge (A \downharpoonleft B) &= (a \downharpoonleft A) \downharpoonleft B + A^* \downharpoonleft (a \wedge B) \end{aligned}$                                                                                                |
| 1205.5935v1_ E00038 (38)  | 015  | ■0.673<br>●N +0 | $\begin{aligned} b_j &= a_j - \left( \frac{a \downharpoonleft a_j}{a \downharpoonleft a_1} \right) a_1. \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                   | $b_j = a_j - \left( \frac{a \downharpoonleft a_j}{a \downharpoonleft a_1} \right) a_1.$                                                                                                                                                                                                                                                                                                                                                                                       | $b_j = a_j - \left( \frac{a \downharpoonleft a_j}{a \downharpoonleft a_1} \right) a_1.$                                                                                                                                                                                                                                                                                                                                                                                       |
| 1205.5935v1_ E00450 (100) | 088  | ■0.677<br>●K +0 | $\begin{aligned} \left( A^{\dagger} \right)^{\perp} &= \left[ I^{\dagger} (A^{\perp})^{\dagger} \right]^{\perp} \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                           | $\left( A^{\dagger} \right)^{\perp} = \left[ I^{\dagger} (A^{\perp})^{\dagger} \right]^{\perp}$                                                                                                                                                                                                                                                                                                                                                                               | $\left( A^{\dagger} \right)^{\perp} = \left[ I^{\dagger} (A^{\perp})^{\dagger} \right]^{\perp}$                                                                                                                                                                                                                                                                                                                                                                               |

Conf. MathPix confidence: 

■ ≥0.9

■ 0.5-0.9

■ <0.5

— no value

LaTeX source MathPix's reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): 

● C component

● W weak

● S stable

● N noise

● K clean

● X crop overran its region

● not measured

**[refined: basis]** beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

| Identifier                   | Page | Conf.                                          | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                 | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                            | Scan image |
|------------------------------|------|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| 1205.5935v1_<br>EQ0247 (240) | 052  | <div><div>■ 0.679</div><div>● X +1</div></div> | $\begin{aligned} & \begin{aligned} & \left( b_1 \wedge \cdots \wedge b_s \right)^{\prime} \colon = b_1^{\prime} \wedge \cdots \wedge b_s^{\prime} \\ & = \left( \boldsymbol{A}_r b_1^{*r} \boldsymbol{A}_r^{-1} \right) \wedge \cdots \wedge \left( \boldsymbol{A}_r^{-1} b_s^{*r} \boldsymbol{A}_r^{-1} \right) . \end{aligned} \end{aligned}$                                                                                              | $\begin{aligned} & \left( b_1 \wedge \cdots \wedge b_s \right)^{\prime} \colon = b_1^{\prime} \wedge \cdots \wedge b_s^{\prime} \\ & = \left( \boldsymbol{A}_r b_1^{*r} \boldsymbol{A}_r^{-1} \right) \wedge \cdots \wedge \left( \boldsymbol{A}_r^{-1} b_s^{*r} \boldsymbol{A}_r^{-1} \right) . \end{aligned}$                                                                                                                     |            |
| 1205.5935v1_<br>EQ0291 (283) | 059  | <div><div>■ 0.683</div><div>●</div></div>      | $\begin{aligned} & a^I a_N = (-1)^{i_r-1} \left( a^{i_1} \wedge \cdots \wedge a^{i_{r-1}} \right) \rfloor a_{i_r^c} \\ & = (-1)^{i_r-1} \left( a^{i_1} \wedge \cdots \wedge a^{i_{r-2}} \right) \rfloor \left( a^{i_{r-1}} \rfloor a_{i_r^c} \right) . \end{aligned}$                                                                                                                                                                        | $\begin{aligned} & a^I a_N = (-1)^{i_r-1} \left( a^{i_1} \wedge \cdots \wedge a^{i_{r-1}} \right) \rfloor a_{i_r^c} \\ & = (-1)^{i_r-1} \left( a^{i_1} \wedge \cdots \wedge a^{i_{r-2}} \right) \rfloor \left( a^{i_{r-1}} \rfloor a_{i_r^c} \right) . \end{aligned}$                                                                                                                                                               |            |
| 1205.5935v1_<br>EQ0079 (77)  | 022  | <div><div>■ 0.684</div><div>● N +0</div></div> | $a \downharpoonleft \left( a_1 \wedge a_2 \wedge \cdots \wedge a_r \right) = \sum_{j=1}^r (-1)^{j-1} \left( a \rfloor a_j \right) a_1 \wedge a_2 \wedge \cdots \wedge \check{a}_j \wedge \cdots \wedge a_r .$                                                                                                                                                                                                                                | $a \rfloor \left( a_1 \wedge a_2 \wedge \cdots \wedge a_r \right) = \sum_{j=1}^r (-1)^{j-1} \left( a \rfloor a_j \right) a_1 \wedge a_2 \wedge \cdots \wedge \check{a}_j \wedge \cdots \wedge a_r .$                                                                                                                                                                                                                                |            |
| 1205.5935v1_<br>EQ0205 (198) | 043  | <div><div>■ 0.684</div><div>● K +0</div></div> | $A_2 \times \left( a_1 \wedge a_2 \wedge \cdots \wedge a_r \right) = \sum_{j=1}^r a_1 \wedge a_2 \wedge \cdots \wedge \left( A_2 \rfloor a_j \right) \wedge \cdots \wedge a_r .$                                                                                                                                                                                                                                                             | $A_2 \times \left( a_1 \wedge a_2 \wedge \cdots \wedge a_r \right) = \sum_{j=1}^r a_1 \wedge a_2 \wedge \cdots \wedge \left( A_2 \rfloor a_j \right) \wedge \cdots \wedge a_r .$                                                                                                                                                                                                                                                    |            |
| 1205.5935v1_<br>EQ0076 (74)  | 021  | <div><div>■ 0.688</div><div>● W +0</div></div> | $\begin{aligned} & a \downharpoonleft \left( A_r B_s \right) = \left( a \rfloor A_r \right) B_s + (-1)^r A_r \left( a \rfloor B_s \right) \\ & = \left( a \wedge A_r \right) B_s - (-1)^r A_r \left( a \wedge B_s \right) \\ & a \wedge \left( A_r B_s \right) = \left( a \wedge A_r \right) B_s - (-1)^r A_r \left( a \rfloor B_s \right) \\ & = \left( a \rfloor A_r \right) B_s + (-1)^r A_r \left( a \wedge B_s \right) . \end{aligned}$ | $\begin{aligned} & a \rfloor \left( A_r B_s \right) = \left( a \rfloor A_r \right) B_s + (-1)^r A_r \left( a \rfloor B_s \right) \\ & = \left( a \wedge A_r \right) B_s - (-1)^r A_r \left( a \wedge B_s \right) \\ & a \wedge \left( A_r B_s \right) = \left( a \wedge A_r \right) B_s - (-1)^r A_r \left( a \rfloor B_s \right) \\ & = \left( a \rfloor A_r \right) B_s + (-1)^r A_r \left( a \wedge B_s \right) . \end{aligned}$ |            |
| 1205.5935v1_<br>EQ0192 (185) | 041  | <div><div>■ 0.699</div><div>● S +0</div></div> | $\begin{aligned} & \left( A^\dagger \right)^\perp = \left[ \left( A^\perp \boldsymbol{I} \right)^\dagger \right]^\perp \\ & = \left[ \boldsymbol{I}^\dagger \left( A^\perp \right)^\dagger \right]^\perp . \end{aligned}$                                                                                                                                                                                                                    | $\begin{aligned} & \left( A^\dagger \right)^\perp = \left[ \left( A^\perp \boldsymbol{I} \right)^\dagger \right]^\perp \\ & = \left[ \boldsymbol{I}^\dagger \left( A^\perp \right)^\dagger \right]^\perp . \end{aligned}$                                                                                                                                                                                                           |            |
| 1205.5935v1_<br>EQ0197 (190) | 042  | <div><div>■ 0.699</div><div>● K +0</div></div> | $A \times \left( B \times C \right) = \left( A \times B \right) \times C + B \times \left( A \times C \right) .$                                                                                                                                                                                                                                                                                                                             | $A \times \left( B \times C \right) = \left( A \times B \right) \times C + B \times \left( A \times C \right) .$                                                                                                                                                                                                                                                                                                                    |            |

**Conf.** MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

**LaTeX source** MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

**Residual** render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● X crop overran its region ● not measured

**[refined: *basis*]** beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Page | Conf.                                                                  | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Scan image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1205.5935v1_<br>EQ0324 (314)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 066  | <div> <div>0.700</div> <div> <div></div> <div>X +1</div> </div> </div> | $\begin{aligned} F(u)=0 \ \& \ \Longrightarrow F(u)\rfloor F(v)=0 \quad \text{for all } v \\ (v)=0 \quad \& \ \text{for all } v \ \& \ \Longrightarrow u \rfloor v = 0 \quad \text{for all } v \\ \downharpoonleft v=0 \quad \& \ \text{for all } v \ \& \ \Longrightarrow u=0 \end{aligned}$                                                                                                                                                                                                                                                                                                | $F(u)=0 \Longrightarrow F(u)\rfloor F(v)=0 \quad \text{for all } v$ $\Longrightarrow u \rfloor v = 0 \quad \text{for all } v$ $\Longrightarrow u = 0$                                                                                                                                                                                                                                                                                                                                                                                                      | $F(u)=0 \Longrightarrow F(u)\rfloor F(v)=0 \quad \text{for all } v$ $\Longrightarrow u \rfloor v = 0 \quad \text{for all } v$ $\Longrightarrow u = 0.$                                                                                                                                                                                                                                                                                                                                                                                                     |
| 1205.5935v1_<br>EQ0399 (383)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 080  | <div> <div>0.705</div> <div> <div></div> <div>N +0</div> </div> </div> | $\frac{l^2}{m}+(\boldsymbol{r} \downharpoonleft \boldsymbol{v})L=k(r+er\cos\theta+\boldsymbol{e} \wedge \boldsymbol{r}),$                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $\frac{l^2}{m}+(\boldsymbol{r} \rfloor \boldsymbol{v})L=k(r+er\cos\theta+\boldsymbol{e} \wedge \boldsymbol{r}),$                                                                                                                                                                                                                                                                                                                                                                                                                                           | $\frac{l^2}{m}+(\boldsymbol{r} \rfloor \boldsymbol{v})L=k(r+er\cos\theta+\boldsymbol{e} \wedge \boldsymbol{r}),$                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 1205.5935v1_<br>EQ0154 (147)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 034  | <div> <div>0.706</div> <div> <div></div> <div>K +0</div> </div> </div> | $A^{\ddagger}=A$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | $A^{\ddagger}=A,$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | $A^{\ddagger}=A,$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 1205.5935v1_<br>EQ0073 (71)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 021  | <div> <div>0.707</div> <div> <div></div> <div></div> </div> </div>     | $\begin{aligned} \left\langle \boldsymbol{A}_r \boldsymbol{B}_s \right\rangle_{s-r} &= a \left\langle \boldsymbol{A}_{r-1} \boldsymbol{B}_s \right\rangle_{s-r+1} \\ \left\langle \boldsymbol{A}_r \boldsymbol{B}_s \right\rangle_{s-r+2j} &= a \left\langle \boldsymbol{A}_{r-1} \boldsymbol{B}_s \right\rangle_{s-r+2j+1} + a \wedge \left\langle \boldsymbol{A}_{r-1} \boldsymbol{B}_s \right\rangle_{s-r+2j-1} \\ \left\langle \boldsymbol{A}_r \boldsymbol{B}_s \right\rangle_{r+s} &= a \wedge \left\langle \boldsymbol{A}_{r-1} \boldsymbol{B}_s \right\rangle_{s+r-1} \end{aligned}$ | $\left\langle \boldsymbol{A}_r \boldsymbol{B}_s \right\rangle_{s-r} = a \left\langle \boldsymbol{A}_{r-1} \boldsymbol{B}_s \right\rangle_{s-r+1}$ $\left\langle \boldsymbol{A}_r \boldsymbol{B}_s \right\rangle_{s-r+2j} = a \left\langle \boldsymbol{A}_{r-1} \boldsymbol{B}_s \right\rangle_{s-r+2j+1} + a \wedge \left\langle \boldsymbol{A}_{r-1} \boldsymbol{B}_s \right\rangle_{s-r+2j-1}$ $\left\langle \boldsymbol{A}_r \boldsymbol{B}_s \right\rangle_{r+s} = a \wedge \left\langle \boldsymbol{A}_{r-1} \boldsymbol{B}_s \right\rangle_{s+r-1}.$ | $\left\langle \boldsymbol{A}_r \boldsymbol{B}_s \right\rangle_{s-r} = a \left\langle \boldsymbol{A}_{r-1} \boldsymbol{B}_s \right\rangle_{s-r+1}$ $\left\langle \boldsymbol{A}_r \boldsymbol{B}_s \right\rangle_{s-r+2j} = a \left\langle \boldsymbol{A}_{r-1} \boldsymbol{B}_s \right\rangle_{s-r+2j+1} + a \wedge \left\langle \boldsymbol{A}_{r-1} \boldsymbol{B}_s \right\rangle_{s-r+2j-1}$ $\left\langle \boldsymbol{A}_r \boldsymbol{B}_s \right\rangle_{r+s} = a \wedge \left\langle \boldsymbol{A}_{r-1} \boldsymbol{B}_s \right\rangle_{s+r-1}.$ |
| 1205.5935v1_<br>EQ0225 (218)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 047  | <div> <div>0.712</div> <div> <div></div> <div>K +0</div> </div> </div> | $e_i e_j = \delta_{ij} + \sum_k \boldsymbol{I} \epsilon_{ijk} e_k$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $e_i e_j = \delta_{ij} + \sum_k \boldsymbol{I} \epsilon_{ijk} e_k,$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | $e_i e_j = \delta_{ij} + \sum_k \boldsymbol{I} \epsilon_{ijk} e_k,$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 1205.5935v1_<br>EQ0339 (328)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 069  | <div> <div>0.717</div> <div> <div></div> <div>X +1</div> </div> </div> | $\begin{aligned} a \, b &= a \downharpoonleft b + a \wedge b \\ &= \mp \sinh(\phi/2) \pm \boldsymbol{B} \cosh(\phi/2) \\ &= \pm \boldsymbol{B} (\cosh(\phi/2) - \boldsymbol{B} \sinh(\phi/2)) \\ &= \pm \boldsymbol{B} \exp(-\boldsymbol{B} \phi/2) \end{aligned}$                                                                                                                                                                                                                                                                                                                           | $ab = a \rfloor b + a \wedge b$ $= \mp \sinh(\phi/2) \pm \boldsymbol{B} \cosh(\phi/2)$ $= \pm \boldsymbol{B} (\cosh(\phi/2) - \boldsymbol{B} \sinh(\phi/2))$ $= \pm \boldsymbol{B} \exp(-\boldsymbol{B} \phi/2)$                                                                                                                                                                                                                                                                                                                                           | $ab = a \rfloor b + a \wedge b$ $= \mp \sinh(\phi/2) \pm \boldsymbol{B} \cosh(\phi/2)$ $= \pm \boldsymbol{B} (\cosh(\phi/2) - \boldsymbol{B} \sinh(\phi/2))$ $= \pm \boldsymbol{B} \exp(-\boldsymbol{B} \phi/2).$                                                                                                                                                                                                                                                                                                                                          |
| <p><b>Conf.</b> MathPix confidence: <span style="color: green;">■</span> ≥0.9 <span style="color: orange;">■</span> 0.5–0.9 <span style="color: red;">■</span> &lt;0.5 — no value</p> <p><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</p> <p><b>Residual</b> render vs scan (inkdrill): <span style="color: red;">●</span> C component <span style="color: orange;">●</span> W weak <span style="color: blue;">●</span> S stable <span style="color: grey;">●</span> N noise <span style="color: green;">●</span> K clean <span style="color: blue;">●</span> X crop overran its region <span style="color: grey;">●</span> not measured</p> <p><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</p> |      |                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Page | Conf.                                             | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                   | Rendered                                                                                                                                                                                                                                                                                                                                                                                   | Scan image                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1285.5935v1_<br>E08072 (70)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 021  | <div> <div>■ 0.720</div> <div>● N +0</div> </div> | $\begin{aligned} & \boldsymbol{B}_s = \left( \sum_{j=0}^{r-1} \left\lfloor a \right\rfloor \boldsymbol{B}_{s-r+2j+1} + a \wedge \boldsymbol{B}_{s-r+2j+1} \right) \\ & \boldsymbol{B}_s = \left( \left\lfloor a \right\rfloor \boldsymbol{B}_{s-r+1} + \sum_{j=1}^{r-1} \left\lfloor a \right\rfloor \boldsymbol{B}_{s-r+2j+1} + a \wedge \boldsymbol{B}_{s-r+2j+1} \right) + a \wedge \boldsymbol{B}_{s+r-1} . \end{aligned}$                 | $\boldsymbol{A}_r \boldsymbol{B}_s = \sum_{j=0}^{r-1} \left[ \left\lfloor a \right\rfloor \langle \boldsymbol{A}_{r-1} \boldsymbol{B}_s \rangle_{s-r+2j+1} + a \wedge \langle \boldsymbol{A}_{r-1} \boldsymbol{B}_s \rangle_{s-r+2j+1} \right]$                                                                                                                                            | $\boldsymbol{A}_r \boldsymbol{B}_s = \sum_{j=0}^{r-1} \left[ \left\lfloor a \right\rfloor \langle \boldsymbol{A}_{r-1} \boldsymbol{B}_s \rangle_{s-r+2j+1} + a \wedge \langle \boldsymbol{A}_{r-1} \boldsymbol{B}_s \rangle_{s-r+2j+1} \right] + a \wedge \langle \boldsymbol{A}_{r-1} \boldsymbol{B}_s \rangle_{s+r-1} .$                                                                 |
| 1285.5935v1_<br>E08096 (92)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 025  | <div> <div>■ 0.721</div> <div>● X +1</div> </div> | $\begin{aligned} & \left\lfloor A_r \right\rfloor \left( B_s \right) = \left\lfloor A_r \right\rfloor \left( B_s \right) \\ & \left\lfloor A_r \right\rfloor \left( B_s \right) = \left\lfloor A_r \right\rfloor \left( B_s \right) \\ & \left\lfloor A_r \right\rfloor \left( B_s \right) = \left\lfloor A_r \right\rfloor \left( B_s \right) \end{aligned}$                                                                                  | $\begin{aligned} A_r \lfloor B_s \rfloor C_t &= \langle A_r B_s C_t \rangle_{t-(r+s)} \\ &= \langle \langle A_r B_s \rangle_{r+s} C_t \rangle_{t-(r+s)} \\ &= \langle A_r B_s \rangle_{r+s} \lfloor C_t \rfloor \\ &= (A_r \wedge B_s) \lfloor C_t \rfloor \end{aligned}$                                                                                                                  | $\begin{aligned} A_r \lfloor (B_s \rfloor C_t) &= \langle A_r B_s C_t \rangle_{t-(r+s)} \\ &= \langle \langle A_r B_s \rangle_{r+s} C_t \rangle_{t-(r+s)} \\ &= \langle A_r B_s \rangle_{r+s} \lfloor C_t \rfloor \\ &= (A_r \wedge B_s) \lfloor C_t \rfloor . \end{aligned}$                                                                                                              |
| 1285.5935v1_<br>E08284 (276)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 058  | <div> <div>■ 0.724</div> <div>●</div> </div>      | $\begin{aligned} & \left( \sum_{i=1}^n a_i \right) \wedge \left( \sum_{j=1}^n a_j \right) = \sum_{i=1}^n a_i \wedge \left( \sum_{j=1}^n a_j \right) \\ & \left( \sum_{i=1}^n a_i \right) \wedge \left( \sum_{j=1}^n a_j \right) = \sum_{i=1}^n a_i \wedge \left( \sum_{j=1}^n a_j \right) \\ & \left( \sum_{i=1}^n a_i \right) \wedge \left( \sum_{j=1}^n a_j \right) = \sum_{i=1}^n a_i \wedge \left( \sum_{j=1}^n a_j \right) \end{aligned}$ | $\begin{aligned} \sum_i a^i \wedge (a_i \rfloor A_r) &= \sum_i a^i \wedge [a_i] (a^{j_1} \wedge \dots \wedge a^{j_r}) \\ &= \sum_{i,k} (-1)^{k-1} (a_i \rfloor a^{j_k}) a^i \wedge a^{j_1} \wedge \dots \wedge \check{a}^{j_k} \wedge \dots \wedge a^{j_r} \\ &= \sum_k (-1)^{k-1} a^{j_k} \wedge a^{j_1} \wedge \dots \wedge \check{a}^{j_k} \wedge \dots \wedge a^{j_r} . \end{aligned}$ | $\begin{aligned} \sum_i a^i \wedge (a_i \rfloor A_r) &= \sum_i a^i \wedge [a_i] (a^{j_1} \wedge \dots \wedge a^{j_r}) \\ &= \sum_{i,k} (-1)^{k-1} (a_i \rfloor a^{j_k}) a^i \wedge a^{j_1} \wedge \dots \wedge \check{a}^{j_k} \wedge \dots \wedge a^{j_r} \\ &= \sum_k (-1)^{k-1} a^{j_k} \wedge a^{j_1} \wedge \dots \wedge \check{a}^{j_k} \wedge \dots \wedge a^{j_r} . \end{aligned}$ |
| <p><b>Conf.</b> MathPix confidence: ■ ≥0.9   ■ 0.5-0.9   ■ &lt;0.5   — no value</p> <p><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</p> <p><b>Residual</b> render vs scan (inkdrill): ● C component   ● W weak   ● S stable   ● N noise   ● K clean   ● X crop overran its region   ● not measured</p> <p><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</p> |      |                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                            |

| Identifier                   | Page | Conf.                                                  | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Scan image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------|------|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1285.5935v1_<br>EQ0423 (34)  | 085  | <div><div></div>0.724</div> <div><div></div>W +0</div> | $\begin{aligned} & \& \left.\left.\left.P_{\boldsymbol{A}}\right.\right.\right\}(\boldsymbol{B})=\boldsymbol{B}\left\lfloor\boldsymbol{A}\right\rfloor\boldsymbol{A}^{-1} \\ & \boldsymbol{A}\left\lfloor\boldsymbol{A}\right\rfloor\boldsymbol{A}^{-1} \\ & =\left(\boldsymbol{B}\left\lfloor\boldsymbol{A}\right\rfloor\boldsymbol{A}^{-1}\right)\left\lfloor\boldsymbol{A}\right\rfloor\boldsymbol{A}^{-1} \\ & \& \boldsymbol{R}_{\boldsymbol{A}}\left\{\boldsymbol{A}\right\}(\boldsymbol{B})=\boldsymbol{B}\wedge\boldsymbol{A} \\ & \boldsymbol{A}\left\lfloor\boldsymbol{A}\right\rfloor\boldsymbol{A}^{-1} \\ & =\boldsymbol{B}\wedge\boldsymbol{A}\left\lfloor\boldsymbol{A}\right\rfloor\boldsymbol{A}^{-1} \\ & \left.\right.\right\} \end{aligned}$ | $\begin{aligned} P_{\boldsymbol{A}_r}(B) &= B \rfloor \boldsymbol{A}_r \boldsymbol{A}_r^{-1} = (B \rfloor \boldsymbol{A}_r) \rfloor \boldsymbol{A}_r^{-1} \\ R_{\boldsymbol{A}_r}(B) &= B \wedge \boldsymbol{A}_r \boldsymbol{A}_r^{-1} = B \wedge \boldsymbol{A}_r \rfloor \boldsymbol{A}_r^{-1} \end{aligned}$                                                                                                                                                                                                                                   | $\begin{aligned} P_{\boldsymbol{A}_r}(B) &= B \rfloor \boldsymbol{A}_r \boldsymbol{A}_r^{-1} = (B \rfloor \boldsymbol{A}_r) \rfloor \boldsymbol{A}_r^{-1} \\ R_{\boldsymbol{A}_r}(B) &= B \wedge \boldsymbol{A}_r \boldsymbol{A}_r^{-1} = B \wedge \boldsymbol{A}_r \rfloor \boldsymbol{A}_r^{-1} \end{aligned}$                                                                                                                                                                                                                                    |
| 1285.5935v1_<br>EQ0177 (170) | 037  | <div><div></div>0.725</div> <div><div></div>N +0</div> | $\begin{aligned} & 1 \& =\boldsymbol{B}^{\dagger} \boldsymbol{A}^{\dagger} \boldsymbol{A} \boldsymbol{B} \& \& = \boldsymbol{A} ^2 \boldsymbol{B}^{\dagger} \boldsymbol{B}, \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $\begin{aligned} 1 &= B^{\dagger} A^{\dagger} A B \\ &=  A ^2 B^{\dagger} B, \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                        | $\begin{aligned} 1 &= B^{\dagger} A^{\dagger} A B \\ &=  A ^2 B^{\dagger} B, \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 1285.5935v1_<br>EQ0088 (85)  | 024  | <div><div></div>0.734</div> <div><div></div>K +0</div> | $\boldsymbol{a}_1 \wedge \boldsymbol{a}_2 \wedge \cdots \wedge \boldsymbol{a}_r = \left\langle \boldsymbol{a}_1 \boldsymbol{a}_2 \cdots \boldsymbol{a}_r \right\rangle_r$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | $\boldsymbol{a}_1 \wedge \boldsymbol{a}_2 \wedge \cdots \wedge \boldsymbol{a}_r = \left\langle \boldsymbol{a}_1 \boldsymbol{a}_2 \cdots \boldsymbol{a}_r \right\rangle_r.$                                                                                                                                                                                                                                                                                                                                                                         | $\boldsymbol{a}_1 \wedge \boldsymbol{a}_2 \wedge \cdots \wedge \boldsymbol{a}_r = \left\langle \boldsymbol{a}_1 \boldsymbol{a}_2 \cdots \boldsymbol{a}_r \right\rangle_r.$                                                                                                                                                                                                                                                                                                                                                                          |
| 1285.5935v1_<br>EQ0462 (117) | 089  | <div><div></div>0.740</div> <div><div></div>K +0</div> | $\boldsymbol{A}_2 \times \left( \boldsymbol{a}_1 \wedge \boldsymbol{a}_2 \wedge \cdots \wedge \boldsymbol{a}_r \right) = \sum_{j=1}^r \boldsymbol{a}_1 \wedge \boldsymbol{a}_2 \wedge \cdots \wedge \left( \boldsymbol{A}_2 \rfloor \boldsymbol{a}_j \right) \wedge \cdots \wedge \boldsymbol{a}_r$                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $\boldsymbol{A}_2 \times \left( \boldsymbol{a}_1 \wedge \boldsymbol{a}_2 \wedge \cdots \wedge \boldsymbol{a}_r \right) = \sum_{j=1}^r \boldsymbol{a}_1 \wedge \boldsymbol{a}_2 \wedge \cdots \wedge \left( \boldsymbol{A}_2 \rfloor \boldsymbol{a}_j \right) \wedge \cdots \wedge \boldsymbol{a}_r$                                                                                                                                                                                                                                                | $\boldsymbol{A}_2 \times \left( \boldsymbol{a}_1 \wedge \boldsymbol{a}_2 \wedge \cdots \wedge \boldsymbol{a}_r \right) = \sum_{j=1}^r \boldsymbol{a}_1 \wedge \boldsymbol{a}_2 \wedge \cdots \wedge \left( \boldsymbol{A}_2 \rfloor \boldsymbol{a}_j \right) \wedge \cdots \wedge \boldsymbol{a}_r$                                                                                                                                                                                                                                                 |
| 1285.5935v1_<br>EQ0171 (164) | 037  | <div><div></div>0.740</div> <div><div></div>X +1</div> | $\begin{aligned} &  \boldsymbol{A} ^2 \& =\boldsymbol{A} * \boldsymbol{A} =\left\langle \boldsymbol{A}^{\dagger} \boldsymbol{A} \right\rangle \\ & \boldsymbol{A}\left\lfloor\boldsymbol{A}\right\rfloor\boldsymbol{A}^{-1} \\ & =\left\langle \boldsymbol{a}_1 \boldsymbol{a}_2 \cdots \boldsymbol{a}_r \right\rangle^{\dagger} \boldsymbol{a}_1 \boldsymbol{a}_2 \cdots \boldsymbol{a}_r \\ & =\left\langle \boldsymbol{a}_r \cdots \boldsymbol{a}_2 \boldsymbol{a}_1 \boldsymbol{a}_2 \cdots \boldsymbol{a}_r \right\rangle \\ & =\boldsymbol{a}_1^2 \boldsymbol{a}_2^2 \cdots \boldsymbol{a}_r^2 \end{aligned}$                                                                                                                                              | $\begin{aligned}  \boldsymbol{A} ^2 &= \boldsymbol{A} * \boldsymbol{A} = \left\langle \boldsymbol{A}^{\dagger} \boldsymbol{A} \right\rangle \\ &= \left\langle \left( \boldsymbol{a}_1 \boldsymbol{a}_2 \cdots \boldsymbol{a}_r \right)^{\dagger} \boldsymbol{a}_1 \boldsymbol{a}_2 \cdots \boldsymbol{a}_r \right\rangle \\ &= \left\langle \boldsymbol{a}_r \cdots \boldsymbol{a}_2 \boldsymbol{a}_1 \boldsymbol{a}_2 \cdots \boldsymbol{a}_r \right\rangle \\ &= \boldsymbol{a}_1^2 \boldsymbol{a}_2^2 \cdots \boldsymbol{a}_r^2 \end{aligned}$ | $\begin{aligned}  \boldsymbol{A} ^2 &= \boldsymbol{A} * \boldsymbol{A} = \left\langle \boldsymbol{A}^{\dagger} \boldsymbol{A} \right\rangle \\ &= \left\langle \left( \boldsymbol{a}_1 \boldsymbol{a}_2 \cdots \boldsymbol{a}_r \right)^{\dagger} \boldsymbol{a}_1 \boldsymbol{a}_2 \cdots \boldsymbol{a}_r \right\rangle \\ &= \left\langle \boldsymbol{a}_r \cdots \boldsymbol{a}_2 \boldsymbol{a}_1 \boldsymbol{a}_2 \cdots \boldsymbol{a}_r \right\rangle \\ &= \boldsymbol{a}_1^2 \boldsymbol{a}_2^2 \cdots \boldsymbol{a}_r^2. \end{aligned}$ |
| 1285.5935v1_<br>EQ0081 (79)  | 022  | <div><div></div>0.740</div> <div><div></div>N +0</div> | $\begin{aligned} & \boldsymbol{a}_1 \boldsymbol{a}_2 \cdots \boldsymbol{a}_r = \boldsymbol{a}_1 \rfloor \left( \boldsymbol{a}_2 \cdots \boldsymbol{a}_r \right) + \boldsymbol{a}_1 \wedge \left( \boldsymbol{a}_2 \cdots \boldsymbol{a}_r \right) \\ & = \sum_{j=2}^r (-1)^{j-2} \left( \boldsymbol{a}_1 \rfloor \boldsymbol{a}_j \right) \boldsymbol{a}_2 \cdots \check{\boldsymbol{a}}_j \cdots \boldsymbol{a}_r + \boldsymbol{a}_1 \wedge \left( \boldsymbol{a}_2 \cdots \boldsymbol{a}_r \right), \end{aligned}$                                                                                                                                                                                                                                             | $\begin{aligned} \boldsymbol{a}_1 \boldsymbol{a}_2 \cdots \boldsymbol{a}_r &= \boldsymbol{a}_1 \rfloor \left( \boldsymbol{a}_2 \cdots \boldsymbol{a}_r \right) + \boldsymbol{a}_1 \wedge \left( \boldsymbol{a}_2 \cdots \boldsymbol{a}_r \right) \\ &= \sum_{j=2}^r (-1)^{j-2} \left( \boldsymbol{a}_1 \rfloor \boldsymbol{a}_j \right) \boldsymbol{a}_2 \cdots \check{\boldsymbol{a}}_j \cdots \boldsymbol{a}_r + \boldsymbol{a}_1 \wedge \left( \boldsymbol{a}_2 \cdots \boldsymbol{a}_r \right), \end{aligned}$                                 | $\begin{aligned} \boldsymbol{a}_1 \boldsymbol{a}_2 \cdots \boldsymbol{a}_r &= \boldsymbol{a}_1 \rfloor \left( \boldsymbol{a}_2 \cdots \boldsymbol{a}_r \right) + \boldsymbol{a}_1 \wedge \left( \boldsymbol{a}_2 \cdots \boldsymbol{a}_r \right) \\ &= \sum_{j=2}^r (-1)^{j-2} \left( \boldsymbol{a}_1 \rfloor \boldsymbol{a}_j \right) \boldsymbol{a}_2 \cdots \check{\boldsymbol{a}}_j \cdots \boldsymbol{a}_r + \boldsymbol{a}_1 \wedge \left( \boldsymbol{a}_2 \cdots \boldsymbol{a}_r \right), \end{aligned}$                                  |
| 1285.5935v1_<br>EQ0051 (50)  | 018  | <div><div></div>0.745</div> <div><div></div></div>     | $\boldsymbol{a} \rfloor \boldsymbol{\lambda} = \boldsymbol{\lambda} \rfloor \boldsymbol{a} = 0.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $\boldsymbol{a} \rfloor \boldsymbol{\lambda} = \boldsymbol{\lambda} \rfloor \boldsymbol{a} = 0.$                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $\boldsymbol{a} \rfloor \boldsymbol{\lambda} = \boldsymbol{\lambda} \rfloor \boldsymbol{a} = 0.$                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 1285.5935v1_<br>EQ0243 (236) | 051  | <div><div></div>0.749</div> <div><div></div>N +0</div> | $\begin{aligned} & \boldsymbol{A} \rfloor \boldsymbol{B}_s = P_{\boldsymbol{B}_s}(\boldsymbol{A}) \boldsymbol{B}_s \\ & \boldsymbol{A} \wedge \boldsymbol{B}_s = R_{\boldsymbol{B}_s}(\boldsymbol{A}) \boldsymbol{B}_s. \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $\begin{aligned} \boldsymbol{A} \rfloor \boldsymbol{B}_s &= P_{\boldsymbol{B}_s}(\boldsymbol{A}) \boldsymbol{B}_s \\ \boldsymbol{A} \wedge \boldsymbol{B}_s &= R_{\boldsymbol{B}_s}(\boldsymbol{A}) \boldsymbol{B}_s. \end{aligned}$                                                                                                                                                                                                                                                                                                               | $\begin{aligned} \boldsymbol{A} \rfloor \boldsymbol{B}_s &= P_{\boldsymbol{B}_s}(\boldsymbol{A}) \boldsymbol{B}_s \\ \boldsymbol{A} \wedge \boldsymbol{B}_s &= R_{\boldsymbol{B}_s}(\boldsymbol{A}) \boldsymbol{B}_s. \end{aligned}$                                                                                                                                                                                                                                                                                                                |

**Conf.** MathPix confidence: 

≥0.9

0.5–0.9

<0.5

 — no value

**LaTeX source** MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

**Residual** render vs scan (inkdrill): 

C component

W weak

S stable

N noise

K clean

X crop overran its region

not measured

**[refined: *basis*]** beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.



| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Page | Conf.                                      | LaTeX source                                                                                                                                                                                                                                                                        | Rendered                                                                                                                                                                                                                                                                         | Scan image                                                                                                                                                                                                                                                                             |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1285.5935v1_<br>E00321 (311)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 065  | <div> <div>0.768</div> <div> </div> </div> | $\begin{aligned} a \circ b &= a \downharpoonleft (b \downharpoonleft A_2) \\ &= (a \wedge b) \downharpoonleft A_2 \\ &= -(b \wedge a) \downharpoonleft A_2 \\ &= -b \downharpoonleft (a \downharpoonleft A_2) \\ &= -b \circ a. \end{aligned}$                                      | $\begin{aligned} a \circ b &= a \downharpoonleft (b \downharpoonleft A_2) \\ &= (a \wedge b) \downharpoonleft A_2 \\ &= -(b \wedge a) \downharpoonleft A_2 \\ &= -b \downharpoonleft (a \downharpoonleft A_2) \\ &= -b \circ a. \end{aligned}$                                   | $\begin{aligned} a \circ b &= a \downharpoonleft (b \downharpoonleft A_2) \\ &= (a \wedge b) \downharpoonleft A_2 \\ &= -(b \wedge a) \downharpoonleft A_2 \\ &= -b \downharpoonleft (a \downharpoonleft A_2) \\ &= -b \circ a. \end{aligned}$                                         |
| 1285.5935v1_<br>E00343 (332)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 071  | <div> <div>0.768</div> <div> </div> </div> | $A = \sum_{s,t} \langle A_s A_t \rangle_{s+t}.$                                                                                                                                                                                                                                     | $A = \sum_{s,t} \langle A_s A_t \rangle_{s+t}.$                                                                                                                                                                                                                                  | $A = \sum_{s,t} \langle A_s A_t \rangle_{s+t}.$                                                                                                                                                                                                                                        |
| 1285.5935v1_<br>E00438 (69)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 086  | <div> <div>0.769</div> <div> </div> </div> | $\begin{aligned} & \& (A \downharpoonleft B)^{\ddagger} = B^{\ddagger} \downharpoonleft A^{\ddagger} \\ & (A \downharpoonleft B)^{\ddagger} = B^{\ddagger} \downharpoonleft A^{\ddagger} \\ & (A \wedge B)^{\ddagger} = B^{\ddagger} \wedge A^{\ddagger} \end{aligned}$             | $\begin{aligned} (A \downharpoonleft B)^{\ddagger} &= B^{\ddagger} \downharpoonleft A^{\ddagger} \\ (A \downharpoonleft B)^{\ddagger} &= B^{\ddagger} \downharpoonleft A^{\ddagger} \\ (A \wedge B)^{\ddagger} &= B^{\ddagger} \wedge A^{\ddagger} \end{aligned}$                | $\begin{aligned} (A \downharpoonleft B)^{\ddagger} &= B^{\ddagger} \downharpoonleft A^{\ddagger} \\ (A \downharpoonleft B)^{\ddagger} &= B^{\ddagger} \downharpoonleft A^{\ddagger} \\ (A \wedge B)^{\ddagger} &= B^{\ddagger} \wedge A^{\ddagger} \end{aligned}$                      |
| 1285.5935v1_<br>E00098 (94)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 025  | <div> <div>0.774</div> <div> </div> </div> | $\left. P_u(v) = (v \downharpoonleft u) \downharpoonleft u^{-1} \right.$                                                                                                                                                                                                            | $P_u(v) = (v \downharpoonleft u) \downharpoonleft u^{-1}.$                                                                                                                                                                                                                       | $P_u(v) = (v \downharpoonleft u) \downharpoonleft u^{-1}.$                                                                                                                                                                                                                             |
| 1285.5935v1_<br>E00086 (83)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 024  | <div> <div>0.786</div> <div> </div> </div> | $\begin{aligned} & A \downharpoonleft (B \downarrow C) = (A \downharpoonleft B) \downharpoonleft C \\ & A \downharpoonleft (B \downharpoonleft C) = (A \wedge B) \downharpoonleft C \\ & A \downharpoonleft (B \wedge C) = (A \downharpoonleft B) \downharpoonleft C \end{aligned}$ | $\begin{aligned} A \downharpoonleft (B \downarrow C) &= (A \downharpoonleft B) \downharpoonleft C \\ A \downharpoonleft (B \downharpoonleft C) &= (A \wedge B) \downharpoonleft C \\ A \downharpoonleft (B \wedge C) &= (A \downharpoonleft B) \downharpoonleft C \end{aligned}$ | $\begin{aligned} A \downharpoonleft (B \downharpoonleft C) &= (A \downharpoonleft B) \downharpoonleft C \\ A \downharpoonleft (B \downharpoonleft C) &= (A \wedge B) \downharpoonleft C \\ A \downharpoonleft (B \wedge C) &= (A \downharpoonleft B) \downharpoonleft C \end{aligned}$ |
| 1285.5935v1_<br>E00206 (199)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 043  | <div> <div>0.792</div> <div> </div> </div> | $A_2 \times (a_1 \wedge a_2 \wedge \cdots \wedge a_r) = A_2 \times (\boldsymbol{B}_{r-1} \wedge a_r)$                                                                                                                                                                               | $A_2 \times (a_1 \wedge a_2 \wedge \cdots \wedge a_r) = A_2 \times (\boldsymbol{B}_{r-1} \wedge a_r)$                                                                                                                                                                            | $A_2 \times (a_1 \wedge a_2 \wedge \cdots \wedge a_r) = A_2 \times (\boldsymbol{B}_{r-1} \wedge a_r)$                                                                                                                                                                                  |
| 1285.5935v1_<br>E00170 (163)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 036  | <div> <div>0.801</div> <div> </div> </div> | $ A ^2 =  A^* ^2 =  A^\dagger ^2 =  A^\ddagger ^2.$                                                                                                                                                                                                                                 | $ A ^2 =  A^* ^2 =  A^\dagger ^2 =  A^\ddagger ^2.$                                                                                                                                                                                                                              | $ A ^2 =  A^* ^2 =  A^\dagger ^2 =  A^\ddagger ^2.$                                                                                                                                                                                                                                    |
| 1285.5935v1_<br>E00424 (38)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 085  | <div> <div>0.803</div> <div> </div> </div> | $\begin{aligned} & \lambda^* = \lambda \\ & a^* = -a \\ & (AB)^* = A^* B^* \\ & (A+B)^* = A^* + B^* \end{aligned}$                                                                                                                                                                  | $\begin{aligned} \lambda^* &= \lambda \\ a^* &= -a \\ (AB)^* &= A^* B^* \\ (A+B)^* &= A^* + B^* \end{aligned}$                                                                                                                                                                   | $\begin{aligned} \lambda^* &= \lambda \\ a^* &= -a \\ (AB)^* &= A^* B^* \\ (A+B)^* &= A^* + B^* \end{aligned}$                                                                                                                                                                         |
| <p><b>Conf.</b> MathPix confidence: <span style="color: green;">■</span> <math>\geq 0.9</math> <span style="color: orange;">■</span> <math>0.5-0.9</math> <span style="color: red;">■</span> <math>&lt; 0.5</math> — no value</p> <p><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</p> <p><b>Residual</b> render vs scan (inkdrill): <span style="color: red;">●</span> C component <span style="color: orange;">●</span> W weak <span style="color: blue;">●</span> S stable <span style="color: grey;">●</span> N noise <span style="color: green;">●</span> K clean <span style="color: blue;">●</span> X crop overran its region <span style="color: grey;">●</span> not measured</p> <p><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</p> |      |                                            |                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                        |

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Page | Conf.           | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Scan image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1205.5935v1_ E00276 (268)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 057  | ■0.815<br>●     | \begin{aligned} & \left.\left.=\left(a^{i_r}\right) \wedge \cdots \wedge a^{i_2}\right)\left[\left(a^{i_1}\right)\left(a_{i_1} \wedge a_{i_2} \wedge \cdots \wedge a_{i_r}\right)\right]\right. \\ & \left.\left(a^{i_r} \wedge \cdots \wedge a^{i_2}\right)\left[\sum_{j=1}^r(-1)^{j-1} a^{i_1} \downarrow a_{i_j} a_{i_1} \wedge a_{i_2} \wedge \cdots \wedge \check{a}_{i_j} \wedge \cdots \wedge a_{i_r}\right]\right] \\ & \left(a^{i_r} \wedge \cdots \wedge a^{i_2}\right)\left(a_{i_2} \wedge \cdots \wedge a_{i_r}\right) \end{aligned}                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | $\begin{aligned}&=\left(a^{i_r} \wedge \cdots \wedge a^{i_2}\right)\left[\left.a^{i_1}\right]\left(a_{i_1} \wedge a_{i_2} \wedge \cdots \wedge a_{i_r}\right)\right] \\&=\left(a^{i_r} \wedge \cdots \wedge a^{i_2}\right)\left[\sum_{j=1}^r(-1)^{j-1} a^{i_1} \downarrow a_{i_j} a_{i_1} \wedge a_{i_2} \wedge \cdots \wedge \check{a}_{i_j} \wedge \cdots \wedge a_{i_r}\right] \\&=\left(a^{i_r} \wedge \cdots \wedge a^{i_2}\right)\left(a_{i_2} \wedge \cdots \wedge a_{i_r}\right)\end{aligned}$ | $\begin{aligned}&=\left(a^{i_r} \wedge \cdots \wedge a^{i_2}\right)\left[\left.a^{i_1}\right]\left(a_{i_1} \wedge a_{i_2} \wedge \cdots \wedge a_{i_r}\right)\right] \\&=\left(a^{i_r} \wedge \cdots \wedge a^{i_2}\right)\left[\sum_{j=1}^r(-1)^{j-1} a^{i_1} \downarrow a_{i_j} a_{i_1} \wedge a_{i_2} \wedge \cdots \wedge \check{a}_{i_j} \wedge \cdots \wedge a_{i_r}\right] \\&=\left(a^{i_r} \wedge \cdots \wedge a^{i_2}\right)\left(a_{i_2} \wedge \cdots \wedge a_{i_r}\right),\end{aligned}$ |
| 1205.5935v1_ E00252 (245)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 053  | ■0.817<br>●W +0 | \begin{aligned} & \boldsymbol{\mathrm{B}}_{\boldsymbol{\mathrm{s}}}^{\prime} \wedge \cdots \wedge \boldsymbol{\mathrm{b}}_{\boldsymbol{\mathrm{s}}}^{\prime} \\ & =\boldsymbol{\mathrm{b}}_1^{\prime} \wedge \cdots \wedge \boldsymbol{\mathrm{b}}_s^{\prime} \\ & =\left(\boldsymbol{\mathrm{A}}_{\boldsymbol{\mathrm{r}}} b_1^{* r} \boldsymbol{\mathrm{A}}_{\boldsymbol{\mathrm{r}}}^{-1}\right) \wedge \cdots \wedge\left(\boldsymbol{\mathrm{A}}_{\boldsymbol{\mathrm{r}}} b_s^{* r} \boldsymbol{\mathrm{A}}_{\boldsymbol{\mathrm{r}}}^{-1}\right) \\ & =\boldsymbol{\mathrm{A}}_{\boldsymbol{\mathrm{r}}}\left(b_1^{* r} \wedge \cdots \wedge b_s^{* r}\right) \boldsymbol{\mathrm{A}}_{\boldsymbol{\mathrm{r}}}^{-1} \\ & =\boldsymbol{\mathrm{A}}_{\boldsymbol{\mathrm{r}}}\left(b_1 \wedge \cdots \wedge b_s\right)^{* r} \boldsymbol{\mathrm{A}}_{\boldsymbol{\mathrm{r}}}^{-1} \\ & =\boldsymbol{\mathrm{A}}_{\boldsymbol{\mathrm{r}}} \boldsymbol{\mathrm{B}}_s^{* r} \boldsymbol{\mathrm{A}}_{\boldsymbol{\mathrm{r}}}^{-1} . \end{aligned} | $\begin{aligned} B'_s&:=b'_1\wedge\dots\wedge b'_s\\&=\left(A_rb_1^{*r}A_r^{-1}\right)\wedge\dots\wedge\left(A_rb_s^{*r}A_r^{-1}\right)\\&=A_r\left(b_1^{*r}\wedge\dots\wedge b_s^{*r}\right)A_r^{-1}\\&=A_r\left(b_1\wedge\dots\wedge b_s\right)^{*r}A_r^{-1}\\&=A_rB_s^{*r}A_r^{-1}.\end{aligned}$                                                                                                                                                                                                   | $\begin{aligned} B'_s&:=b'_1\wedge\dots\wedge b'_s\\&=\left(A_rb_1^{*r}A_r^{-1}\right)\wedge\dots\wedge\left(A_rb_s^{*r}A_r^{-1}\right)\\&=A_r\left(b_1^{*r}\wedge\dots\wedge b_s^{*r}\right)A_r^{-1}\\&=A_r\left(b_1\wedge\dots\wedge b_s\right)^{*r}A_r^{-1}\\&=A_rB_s^{*r}A_r^{-1}.\end{aligned}$                                                                                                                                                                                                    |
| 1205.5935v1_ E00118 (112)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 031  | ■0.817<br>●N +0 | \begin{aligned} & \lambda^* := \lambda \\ & a^* := -a \\ & (AB)^* := A^* B^* \\ & (A+B)^* := A^* + B^*. \end{aligned}                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $\begin{aligned} \lambda^*&:=\lambda\\a^*&:=-a\\(AB)^*&:=A^*B^*\\(A+B)^*&:=A^*+B^*.\end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                      | $\begin{aligned} \lambda^*&:=\lambda\\a^*&:=-a\\(AB)^*&:=A^*B^*\\(A+B)^*&:=A^*+B^*.\end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Conf.</b> MathPix confidence: ■≥0.9   ■0.5-0.9   ■<0.5   — no value<br><b>LaTeX source</b> MathPix's reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ<br><b>Residual</b> render vs scan (inkdrill): ●C component   ●W weak   ●S stable   ●N noise   ●K clean   ●X crop overran its region   ●not measured<br><b>[refined: basis]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here. |      |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |



| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Page | Conf.                                                                  | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Scan image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1205.5935v1_<br>EQ0155 (148)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 035  | <div> <div>0.860</div> <div> <div></div> <div>N +0</div> </div> </div> | $\langle A\rangle_{r}^{\dagger}=\left\langle A^{\dagger}\right\rangle _{r}\right\rangle _{r}\text{ , }$                                                                                                                                                                                                                                                                                                                                                                                                            | $\left\langle A\right\rangle _{r}^{\dagger}=\left\langle A^{\dagger}\right\rangle _{r}\text{ , }$                                                                                                                                                                                                                                                                                                                                                                                                                    | $\left\langle A\right\rangle _{r}^{\dagger}=\left\langle A^{\dagger}\right\rangle _{r}\text{ , }$                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 1205.5935v1_<br>EQ0082 (80)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 023  | <div> <div>0.862</div> <div> <div></div> <div>X +1</div> </div> </div> | $\begin{aligned} A\rfloor B&:=\sum_{r,s}\left\langle A_rB_s\right\rangle _{s-r} \\ A\lfloor B&:=\sum_{r,s}\left\langle A_rB_s\right\rangle _{r-s} \\ A\wedge B&:=\sum_{r,s}\left\langle A_rB_s\right\rangle _{r+s} \end{aligned}$                                                                                                                                                                                                                                                                                  | $\begin{aligned} A\rfloor B&:=\sum_{r,s}\left\langle A_rB_s\right\rangle _{s-r} \\ A\lfloor B&:=\sum_{r,s}\left\langle A_rB_s\right\rangle _{r-s} \\ A\wedge B&:=\sum_{r,s}\left\langle A_rB_s\right\rangle _{r+s} \end{aligned}$                                                                                                                                                                                                                                                                                    | $\begin{aligned} A\rfloor B&:=\sum_{r,s}\left\langle A_rB_s\right\rangle _{s-r} \\ A\lfloor B&:=\sum_{r,s}\left\langle A_rB_s\right\rangle _{r-s} \\ A\wedge B&:=\sum_{r,s}\left\langle A_rB_s\right\rangle _{r+s} . \end{aligned}$                                                                                                                                                                                                                                                                                  |
| 1205.5935v1_EQ0048                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 017  | <div> <div>0.863</div> <div> <div></div> <div></div> </div> </div>     | $\begin{aligned} &\frac{1}{2}\left(a_1a_2\cdots a_r-(-1)^ra_1a_2\cdots a_ra\right) \\ &= \frac{1}{2}(-1)^{r-1}\left(a_1a_2\cdots a_{r-1}aa_r+a_1a_2\cdots a_{r-1}a_ra\right) + \sum_{j=1}^{r-1}(-1)^{j-1}\left(a\rfloor a_j\right)a_1a_2\cdots \check{a}_j\cdots a_{r-1}a_r \\ &= (-1)^{r-1}\left(a\rfloor a_r\right)a_1a_2\cdots a_{r-1} + \sum_{j=1}^{r-1}(-1)^{j-1}\left(a\rfloor a_j\right)a_1a_2\cdots \check{a}_j\cdots a_{r-1}a_r \end{aligned}$                                                            | $\begin{aligned} &\frac{1}{2}\left[aa_1a_2\cdots a_r-(-1)^ra_1a_2\cdots a_ra\right] \\ &= \frac{1}{2}(-1)^{r-1}\left(a_1a_2\cdots a_{r-1}aa_r+a_1a_2\cdots a_{r-1}a_ra\right) + \sum_{j=1}^{r-1}(-1)^{j-1}\left(a\rfloor a_j\right)a_1a_2\cdots \check{a}_j\cdots a_{r-1}a_r \\ &= (-1)^{r-1}\left(a\rfloor a_r\right)a_1a_2\cdots a_{r-1} + \sum_{j=1}^{r-1}(-1)^{j-1}\left(a\rfloor a_j\right)a_1a_2\cdots \check{a}_j\cdots a_{r-1}a_r \end{aligned}$                                                             | $\begin{aligned} &\frac{1}{2}\left[aa_1a_2\cdots a_r-(-1)^ra_1a_2\cdots a_ra\right] \\ &= \frac{1}{2}(-1)^{r-1}\left(a_1a_2\cdots a_{r-1}aa_r+a_1a_2\cdots a_{r-1}a_ra\right) + \sum_{j=1}^{r-1}(-1)^{j-1}\left(a\rfloor a_j\right)a_1a_2\cdots \check{a}_j\cdots a_{r-1}a_r \\ &= (-1)^{r-1}\left(a\rfloor a_r\right)a_1a_2\cdots a_{r-1} + \sum_{j=1}^{r-1}(-1)^{j-1}\left(a\rfloor a_j\right)a_1a_2\cdots \check{a}_j\cdots a_{r-1}a_r \end{aligned}$                                                             |
| 1205.5935v1_<br>EQ0359 (347)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 074  | <div> <div>0.879</div> <div> <div></div> <div>C -3</div> </div> </div> | $\begin{aligned} &\left\langle \left[\overline{F}\right]\left(\left\{\left[F\right]\left(A^{\dagger}\right)\right\}\rfloor B\right)\right\rangle \\ &= \left[\overline{F}\right]\left(\left\langle \left\{\left[F\right]\left(A^{\dagger}\right)\right\}\rfloor B\right\rangle\right) \\ &= \left\langle \left\{\left[F\right]\left(A^{\dagger}\right)\right\}\rfloor B\right\rangle \\ &= \left\langle \left\{\left[F\right](A)\right\}^{\dagger}\rfloor B\right\rangle \\ &= \left[F\right](A)*B. \end{aligned}$ | $\begin{aligned} &= \left\langle \left[\overline{F}\right]\left(\left\{\left[F\right]\left(A^{\dagger}\right)\right\}\rfloor B\right)\right\rangle \\ &= \left[\overline{F}\right]\left(\left\langle \left\{\left[F\right]\left(A^{\dagger}\right)\right\}\rfloor B\right\rangle\right) \\ &= \left\langle \left\{\left[F\right]\left(A^{\dagger}\right)\right\}\rfloor B\right\rangle \\ &= \left\langle \left\{\left[F\right](A)\right\}^{\dagger}\rfloor B\right\rangle \\ &= \left[F\right](A)*B. \end{aligned}$ | $\begin{aligned} &= \left\langle \left[\overline{F}\right]\left(\left\{\left[F\right]\left(A^{\dagger}\right)\right\}\rfloor B\right)\right\rangle \\ &= \left[\overline{F}\right]\left(\left\langle \left\{\left[F\right]\left(A^{\dagger}\right)\right\}\rfloor B\right\rangle\right) \\ &= \left\langle \left\{\left[F\right]\left(A^{\dagger}\right)\right\}\rfloor B\right\rangle \\ &= \left\langle \left\{\left[F\right](A)\right\}^{\dagger}\rfloor B\right\rangle \\ &= \left[F\right](A)*B. \end{aligned}$ |
| 1205.5935v1_<br>EQ0355 (344)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 073  | <div> <div>0.880</div> <div> <div></div> <div>C -5</div> </div> </div> | $\begin{aligned} &A\rfloor\left\{\left[\overline{F}\right](B)\right\}=\left[\overline{F}\right]\left(\left\{\left[F\right](A)\right\}\rfloor B\right) \\ &\left\{\left[F\right](A)\right\}\rfloor B=\left[F\right]\left(A\rfloor\left\{\left[\overline{F}\right](B)\right\}\right) \end{aligned}$                                                                                                                                                                                                                  | $\begin{aligned} &A\rfloor\left\{\left[\overline{F}\right](B)\right\}=\left[\overline{F}\right]\left(\left\{\left[F\right](A)\right\}\rfloor B\right) \\ &\left\{\left[F\right](A)\right\}\rfloor B=\left[F\right]\left(A\rfloor\left\{\left[\overline{F}\right](B)\right\}\right) \end{aligned}$                                                                                                                                                                                                                    | $\begin{aligned} &A\rfloor\left\{\left[\overline{F}\right](B)\right\}=\left[\overline{F}\right]\left(\left\{\left[F\right](A)\right\}\rfloor B\right) \\ &\left\{\left[F\right](A)\right\}\rfloor B=\left[F\right]\left(A\rfloor\left\{\left[\overline{F}\right](B)\right\}\right) \end{aligned}$                                                                                                                                                                                                                    |
| 1205.5935v1_<br>EQ0263 (256)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 055  | <div> <div>0.887</div> <div> <div></div> <div>N +0</div> </div> </div> | $a^i\rfloor a_j=\delta_j^i.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $a^i\rfloor a_j=\delta_j^i.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | $a^i\rfloor a_j=\delta_j^i.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <p><b>Conf.</b> MathPix confidence: <span style="color: green;">■</span> ≥0.9 <span style="color: orange;">■</span> 0.5–0.9 <span style="color: red;">■</span> &lt;0.5 — no value</p> <p><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</p> <p><b>Residual</b> render vs scan (inkdrill): <span style="color: red;">●</span> C component <span style="color: orange;">●</span> W weak <span style="color: blue;">●</span> S stable <span style="color: grey;">●</span> N noise <span style="color: green;">●</span> K clean <span style="color: purple;">●</span> X crop overran its region <span style="color: grey;">●</span> not measured</p> <p><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</p> |      |                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |



| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Page | Conf.                                         | LaTeX source                                                                                                                                                                                                                                                                                                                      | Rendered                                                                                                                                                                                                                                                                                                                         | Scan image                                                                                                                                                                                                                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1285.5935v1_<br>EQ0108 (102)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 027  | <div> <div>0.896</div> <div>K +0</div> </div> | $\sum_{j=1}^r \left(i_j - j\right) = \sum_{j=1}^r \left(i_j - 1\right) - \frac{r(r-1)}{2}$                                                                                                                                                                                                                                        | $\sum_{j=1}^r (i_j - j) = \sum_{j=1}^r (i_j - 1) - \frac{r(r-1)}{2},$                                                                                                                                                                                                                                                            | $\sum_{j=1}^r (i_j - j) = \sum_{j=1}^r (i_j - 1) - \frac{r(r-1)}{2},$                                                                                                                                                                                                                                                            |
| 1285.5935v1_<br>EQ0254 (247)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 053  | <div> <div>0.897</div> <div>X +1</div> </div> | $\begin{aligned} \boldsymbol{I}^{\prime} &= \boldsymbol{A}_r \boldsymbol{I}^{*r} \boldsymbol{A}_r^{-1} \\ &= (-1)^{nr} \boldsymbol{A}_r \boldsymbol{I} \boldsymbol{A}_r^{-1} \\ &= (-1)^{nr} (-1)^{r(n-1)} \boldsymbol{I} \boldsymbol{A}_r \boldsymbol{A}_r^{-1} \\ &= (-1)^r \boldsymbol{I} \end{aligned}$                       | $\begin{aligned} \boldsymbol{I}' &= \boldsymbol{A}_r \boldsymbol{I}^{*r} \boldsymbol{A}_r^{-1} \\ &= (-1)^{nr} \boldsymbol{A}_r \boldsymbol{I} \boldsymbol{A}_r^{-1} \\ &= (-1)^{nr} (-1)^{r(n-1)} \boldsymbol{I} \boldsymbol{A}_r \boldsymbol{A}_r^{-1} \\ &= (-1)^r \boldsymbol{I} \end{aligned}$                              | $\begin{aligned} \boldsymbol{I}' &= \boldsymbol{A}_r \boldsymbol{I}^{*r} \boldsymbol{A}_r^{-1} \\ &= (-1)^{nr} \boldsymbol{A}_r \boldsymbol{I} \boldsymbol{A}_r^{-1} \\ &= (-1)^{nr} (-1)^{r(n-1)} \boldsymbol{I} \boldsymbol{A}_r \boldsymbol{A}_r^{-1} \\ &= (-1)^r \boldsymbol{I} \end{aligned}$                              |
| 1285.5935v1_<br>EQ0322 (312)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 065  | <div> <div>0.898</div> <div></div> </div>     | $\begin{aligned} A_2 &= \frac{1}{2} \sum_i a^i \wedge (a_i \rfloor A_2) \\ &= \frac{1}{2} \sum_i a^i \wedge F(a_i). \end{aligned}$                                                                                                                                                                                                | $\begin{aligned} A_2 &= \frac{1}{2} \sum_i a^i \wedge (a_i \rfloor A_2) \\ &= \frac{1}{2} \sum_i a^i \wedge F(a_i). \end{aligned}$                                                                                                                                                                                               | $\begin{aligned} A_2 &= \frac{1}{2} \sum_i a^i \wedge (a_i \rfloor A_2) \\ &= \frac{1}{2} \sum_i a^i \wedge F(a_i). \end{aligned}$                                                                                                                                                                                               |
| 1285.5935v1_<br>EQ0221 (214)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 046  | <div> <div>0.898</div> <div>X +1</div> </div> | $\pm  a_{\perp}   a_{\parallel}   b  \boldsymbol{I} = \pm  a_{\parallel}   \boldsymbol{B}  \boldsymbol{I}$                                                                                                                                                                                                                        | $\pm  a_{\perp}   a_{\parallel}   b  \boldsymbol{I} = \pm  a_{\parallel}   \boldsymbol{B}  \boldsymbol{I}$                                                                                                                                                                                                                       | $\pm  a_{\perp}   a_{\parallel}   b  \boldsymbol{I} = \pm  a_{\parallel}   \boldsymbol{B}  \boldsymbol{I}$                                                                                                                                                                                                                       |
| 1285.5935v1_<br>EQ0419 (27)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 084  | <div> <div>0.899</div> <div>N +0</div> </div> | $B_r \rfloor (a_1 \wedge a_2 \wedge \cdots \wedge a_s) = \sum (-1)^{\sum_{j=1}^r (i_j - j)} (B_r \rfloor a_{i_1} \wedge a_{i_2} \wedge \cdots \wedge a_{i_r}) a_{i_{r+1}} \wedge \cdots \wedge a_{i_s}$                                                                                                                           | $B_r \rfloor (a_1 \wedge a_2 \wedge \cdots \wedge a_s) = \sum (-1)^{\sum_{j=1}^r (i_j - j)} (B_r \rfloor a_{i_1} \wedge a_{i_2} \wedge \cdots \wedge a_{i_r}) a_{i_{r+1}} \wedge \cdots \wedge a_{i_s}$                                                                                                                          | $B_r \rfloor (a_1 \wedge a_2 \wedge \cdots \wedge a_s) = \sum (-1)^{\sum_{j=1}^r (i_j - j)} (B_r \rfloor a_{i_1} \wedge a_{i_2} \wedge \cdots \wedge a_{i_r}) a_{i_{r+1}} \wedge \cdots \wedge a_{i_s}$                                                                                                                          |
| 1285.5935v1_<br>EQ0241 (234)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 050  | <div> <div>0.902</div> <div>K +0</div> </div> | $\begin{aligned} P_{\boldsymbol{I}}(B) &= B \\ R_{\boldsymbol{I}}(B) &= \langle B \rangle. \end{aligned}$                                                                                                                                                                                                                         | $\begin{aligned} P_{\boldsymbol{I}}(B) &= B \\ R_{\boldsymbol{I}}(B) &= \langle B \rangle. \end{aligned}$                                                                                                                                                                                                                        | $\begin{aligned} P_{\boldsymbol{I}}(B) &= B \\ R_{\boldsymbol{I}}(B) &= \langle B \rangle. \end{aligned}$                                                                                                                                                                                                                        |
| 1285.5935v1_<br>EQ0382 (368)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 078  | <div> <div>0.903</div> <div>W +2</div> </div> | $\begin{aligned} G F(A) &= G[F(A)] \\ &= \bar{F}[F(A)] \boldsymbol{I} \boldsymbol{I}^{-1} \\ &= [A] \bar{F}(\boldsymbol{I}) \boldsymbol{I} \boldsymbol{I}^{-1} \\ &= \det(\bar{F})(A \rfloor \boldsymbol{I}) \boldsymbol{I} \boldsymbol{I}^{-1} \\ &= \det(F) A \boldsymbol{I} \boldsymbol{I}^{-1} \\ &= \det(F) A \end{aligned}$ | $\begin{aligned} GF(A) &= G[F(A)] \\ &= \bar{F}[F(A)] \boldsymbol{I} \boldsymbol{I}^{-1} \\ &= [A] \bar{F}(\boldsymbol{I}) \boldsymbol{I} \boldsymbol{I}^{-1} \\ &= \det(\bar{F})(A \rfloor \boldsymbol{I}) \boldsymbol{I} \boldsymbol{I}^{-1} \\ &= \det(F) A \boldsymbol{I} \boldsymbol{I}^{-1} \\ &= \det(F) A \end{aligned}$ | $\begin{aligned} GF(A) &= G[F(A)] \\ &= \bar{F}[F(A)] \boldsymbol{I} \boldsymbol{I}^{-1} \\ &= [A] \bar{F}(\boldsymbol{I}) \boldsymbol{I} \boldsymbol{I}^{-1} \\ &= \det(\bar{F})(A \rfloor \boldsymbol{I}) \boldsymbol{I} \boldsymbol{I}^{-1} \\ &= \det(F) A \boldsymbol{I} \boldsymbol{I}^{-1} \\ &= \det(F) A \end{aligned}$ |
| <p><b>Conf.</b> MathPix confidence: <span style="color: green;">■</span> ≥0.9 <span style="color: orange;">■</span> 0.5–0.9 <span style="color: red;">■</span> &lt;0.5 — no value</p> <p><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</p> <p><b>Residual</b> render vs scan (inkdrill): <span style="color: red;">●</span> C component <span style="color: orange;">●</span> W weak <span style="color: blue;">●</span> S stable <span style="color: grey;">●</span> N noise <span style="color: green;">●</span> K clean <span style="color: blue;">●</span> X crop overran its region <span style="color: grey;">●</span> not measured</p> <p><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</p> |      |                                               |                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |

| Identifier               | Page | Conf.                                       | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                  | Scan image                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------|------|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1205.5935v1_E00335 (324) | 068  | <div><div></div>0.905</div> <div>W +0</div> | $\begin{aligned} a\,b\,\&=a\,\downharpoonleft b+a\,\wedge b\,\backslash\backslash \\ \&=\cos\left(\theta/2\right)\boldsymbol{B}\,\sin\left(\theta/2\right)\,\backslash\backslash \\ \&=\exp\left(-\boldsymbol{B}\theta/2\right)\end{aligned}$                                                                                                                                                                                                                           | $\begin{aligned} ab&=a\downarrow b+a\wedge b \\ &=\cos(\theta/2)-\boldsymbol{B}\sin(\theta/2) \\ &=\exp(-\boldsymbol{B}\theta/2) \end{aligned}$                                                                                                                                                                                                                                                                           | $\begin{aligned} ab&=a\downarrow b+a\wedge b \\ &=\cos(\theta/2)-\boldsymbol{B}\sin(\theta/2) \\ &=\exp(-\boldsymbol{B}\theta/2) \end{aligned}$                                                                                                                                                                                                                                                                           |
| 1205.5935v1_E00426 (47)  | 085  | <div><div></div>0.905</div> <div>W +0</div> | $\begin{aligned} (A\rfloor B)^*&=A^*\leftarrow B^* \\ (A\lfloor B)^*&=A^*\llcorner B^* \\ (A\wedge B)^*&=A^*\wedge B^* \end{aligned}$                                                                                                                                                                                                                                                                                                                                   | $\begin{aligned} (A\rfloor B)^*&=A^*\leftarrow B^* \\ (A\lfloor B)^*&=A^*\llcorner B^* \\ (A\wedge B)^*&=A^*\wedge B^* \end{aligned}$                                                                                                                                                                                                                                                                                     | $\begin{aligned} (A\rfloor B)^*&=A^*\leftarrow B^* \\ (A\lfloor B)^*&=A^*\llcorner B^* \\ (A\wedge B)^*&=A^*\wedge B^* \end{aligned}$                                                                                                                                                                                                                                                                                     |
| 1205.5935v1_E00106       | 026  | <div><div></div>0.912</div> <div>N +0</div> | $C\left(a_{i_1},\ldots,a_{i_r}\right)=\sum_{j=1}^r\left(-1\right)^{\sum_{k=1}^r\left(i_k-k\right)+r-j}\boldsymbol{B}_{\boldsymbol{r}-1}\downharpoonleft\left(a_{i_1}\wedge\cdots\wedge\check{a}_{i_j}\wedge\cdots\wedge a_{i_r}\right)a\downarrow a_{i_j}$                                                                                                                                                                                                              | $C\left(a_{i_1},\ldots,a_{i_r}\right)=\sum_{j=1}^r\left(-1\right)^{\sum_{k=1}^r\left(i_k-k\right)+r-j}\boldsymbol{B}_{\boldsymbol{r}-1}\downarrow\left(a_{i_1}\wedge\cdots\wedge\check{a}_{i_j}\wedge\cdots\wedge a_{i_r}\right)a\downarrow a_{i_j}$                                                                                                                                                                      | $C(a_{i_1},\ldots,a_{i_r})=\sum_{j=1}^r(-1)^{\sum_{k=1}^r(i_k-k)+r-j}\boldsymbol{B}_{\boldsymbol{r}-1}\downarrow(a_{i_1}\wedge\cdots\wedge\check{a}_{i_j}\wedge\cdots\wedge a_{i_r})a\downarrow a_{i_j}$                                                                                                                                                                                                                  |
| 1205.5935v1_E00388 (374) | 079  | <div><div></div>0.915</div> <div></div>     | $L=-L^{\dagger}=-m\,r^2\dot{\hat{\boldsymbol{r}}}\hat{\boldsymbol{r}}$                                                                                                                                                                                                                                                                                                                                                                                                  | $L=-L^{\dagger}=-mr^2\dot{\hat{\boldsymbol{r}}}\hat{\boldsymbol{r}}$                                                                                                                                                                                                                                                                                                                                                      | $L=-L^{\dagger}=-mr^2\dot{\hat{\boldsymbol{r}}}\hat{\boldsymbol{r}}$                                                                                                                                                                                                                                                                                                                                                      |
| 1205.5935v1_E00194 (187) | 041  | <div><div></div>0.917</div> <div>N +0</div> | $\begin{aligned} A^{\perp}\ast B^{\perp}&=\left(A\boldsymbol{I}^{-1}\right)\ast\left(B\boldsymbol{I}^{-1}\right) \\ &=\left \boldsymbol{I}^{-1}\right ^2A\ast B \\ &=\left \boldsymbol{I}\right ^{-2}A\ast B. \end{aligned}$                                                                                                                                                                                                                                            | $\begin{aligned} A^{\perp}\ast B^{\perp}&=\left(A\boldsymbol{I}^{-1}\right)\ast\left(B\boldsymbol{I}^{-1}\right) \\ &=\left \boldsymbol{I}^{-1}\right ^2A\ast B \\ &=\left \boldsymbol{I}\right ^{-2}A\ast B. \end{aligned}$                                                                                                                                                                                              | $\begin{aligned} A^{\perp}\ast B^{\perp}&=\left(A\boldsymbol{I}^{-1}\right)\ast\left(B\boldsymbol{I}^{-1}\right) \\ &=\left \boldsymbol{I}^{-1}\right ^2A\ast B \\ &=\left \boldsymbol{I}\right ^{-2}A\ast B. \end{aligned}$                                                                                                                                                                                              |
| 1205.5935v1_E00063 (62)  | 019  | <div><div></div>0.919</div> <div>X +1</div> | $\begin{aligned} &a\,\wedge e_1\,\wedge e_2\,\wedge\cdots\wedge e_r \\ &=\frac{1}{r+1}\left[a\,e_1\,e_2\,\cdots\,e_r+\sum_{j=1}^r\left(\frac{1}{2}a\,e_1\,e_2\,\cdots\,e_r+\frac{1}{2}(-1)^re_1\,e_2\,\cdots\,e_ra+\right.\right. \\ &\quad\left.\left.(-1)^j(a\rfloor e_j)\,e_1\,e_2\,\cdots\,\check{e}_j\,\cdots\,e_r\right)\right] \\ &=\left(\frac{1}{r+1}+\frac{r}{2r+2}\right)a\,e_1\,e_2\,\cdots\,e_r+\frac{r}{2r+2}(-1)^re_1\,e_2\,\cdots\,e_ra- \end{aligned}$ | $\begin{aligned} &a\wedge e_1\wedge e_2\wedge\cdots\wedge e_r \\ &=\frac{1}{r+1}\left[ae_1e_2\cdots e_r+\sum_{j=1}^r\left(\frac{1}{2}ae_1e_2\cdots e_r+\frac{1}{2}(-1)^re_1e_2\cdots e_ra+\right.\right. \\ &\quad\left.\left.(-1)^j(a\rfloor e_j)\,e_1e_2\cdots\check{e}_j\cdots e_r\right)\right] \\ &=\left(\frac{1}{r+1}+\frac{r}{2r+2}\right)ae_1e_2\cdots e_r+\frac{r}{2r+2}(-1)^re_1e_2\cdots e_ra- \end{aligned}$ | $\begin{aligned} &a\wedge e_1\wedge e_2\wedge\cdots\wedge e_r \\ &=\frac{1}{r+1}\left[ae_1e_2\cdots e_r+\sum_{j=1}^r\left(\frac{1}{2}ae_1e_2\cdots e_r+\frac{1}{2}(-1)^re_1e_2\cdots e_ra+\right.\right. \\ &\quad\left.\left.(-1)^j(a\rfloor e_j)\,e_1e_2\cdots\check{e}_j\cdots e_r\right)\right] \\ &=\left(\frac{1}{r+1}+\frac{r}{2r+2}\right)ae_1e_2\cdots e_r+\frac{r}{2r+2}(-1)^re_1e_2\cdots e_ra- \end{aligned}$ |
| 1205.5935v1_E00020 (20)  | 006  | <div><div></div>0.921</div> <div>X +1</div> | $\begin{aligned} v&=v\left(nn^{-1}\right) \\ &=(vn)n^{-1} \\ &=(v\cdot n)n^{-1}+(v\wedge n)n^{-1} \end{aligned}$                                                                                                                                                                                                                                                                                                                                                        | $\begin{aligned} v&=v\left(nn^{-1}\right) \\ &=(vn)n^{-1} \\ &=(v\cdot n)n^{-1}+(v\wedge n)n^{-1} \end{aligned}$                                                                                                                                                                                                                                                                                                          | $\begin{aligned} v&=v\left(nn^{-1}\right) \\ &=(vn)n^{-1} \\ &=(v\cdot n)n^{-1}+(v\wedge n)n^{-1}. \end{aligned}$                                                                                                                                                                                                                                                                                                         |

**Conf.** MathPix confidence: 

≥0.9

0.5–0.9

<0.5

— no value

**LaTeX source** MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

**Residual** render vs scan (inkdrill): 

C component

W weak

S stable

N noise

K clean

X crop overran its region

not measured

**[refined: *basis*]** beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.



| Identifier               | Page | Conf.                                         | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Scan image                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------|------|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1205.5935v1_EQ0211 (204) | 044  | <div><div>■0.950</div><div>● X +1</div></div> | $\begin{aligned} Z^{\{\prime\}} &= e_{\{1\}} z^{\{\prime\}} \quad \& \& = -e_{\{1\}} \left( e_{\{2\}} z_{\{2\}} \right) \quad \& \& = -e_{\{1\}} e_{\{2\}} \left( e_1 Z \right) e_2 \\ &= x - \boldsymbol{I} y \quad \& \& = x - \boldsymbol{I} y \quad \& \& = Z^{\dagger} \end{aligned}$                                                                                                                                                                                      | $\begin{aligned} Z' &= e_1 z' \\ &= -e_1 \left( e_2 z e_2 \right) \\ &= -e_1 e_2 \left( e_1 Z \right) e_2 \\ &= -\boldsymbol{I} e_1 \left( x + \boldsymbol{I} y \right) e_2 \\ &= x - \boldsymbol{I} y \\ &= Z^\dagger \end{aligned}$                                                                                                                                                                                                                                     | $\begin{aligned} Z' &= e_1 z' \\ &= -e_1 \left( e_2 z e_2 \right) \\ &= -e_1 e_2 \left( e_1 Z \right) e_2 \\ &= -\boldsymbol{I} e_1 \left( x + \boldsymbol{I} y \right) e_2 \\ &= x - \boldsymbol{I} y \\ &= Z^\dagger, \end{aligned}$                                                                                                                                                                                                                                     |
| 1205.5935v1_EQ0062 (61)  | 019  | <div><div>■0.953</div><div>●</div></div>      | $\begin{aligned} &\& \& \frac{1}{r!} \sum_{\pi_j} (\operatorname{sgn} \pi_j) e_j a_{\pi_j(1)} a_{\pi_j(2)} \cdots a_{\pi_j(r)} \\ &= \frac{1}{2} (-1)^{j-1} a e_j e_1 e_2 \cdots \check{e}_j \cdots e_r + (-1)^j (a \rfloor e_j) e_1 e_2 \cdots \check{e}_j \cdots e_r + \frac{1}{2} (-1)^r e_1 e_2 \cdots e_r a \\ &= \frac{1}{2} a e_1 e_2 \cdots e_r + \frac{1}{2} (-1)^r e_1 e_2 \cdots e_r a + (-1)^j (a \rfloor e_j) e_1 e_2 \cdots \check{e}_j \cdots e_r \end{aligned}$ | $\begin{aligned} &\frac{1}{r!} \sum_{\pi_j} (\operatorname{sgn} \pi_j) e_j a_{\pi_j(1)} a_{\pi_j(2)} \cdots a_{\pi_j(r)} \\ &= \frac{1}{2} (-1)^{j-1} a e_j e_1 e_2 \cdots \check{e}_j \cdots e_r + (-1)^j (a \rfloor e_j) e_1 e_2 \cdots \check{e}_j \cdots e_r + \frac{1}{2} (-1)^r e_1 e_2 \cdots e_r a \\ &= \frac{1}{2} a e_1 e_2 \cdots e_r + \frac{1}{2} (-1)^r e_1 e_2 \cdots e_r a + (-1)^j (a \rfloor e_j) e_1 e_2 \cdots \check{e}_j \cdots e_r \end{aligned}$ | $\begin{aligned} &\frac{1}{r!} \sum_{\pi_j} (\operatorname{sgn} \pi_j) e_j a_{\pi_j(1)} a_{\pi_j(2)} \cdots a_{\pi_j(r)} \\ &= \frac{1}{2} (-1)^{j-1} a e_j e_1 e_2 \cdots \check{e}_j \cdots e_r + (-1)^j (a \rfloor e_j) e_1 e_2 \cdots \check{e}_j \cdots e_r + \frac{1}{2} (-1)^r e_1 e_2 \cdots e_r a \\ &= \frac{1}{2} a e_1 e_2 \cdots e_r + \frac{1}{2} (-1)^r e_1 e_2 \cdots e_r a + (-1)^j (a \rfloor e_j) e_1 e_2 \cdots \check{e}_j \cdots e_r. \end{aligned}$ |
| 1205.5935v1_EQ0156 (149) | 035  | <div><div>■0.955</div><div>● N +0</div></div> | $\left( A^{\{-1\}} \right)^{\dagger\dagger} = \left( A^{\{\dagger\dagger\}} \right)^{\{-1\}}$                                                                                                                                                                                                                                                                                                                                                                                   | $\left( A^{-1} \right)^\ddagger = \left( A^\ddagger \right)^{-1},$                                                                                                                                                                                                                                                                                                                                                                                                        | $\left( A^{-1} \right)^\ddagger = \left( A^\ddagger \right)^{-1},$                                                                                                                                                                                                                                                                                                                                                                                                         |
| 1205.5935v1_EQ0358       | 073  | <div><div>■0.955</div><div>● C -3</div></div> | $\left. A * [\bar{F}] (B) \right\rfloor \left. A^{\dagger} \right\rfloor \left. [\bar{F}] (B) \right\rfloor$                                                                                                                                                                                                                                                                                                                                                                    | $A * [\bar{F}] (B) = \left\langle A^\dagger \right  \left\{ [\bar{F}] (B) \right\}$                                                                                                                                                                                                                                                                                                                                                                                       | $A * [\overline{F}] (B) = \left\langle A^\dagger \right  \left\{ [\overline{F}] (B) \right\}$                                                                                                                                                                                                                                                                                                                                                                              |
| 1205.5935v1_EQ0461 (115) | 089  | <div><div>■0.956</div><div>● W +0</div></div> | $\begin{aligned} &\left. A_2 \times (B \rfloor C) \right\rfloor \left. (A_2 \times B) \right\rfloor C + B \rfloor (A_2 \times C) \\ &\left. A_2 \times (B \rfloor C) \right\rfloor \left. (A_2 \times B) \right\rfloor C + B \rfloor (A_2 \times C) \\ &\left. A_2 \times (B \wedge C) \right\rfloor \left. (A_2 \times B) \right\rfloor C + B \wedge (A_2 \times C) \end{aligned}$                                                                                             | $\begin{aligned} A_2 \times (B \rfloor C) &= (A_2 \times B) \rfloor C + B \rfloor (A_2 \times C) \\ A_2 \times (B \rfloor C) &= (A_2 \times B) \rfloor C + B \rfloor (A_2 \times C) \\ A_2 \times (B \wedge C) &= (A_2 \times B) \wedge C + B \wedge (A_2 \times C) \end{aligned}$                                                                                                                                                                                        | $\begin{aligned} A_2 \times (B \rfloor C) &= (A_2 \times B) \rfloor C + B \rfloor (A_2 \times C) \\ A_2 \times (B \rfloor C) &= (A_2 \times B) \rfloor C + B \rfloor (A_2 \times C) \\ A_2 \times (B \wedge C) &= (A_2 \times B) \wedge C + B \wedge (A_2 \times C) \end{aligned}$                                                                                                                                                                                         |
| 1205.5935v1_EQ0449 (99)  | 088  | <div><div>■0.956</div><div>● K +0</div></div> | $\left( A^{\{\dagger\}} \right)^{\perp} = \left[ \boldsymbol{I}^{\dagger} \left( A^{\perp} \right)^{\dagger} \right]^{\perp}$                                                                                                                                                                                                                                                                                                                                                   | $\left( A^\dagger \right)^\perp = \left[ \boldsymbol{I}^\dagger \left( A^\perp \right)^\dagger \right]^\perp$                                                                                                                                                                                                                                                                                                                                                             | $\left( A^\dagger \right)^\perp = \left[ \boldsymbol{I}^\dagger \left( A^\perp \right)^\dagger \right]^\perp$                                                                                                                                                                                                                                                                                                                                                              |
| 1205.5935v1_EQ0195 (108) | 042  | <div><div>■0.960</div><div>● K +0</div></div> | $A \times B := \frac{1}{2} (A B - B A) \text{ . }$                                                                                                                                                                                                                                                                                                                                                                                                                              | $A \times B := \frac{1}{2} (AB - BA).$                                                                                                                                                                                                                                                                                                                                                                                                                                    | $A \times B := \frac{1}{2} (AB - BA).$                                                                                                                                                                                                                                                                                                                                                                                                                                     |

**Conf.** MathPix confidence: 

■≥0.9

■0.5–0.9

■<0.5

 — no value

**LaTeX source** MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

**Residual** render vs scan (inkdrill): 

●C component

●W weak

●S stable

●N noise

●K clean

●X crop overran its region

●not measured

**[refined: *basis*]** beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.



| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Page | Conf.                                        | LaTeX source                                                                                                                                                                                                                                                                              | Rendered                                                                                                                                                                                                                                                           | Scan image                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1205.5935v1_<br>EQ0237 (230)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 050  | <div><div>■0.973</div><div>●W +0</div></div> | $\begin{aligned} & P_{\boldsymbol{A}}(b_1 \wedge \cdots \wedge b_s) := P_{\boldsymbol{A}}(b_1) \wedge \cdots \wedge P_{\boldsymbol{A}}(b_s) \\ & R_{\boldsymbol{A}}(b_1 \wedge \cdots \wedge b_s) := R_{\boldsymbol{A}}(b_1) \wedge \cdots \wedge R_{\boldsymbol{A}}(b_s). \end{aligned}$ | $P_{\boldsymbol{A}_r}(b_1 \wedge \cdots \wedge b_s) := P_{\boldsymbol{A}_r}(b_1) \wedge \cdots \wedge P_{\boldsymbol{A}_r}(b_s)$ $R_{\boldsymbol{A}_r}(b_1 \wedge \cdots \wedge b_s) := R_{\boldsymbol{A}_r}(b_1) \wedge \cdots \wedge R_{\boldsymbol{A}_r}(b_s).$ | $P_{\boldsymbol{A}_r}(b_1 \wedge \cdots \wedge b_s) := P_{\boldsymbol{A}_r}(b_1) \wedge \cdots \wedge P_{\boldsymbol{A}_r}(b_s)$ $R_{\boldsymbol{A}_r}(b_1 \wedge \cdots \wedge b_s) := R_{\boldsymbol{A}_r}(b_1) \wedge \cdots \wedge R_{\boldsymbol{A}_r}(b_s).$ |
| 1205.5935v1_<br>EQ0075 (73)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 021  | <div><div>■0.973</div><div>●N +0</div></div> | $\begin{aligned} & \lfloor (A_{\{r\}} B_{\{s\}}) \wedge \frac{1}{2} (a A_{\{r\}} B_{\{s\}} - (-1)^{r+s} A_{\{r\}} B_{\{s\}} a) \\ & \wedge \lfloor (A_{\{r\}} B_{\{s\}}) \wedge \frac{1}{2} (a A_{\{r\}} B_{\{s\}} + (-1)^{r+s} A_{\{r\}} B_{\{s\}} a) \end{aligned}$                     | $a \rfloor (A_r B_s) = \frac{1}{2} (a A_r B_s - (-1)^{r+s} A_r B_s a)$ $a \wedge (A_r B_s) = \frac{1}{2} (a A_r B_s + (-1)^{r+s} A_r B_s a)$                                                                                                                       | $a \rfloor (A_r B_s) = \frac{1}{2} (a A_r B_s - (-1)^{r+s} A_r B_s a)$ $a \wedge (A_r B_s) = \frac{1}{2} (a A_r B_s + (-1)^{r+s} A_r B_s a)$                                                                                                                       |
| 1205.5935v1_<br>EQ0446 (95)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 088  | <div><div>■0.973</div><div>●</div></div>     | $\begin{aligned} & (A B)^{\perp} = A B^{\perp} \\ & (A \wedge B)^{\perp} = A \rfloor B^{\perp} \\ & (A \rfloor B)^{\perp} = A \wedge B^{\perp} \end{aligned}$                                                                                                                             | $(AB)^{\perp} = AB^{\perp}$ $(A \wedge B)^{\perp} = A \rfloor B^{\perp}$ $(A \rfloor B)^{\perp} = A \wedge B^{\perp}$                                                                                                                                              | $(AB)^{\perp} = AB^{\perp}$ $(A \wedge B)^{\perp} = A \rfloor B^{\perp}$ $(A \rfloor B)^{\perp} = A \wedge B^{\perp}$                                                                                                                                              |
| 1205.5935v1_<br>EQ0479 (139)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 091  | <div><div>■0.973</div><div>●N +0</div></div> | $\begin{gathered} \operatorname{det} F = F(\boldsymbol{I})^{\perp} \\ F^{-1}(A) = \frac{\bar{F}(A^{-\perp})^{\perp}}{\det(F)} \end{gathered}$                                                                                                                                             | $\det F = F(\boldsymbol{I})^{\perp}$ $F^{-1}(A) = \frac{\bar{F}(A^{-\perp})^{\perp}}{\det(F)}$                                                                                                                                                                     | $\det F = F(\boldsymbol{I})^{\perp}$ $F^{-1}(A) = \frac{\bar{F}(A^{-\perp})^{\perp}}{\det(F)}$                                                                                                                                                                     |
| 1205.5935v1_<br>EQ0223 (216)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 047  | <div><div>■0.975</div><div>●N +0</div></div> | $\begin{aligned} & \boldsymbol{A}_2 \boldsymbol{B}_2 = baac \\ & = a^2 bc \\ & = a^2 b \rfloor c + a^2 b \wedge c. \end{aligned}$                                                                                                                                                         | $\boldsymbol{A}_2 \boldsymbol{B}_2 = baac$ $= a^2 bc$ $= a^2 b \rfloor c + a^2 b \wedge c.$                                                                                                                                                                        | $\boldsymbol{A}_2 \boldsymbol{B}_2 = baac$ $= a^2 bc$ $= a^2 b \rfloor c + a^2 b \wedge c.$                                                                                                                                                                        |
| 1205.5935v1_<br>EQ0456 (106)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 088  | <div><div>■0.975</div><div>●K +0</div></div> | $A \times (B \times C) + B \times (C \times A) + C \times (A \times B) = 0$                                                                                                                                                                                                               | $A \times (B \times C) + B \times (C \times A) + C \times (A \times B) = 0.$                                                                                                                                                                                       | $A \times (B \times C) + B \times (C \times A) + C \times (A \times B) = 0.$                                                                                                                                                                                       |
| 1205.5935v1_<br>EQ0248 (241)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 052  | <div><div>■0.976</div><div>●N +0</div></div> | $A B_{\{s\}} A^{\dagger} = \left\langle A B_{\{s\}} A^{\dagger} \right\rangle_s$                                                                                                                                                                                                          | $AB_s A^{\dagger} = \left\langle AB_s A^{\dagger} \right\rangle_s,$                                                                                                                                                                                                | $AB_s A^{\dagger} = \left\langle AB_s A^{\dagger} \right\rangle_s,$                                                                                                                                                                                                |
| 1205.5935v1_<br>EQ0368 (355)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 076  | <div><div>■0.978</div><div>●K +0</div></div> | $\operatorname{weight}(F(\boldsymbol{A})) =  \lambda  \operatorname{weight}(\boldsymbol{A}),$                                                                                                                                                                                             | $\operatorname{weight}(F(\boldsymbol{A})) =  \lambda  \operatorname{weight}(\boldsymbol{A}),$                                                                                                                                                                      | $\operatorname{weight}(F(\boldsymbol{A})) =  \lambda  \operatorname{weight}(\boldsymbol{A}),$                                                                                                                                                                      |
| 1205.5935v1_<br>EQ0413 (15)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 084  | <div><div>■0.979</div><div>●N +0</div></div> | $\begin{aligned} & \lfloor a = -a \rfloor A^* \\ & A \wedge a = a \wedge A^* \end{aligned}$                                                                                                                                                                                               | $A \rfloor a = -a \rfloor A^*$ $A \wedge a = a \wedge A^*$                                                                                                                                                                                                         | $A \rfloor a = -a \rfloor A^*$ $A \wedge a = a \wedge A^*$                                                                                                                                                                                                         |
| 1205.5935v1_<br>EQ0152 (145)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 034  | <div><div>■0.979</div><div>●N +0</div></div> | $\begin{aligned} & \lambda^{\ddagger} := \lambda \\ & a^{\ddagger} := -a \\ & (AB)^{\ddagger} := B^{\ddagger} A^{\ddagger} \\ & (A+B)^{\ddagger} := A^{\ddagger} + B^{\ddagger}. \end{aligned}$                                                                                           | $\lambda^{\ddagger} := \lambda$ $a^{\ddagger} := -a$ $(AB)^{\ddagger} := B^{\ddagger} A^{\ddagger}$ $(A+B)^{\ddagger} := A^{\ddagger} + B^{\ddagger}.$                                                                                                             | $\lambda^{\ddagger} := \lambda$ $a^{\ddagger} := -a$ $(AB)^{\ddagger} := B^{\ddagger} A^{\ddagger}$ $(A+B)^{\ddagger} := A^{\ddagger} + B^{\ddagger}.$                                                                                                             |
| <div><div>Conf.</div><div>MathPix confidence:</div><div><div>■≥0.9</div><div>■0.5-0.9</div><div>■&lt;0.5</div><div>— no value</div></div></div> <div><div>LaTeX source</div><div>MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</div></div> <div><div>Residual</div><div>render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●X crop overran its region ●not measured</div></div> <div><div>[refined: <i>basis</i>]</div><div>beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</div></div> |      |                                              |                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                    |

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Page | Conf.                                | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Scan image                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1205.5935v1_<br>EQ0140 (133)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 033  | <div>■ 0.981</div> <div>● W +0</div> | $\begin{aligned} & (A\rfloor B)^{\dagger} = B^{\dagger} \& =B^{\dagger} \\\left\lfloor\! \! \! \! A^{\dagger}\right.\right. \\\left\lfloor\! \! \! \! A^{\dagger}\right.\right. \& \left.\left.\right\rfloor\! \! \! \! B^{\dagger} \\\left\lfloor\! \! \! \! A^{\dagger}\right.\right. \& \left.\left.\right\rfloor\! \! \! \! B^{\dagger} \end{aligned}$                                                                                                                                                                                                                                             | $\begin{aligned} (A\rfloor B)^{\dagger} &= B^{\dagger} \lfloor A^{\dagger} \\ (A\lfloor B)^{\dagger} &= B^{\dagger} \rfloor A^{\dagger} \\ (A\wedge B)^{\dagger} &= B^{\dagger} \wedge A^{\dagger}. \end{aligned}$                                                                                                                                                                                                                                                             | $\begin{aligned} (A\rfloor B)^{\dagger} &= B^{\dagger} \lfloor A^{\dagger} \\ (A\lfloor B)^{\dagger} &= B^{\dagger} \rfloor A^{\dagger} \\ (A\wedge B)^{\dagger} &= B^{\dagger} \wedge A^{\dagger}. \end{aligned}$                                                                                                                                                                                                                                                              |
| 1205.5935v1_<br>EQ0364 (352)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 075  | <div>■ 0.981</div> <div>● N +0</div> | $\begin{aligned} & F(\boldsymbol{A}) \& =F\left(a_{\{1\}} \wedge a_{\{2\}}\right) \\\& =F\left(a_{\{1\}}\right) \wedge F\left(a_{\{2\}}\right) \\\& =\lambda_1 \lambda_2 a_{\{1\}} \wedge a_{\{2\}} \\\& =\lambda_1 \lambda_2 \boldsymbol{A}. \end{aligned}$                                                                                                                                                                                                                                                                                                                                           | $\begin{aligned} F(\boldsymbol{A}) &= F\left(a_1 \wedge a_2\right) \\ &= F\left(a_1\right) \wedge F\left(a_2\right) \\ &= \lambda_1 \lambda_2 a_1 \wedge a_2 \\ &= \lambda_1 \lambda_2 \boldsymbol{A}. \end{aligned}$                                                                                                                                                                                                                                                          | $\begin{aligned} F(\boldsymbol{A}) &= F\left(a_1 \wedge a_2\right) \\ &= F\left(a_1\right) \wedge F\left(a_2\right) \\ &= \lambda_1 \lambda_2 a_1 \wedge a_2 \\ &= \lambda_1 \lambda_2 \boldsymbol{A}. \end{aligned}$                                                                                                                                                                                                                                                           |
| 1205.5935v1_<br>EQ0428 (51)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 086  | <div>■ 0.981</div> <div>●</div>      | $\begin{aligned} & \lambda^{\dagger} \& =\lambda \\\& a^{\dagger} \& =a \\\& (A B)^{\dagger} \& =B^{\dagger} A^{\dagger} \\\& (A+B)^{\dagger} \& =A^{\dagger}+B^{\dagger} \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                               | $\begin{aligned} \lambda^{\dagger} &= \lambda \\ a^{\dagger} &= a \\ (AB)^{\dagger} &= B^{\dagger} A^{\dagger} \\ (A+B)^{\dagger} &= A^{\dagger} + B^{\dagger} \end{aligned}$                                                                                                                                                                                                                                                                                                  | $\begin{aligned} \lambda^{\dagger} &= \lambda \\ a^{\dagger} &= a \\ (AB)^{\dagger} &= B^{\dagger} A^{\dagger} \\ (A+B)^{\dagger} &= A^{\dagger} + B^{\dagger} \end{aligned}$                                                                                                                                                                                                                                                                                                   |
| 1205.5935v1_<br>EQ0058 (57)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 019  | <div>■ 0.981</div> <div>● N +0</div> | $\begin{aligned} & a \wedge e_1 \wedge e_2 \wedge \cdots \wedge e_r = \frac{1}{r+1} \left[ \frac{1}{r!} \sum_{\pi} (\operatorname{sgn} \pi) a e_{\pi(1)} e_{\pi(2)} \cdots e_{\pi(r)} + \right. \\ & \quad \left. \sum_{j=1}^r \frac{1}{r!} \sum_{\pi_j} (\operatorname{sgn} \pi_j) e_j a_{\pi_j(1)} a_{\pi_j(2)} \cdots a_{\pi_j(r)} \right] \end{aligned}$                                                                                                                                                                                                                                           | $\begin{aligned} a \wedge e_1 \wedge e_2 \wedge \cdots \wedge e_r &= \frac{1}{r+1} \left[ \frac{1}{r!} \sum_{\pi} (\operatorname{sgn} \pi) a e_{\pi(1)} e_{\pi(2)} \cdots e_{\pi(r)} + \right. \\ & \quad \left. \sum_{j=1}^r \frac{1}{r!} \sum_{\pi_j} (\operatorname{sgn} \pi_j) e_j a_{\pi_j(1)} a_{\pi_j(2)} \cdots a_{\pi_j(r)} \right] \end{aligned}$                                                                                                                    | $\begin{aligned} a \wedge e_1 \wedge e_2 \wedge \cdots \wedge e_r &= \frac{1}{r+1} \left[ \frac{1}{r!} \sum_{\pi} (\operatorname{sgn} \pi) a e_{\pi(1)} e_{\pi(2)} \cdots e_{\pi(r)} + \right. \\ & \quad \left. \sum_{j=1}^r \frac{1}{r!} \sum_{\pi_j} (\operatorname{sgn} \pi_j) e_j a_{\pi_j(1)} a_{\pi_j(2)} \cdots a_{\pi_j(r)} \right] \end{aligned}$                                                                                                                     |
| 1205.5935v1_<br>EQ0361 (349)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 075  | <div>■ 0.982</div> <div>● X +3</div> | $\begin{aligned} & F\left(\boldsymbol{A}_{\{\boldsymbol{r}\}} \boldsymbol{B}_{\{\boldsymbol{s}\}}\right) \& =F\left(a_{\{1\}} a_{\{2\}} \cdots a_{\{r\}} b_{\{1\}} b_{\{2\}} \cdots b_{\{s}\}\right) \\\& =F\left(a_{\{1\}}\right) F\left(a_{\{2\}}\right) \cdots F\left(a_{\{r\}}\right) F\left(b_{\{1\}}\right) F\left(b_{\{2\}}\right) \cdots F\left(b_{\{s}\}\right) \\\& =F\left(a_{\{1\}} a_{\{2\}} \cdots a_{\{r\}}\right) F\left(b_{\{1\}} b_{\{2\}} \cdots b_{\{s}\}\right) \\\& =F\left(\boldsymbol{A}_{\boldsymbol{r}}\right) F\left(\boldsymbol{B}_{\boldsymbol{s}}\right). \end{aligned}$ | $\begin{aligned} F\left(\boldsymbol{A}_{\boldsymbol{r}} \boldsymbol{B}_{\boldsymbol{s}}\right) &= F\left(a_1 a_2 \cdots a_r b_1 b_2 \cdots b_s\right) \\ &= F\left(a_1\right) F\left(a_2\right) \cdots F\left(a_r\right) F\left(b_1\right) F\left(b_2\right) \cdots F\left(b_s\right) \\ &= F\left(a_1 a_2 \cdots a_r\right) F\left(b_1 b_2 \cdots b_s\right) \\ &= F\left(\boldsymbol{A}_{\boldsymbol{r}}\right) F\left(\boldsymbol{B}_{\boldsymbol{s}}\right) \end{aligned}$ | $\begin{aligned} F\left(\boldsymbol{A}_{\boldsymbol{r}} \boldsymbol{B}_{\boldsymbol{s}}\right) &= F\left(a_1 a_2 \cdots a_r b_1 b_2 \cdots b_s\right) \\ &= F\left(a_1\right) F\left(a_2\right) \cdots F\left(a_r\right) F\left(b_1\right) F\left(b_2\right) \cdots F\left(b_s\right) \\ &= F\left(a_1 a_2 \cdots a_r\right) F\left(b_1 b_2 \cdots b_s\right) \\ &= F\left(\boldsymbol{A}_{\boldsymbol{r}}\right) F\left(\boldsymbol{B}_{\boldsymbol{s}}\right). \end{aligned}$ |
| <div> <b>Conf.</b> MathPix confidence: <span>■ ≥0.9</span> <span>■ 0.5–0.9</span> <span>■ &lt;0.5</span> — no value         </div> <div> <b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ         </div> <div> <b>Residual</b> render vs scan (inkdrill): <span>● C component</span> <span>● W weak</span> <span>● S stable</span> <span>● N noise</span> <span>● K clean</span> <span>● X crop overran its region</span> <span>● not measured</span> </div> <div> <b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.         </div> |      |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Page | Conf.            | LaTeX source                                                                                                                                                                                                                                                                                                                        | Rendered                                                                                                                                                                                         | Scan image                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1205.5935v1_<br>E00266 (259)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 055  | ■0.982<br>● X +1 | $\begin{aligned} a_i \downarrow a^j &= (a_1 \wedge \cdots \wedge a_n) a_N^{-1} \delta_i^j \\ &= \delta_i^j. \end{aligned}$                                                                                                                                                                                                          | $a_i \downarrow a^j = (a_1 \wedge \cdots \wedge a_n) a_N^{-1} \delta_i^j$ $= \delta_i^j.$                                                                                                        | $a_i \downarrow a^j = (a_1 \wedge \cdots \wedge a_n) a_N^{-1} \delta_i^j$ $= \delta_i^j.$                                                                                                        |
| 1205.5935v1_<br>E00332 (321)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 068  | ■0.982<br>● K +0 | $a^{\{2\}} b^{\{2\}} = (a \downharpoonleft b)^{\{2\}} - (a \wedge b)^{\{2\}}$                                                                                                                                                                                                                                                       | $a^2 b^2 = (a \downarrow b)^2 - (a \wedge b)^2.$                                                                                                                                                 | $a^2 b^2 = (a \downarrow b)^2 - (a \wedge b)^2.$                                                                                                                                                 |
| 1205.5935v1_<br>E00228 (221)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 047  | ■0.984<br>● K +0 | $\begin{aligned} \boldsymbol{B}_1 \boldsymbol{B}_2 \boldsymbol{B}_3 &= -1, \\ \boldsymbol{B}_1 \boldsymbol{B}_2 \boldsymbol{B}_3 &= -1 \end{aligned}$                                                                                                                                                                               | $\boldsymbol{B}_1 \boldsymbol{B}_2 \boldsymbol{B}_3 = -1,$                                                                                                                                       | $\boldsymbol{B}_1 \boldsymbol{B}_2 \boldsymbol{B}_3 = -1,$                                                                                                                                       |
| 1205.5935v1_<br>E00226 (219)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 047  | ■0.985<br>● X +1 | $\begin{aligned} \boldsymbol{B}_1 \boldsymbol{B}_2 \boldsymbol{B}_3 &= \boldsymbol{e}_1 \boldsymbol{e}_2 \boldsymbol{e}_3, \\ \boldsymbol{B}_1 \boldsymbol{B}_2 \boldsymbol{B}_3 &= \boldsymbol{e}_1 \boldsymbol{e}_2 \boldsymbol{e}_3, \end{aligned} \quad \text{and} \quad \boldsymbol{B}_3 = \boldsymbol{e}_1 \boldsymbol{e}_2.$ | $\boldsymbol{B}_1 = \boldsymbol{e}_2 \boldsymbol{e}_3, \quad \boldsymbol{B}_2 = \boldsymbol{e}_1 \boldsymbol{e}_3, \quad \text{and} \quad \boldsymbol{B}_3 = \boldsymbol{e}_1 \boldsymbol{e}_2.$ | $\boldsymbol{B}_1 = \boldsymbol{e}_2 \boldsymbol{e}_3, \quad \boldsymbol{B}_2 = \boldsymbol{e}_1 \boldsymbol{e}_3, \quad \text{and} \quad \boldsymbol{B}_3 = \boldsymbol{e}_1 \boldsymbol{e}_2.$ |
| 1205.5935v1_<br>E00105 (100)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 026  | ■0.987<br>● X +1 | $\begin{aligned} \epsilon_j &= \sum_{k=1}^{j-1} (i_k - k) + \sum_{k=j+1}^r (i_k - k + 1) + i_j - j \\ &= \sum_{k=1}^r (i_k - k) + r - j \end{aligned}$                                                                                                                                                                              | $\epsilon_j = \sum_{k=1}^{j-1} (i_k - k) + \sum_{k=j+1}^r (i_k - k + 1) + i_j - j$ $= \sum_{k=1}^r (i_k - k) + r - j$                                                                            | $\epsilon_j = \sum_{k=1}^{j-1} (i_k - k) + \sum_{k=j+1}^r (i_k - k + 1) + i_j - j$ $= \sum_{k=1}^r (i_k - k) + r - j.$                                                                           |
| 1205.5935v1_<br>E00218 (211)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 046  | ■0.987<br>● N +0 | $a \times (b \times c) = -a \rfloor (b \wedge c)$                                                                                                                                                                                                                                                                                   | $a \times (b \times c) = -a \rfloor (b \wedge c).$                                                                                                                                               | $a \times (b \times c) = -a \rfloor (b \wedge c).$                                                                                                                                               |
| 1205.5935v1_<br>E00044 (44)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 017  | ■0.987<br>●      | $\begin{aligned} a \rfloor A_+ &= -A_+ \rfloor a \quad \text{while} \quad a \rfloor A_- = A_- \rfloor a. \\ a \rfloor A_- &= A_- \rfloor a \end{aligned}$                                                                                                                                                                           | $a \rfloor A_+ = -A_+ \rfloor a \quad \text{while} \quad a \rfloor A_- = A_- \rfloor a.$                                                                                                         | $a \rfloor A_+ = -A_+ \rfloor a \quad \text{while} \quad a \rfloor A_- = A_- \rfloor a.$                                                                                                         |
| 1205.5935v1_<br>E00298 (290)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 060  | ■0.987<br>● X +1 | $\begin{aligned} P_{\boldsymbol{A}_r} (B_s) &= \sum_I (B_s^\dagger \rfloor a_I) (-1)^{\sum_{j=1}^s (i_j - 1)} a_{I^c} \boldsymbol{A}_r^{-1} \\ &= \sum_I (B_s * a_I) a^I. \end{aligned}$                                                                                                                                            | $P_{\boldsymbol{A}_r} (B_s) = \sum_I (B_s^\dagger \rfloor a_I) (-1)^{\sum_{j=1}^s (i_j - 1)} a_{I^c} \boldsymbol{A}_r^{-1}$ $= \sum_I (B_s * a_I) a^I.$                                          | $P_{\boldsymbol{A}_r} (B_s) = \sum_I (B_s^\dagger \rfloor a_I) (-1)^{\sum_{j=1}^s (i_j - 1)} a_{I^c} \boldsymbol{A}_r^{-1}$ $= \sum_I (B_s * a_I) a^I.$                                          |
| 1205.5935v1_<br>E00157 (150)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 035  | ■0.988<br>● W +0 | $\begin{aligned} (A \rfloor B)^\dagger &= B^\dagger \rfloor A^\dagger \\ (A \rfloor B)^\dagger &= B^\dagger \rfloor A^\dagger \\ (A \wedge B)^\dagger &= B^\dagger \wedge A^\dagger. \end{aligned}$                                                                                                                                 | $(A \rfloor B)^\dagger = B^\dagger \rfloor A^\dagger$ $(A \rfloor B)^\dagger = B^\dagger \rfloor A^\dagger$ $(A \wedge B)^\dagger = B^\dagger \wedge A^\dagger.$                                 | $(A \rfloor B)^\dagger = B^\dagger \rfloor A^\dagger$ $(A \rfloor B)^\dagger = B^\dagger \rfloor A^\dagger$ $(A \wedge B)^\dagger = B^\dagger \wedge A^\dagger.$                                 |
| 1205.5935v1_<br>E00318 (308)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 064  | ■0.988<br>● N +0 | $\begin{aligned} u \in \operatorname{Ker}(F) &\implies F(u) = 0 \\ &\implies \bar{F} F(u) = 0 \\ &\implies F \bar{F}(u) = 0 \\ &\implies \bar{F}(u) \in \operatorname{Ker}(F). \end{aligned}$                                                                                                                                       | $u \in \operatorname{Ker}(F) \implies F(u) = 0$ $\implies \bar{F} F(u) = 0$ $\implies F \bar{F}(u) = 0$ $\implies \bar{F}(u) \in \operatorname{Ker}(F).$                                         | $u \in \operatorname{Ker}(F) \implies F(u) = 0$ $\implies \bar{F} F(u) = 0$ $\implies F \bar{F}(u) = 0$ $\implies \bar{F}(u) \in \operatorname{Ker}(F).$                                         |
| <b>Conf.</b> MathPix confidence: ■ ≥0.9   ■ 0.5–0.9   ■ <0.5   — no value<br><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ<br><b>Residual</b> render vs scan (inkdrill): ●C component   ●W weak   ●S stable   ●N noise   ●K clean   ●X crop overran its region   ●not measured<br><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here. |      |                  |                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                  |                                                                                                                                                                                                  |

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Page | Conf.                                            | LaTeX source                                                                                                                                                                                                                            | Rendered                                                                                                                                                                                            | Scan image                                                                                                                                                                                          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1205.5935v1_<br>EQ0340 (329)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 069  | <div> <div>■0.988</div> <div>● N +0</div> </div> | $\begin{aligned} a\,b\,\&=a\,\downharpoonleft\,b+a\,\wedge\,b\,\,\,\\ \&=\,\pm\!\!\left(1-\frac{\lambda}{2}\,\boldsymbol{B}\right)\,\,\,\\ \&=\,\pm\exp\left(-\boldsymbol{B}\,\lambda/2\right)\end{aligned}$                            | $\begin{aligned} ab&=a\,\downharpoonleft\,b+a\wedge b\\ &=\pm\left(1-\frac{\lambda}{2}\boldsymbol{B}\right)\\ &=\pm\exp(-\boldsymbol{B}\lambda/2) \end{aligned}$                                    | $\begin{aligned} a b &= a \downharpoonleft b + a \wedge b \\ &= \pm \left( 1 - \frac{\lambda}{2} \boldsymbol{B} \right) \\ &= \pm \exp(-\boldsymbol{B} \lambda / 2) \end{aligned}$                  |
| 1205.5935v1_<br>EQ0433 (61)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 086  | <div> <div>■0.988</div> <div>● N +0</div> </div> | $\begin{aligned} \lambda^{\ddagger} &= \lambda \\ a^{\ddagger} &= -a \\ (AB)^{\ddagger} &= B^{\ddagger} A^{\ddagger} \\ (A+B)^{\ddagger} &= A^{\ddagger} + B^{\ddagger} \end{aligned}$                                                  | $\begin{aligned} \lambda^{\ddagger} &= \lambda \\ a^{\ddagger} &= -a \\ (AB)^{\ddagger} &= B^{\ddagger} A^{\ddagger} \\ (A+B)^{\ddagger} &= A^{\ddagger} + B^{\ddagger} \end{aligned}$              | $\begin{aligned} \lambda^{\ddagger} &= \lambda \\ a^{\ddagger} &= -a \\ (AB)^{\ddagger} &= B^{\ddagger} A^{\ddagger} \\ (A+B)^{\ddagger} &= A^{\ddagger} + B^{\ddagger} \end{aligned}$              |
| 1205.5935v1_<br>EQ0296 (288)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 060  | <div> <div>■0.989</div> <div>● K +0</div> </div> | $P_{a_1\wedge\cdots\wedge a_r}(a)=\sum_{j=1}^r(a\,\downharpoonleft\,a_j)a^j.$                                                                                                                                                           | $P_{a_1\wedge\cdots\wedge a_r}(a)=\sum_{j=1}^r(a\,\downharpoonleft\,a_j)a^j.$                                                                                                                       | $P_{a_1\wedge\cdots\wedge a_r}(a)=\sum_{j=1}^r(a\,\downharpoonleft\,a_j)a^j.$                                                                                                                       |
| 1205.5935v1_<br>EQ0053 (52)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 018  | <div> <div>■0.990</div> <div>● X +1</div> </div> | $\begin{aligned} A_r\wedge a &:=\frac{1}{2}\left[A_r a+(-1)^r a A_r\right] \\ &= (-1)^r a\wedge A_r. \end{aligned}$                                                                                                                     | $\begin{aligned} A_r\wedge a &:=\frac{1}{2}\left[A_r a+(-1)^r a A_r\right] \\ &= (-1)^r a\wedge A_r. \end{aligned}$                                                                                 | $\begin{aligned} A_r\wedge a &:=\frac{1}{2}\left[A_r a+(-1)^r a A_r\right] \\ &= (-1)^r a\wedge A_r. \end{aligned}$                                                                                 |
| 1205.5935v1_<br>EQ0468 (125)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 089  | <div> <div>■0.990</div> <div>● N +0</div> </div> | $a^I=(-1)^{\sum_{j=1}^r(i_j-1)}a_{I^c}a_N^{-1}.$                                                                                                                                                                                        | $a^I=(-1)^{\sum_{j=1}^r(i_j-1)}a_{I^c}a_N^{-1}.$                                                                                                                                                    | $a^I=(-1)^{\sum_{j=1}^r(i_j-1)}a_{I^c}a_N^{-1}.$                                                                                                                                                    |
| 1205.5935v1_<br>EQ0401 (385)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 080  | <div> <div>■0.991</div> <div>●</div> </div>      | $r=\frac{l^2/mk}{1+e\cos\theta},$                                                                                                                                                                                                       | $r=\frac{l^2/mk}{1+e\cos\theta},$                                                                                                                                                                   | $r=\frac{l^2/mk}{1+e\cos\theta},$                                                                                                                                                                   |
| 1205.5935v1_<br>EQ0403 (387)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 081  | <div> <div>■0.991</div> <div>● S +0</div> </div> | $\begin{aligned} (L\,\boldsymbol{v})^2 &= L\,\boldsymbol{v}L\,\boldsymbol{v} \\ &= (L\,\boldsymbol{v})^{\dagger}L\,\boldsymbol{v} \\ &= \boldsymbol{v}L^{\dagger}L\,\boldsymbol{v} \\ &= l^2v^2. \end{aligned}$                         | $\begin{aligned} (L\boldsymbol{v})^2 &= L\boldsymbol{v}L\boldsymbol{v} \\ &= (L\boldsymbol{v})^{\dagger}L\boldsymbol{v} \\ &= \boldsymbol{v}L^{\dagger}L\boldsymbol{v} \\ &= l^2v^2. \end{aligned}$ | $\begin{aligned} (L\boldsymbol{v})^2 &= L\boldsymbol{v}L\boldsymbol{v} \\ &= (L\boldsymbol{v})^{\dagger}L\boldsymbol{v} \\ &= \boldsymbol{v}L^{\dagger}L\boldsymbol{v} \\ &= l^2v^2. \end{aligned}$ |
| 1205.5935v1_<br>EQ0337 (326)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 068  | <div> <div>■0.992</div> <div>● N +0</div> </div> | $\begin{aligned} a\,b\,\&=a\,\downharpoonleft\,b+a\,\wedge\,b\,\,\,\\ \&=\,\pm\!\!\left(\cosh\left(\phi/2\right)-\boldsymbol{B}\,\sinh\left(\phi/2\right)\right)\,\,\,\\ \&=\,\pm\exp\left(-\boldsymbol{B}\,\phi/2\right)\end{aligned}$ | $\begin{aligned} ab&=a\,\downharpoonleft\,b+a\wedge b\\ &=\pm(\cosh(\phi/2)-\boldsymbol{B}\sinh(\phi/2))\\ &=\pm\exp(-\boldsymbol{B}\phi/2) \end{aligned}$                                          | $\begin{aligned} ab&=a\,\downharpoonleft\,b+a\wedge b\\ &=\pm(\cosh(\phi/2)-\boldsymbol{B}\sinh(\phi/2))\\ &=\pm\exp(-\boldsymbol{B}\phi/2) \end{aligned}$                                          |
| <p><b>Conf.</b> MathPix confidence: <span>■</span>≥0.9 <span>■</span>0.5–0.9 <span>■</span>&lt;0.5 — no value</p> <p><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</p> <p><b>Residual</b> render vs scan (inkdrill): <span>●</span>C component <span>●</span>W weak <span>●</span>S stable <span>●</span>N noise <span>●</span>K clean <span>●</span>X crop overran its region <span>●</span>not measured</p> <p><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</p> |      |                                                  |                                                                                                                                                                                                                                         |                                                                                                                                                                                                     |                                                                                                                                                                                                     |



| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Page | Conf.                                                          | LaTeX source                                                                                                                                                                                                                                        | Rendered                                                                                                                                                                                                             | Scan image                                                                                                                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1285.5935v1_<br>EQ0066 (65)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 020  | <div><span>■</span> 0.995</div> <div><span>●</span> K +0</div> | $\begin{aligned} a \downharpoonleft A_r &= \left\langle a A_r \right\rangle_{r-1} \\ a A_r &= \left\langle a A_r \right\rangle_{r+1}, \end{aligned}$                                                                                                | $a \downharpoonleft A_r = \langle a A_r \rangle_{r-1}$ $a \wedge A_r = \langle a A_r \rangle_{r+1},$                                                                                                                 | $a \downharpoonleft A_r = \langle a A_r \rangle_{r-1}$ $a \wedge A_r = \langle a A_r \rangle_{r+1},$                                                                                                                 |
| 1285.5935v1_<br>EQ0055 (54)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 018  | <div><span>■</span> 0.996</div> <div><span>●</span> K +0</div> | $\frac{1}{2} \left[ a \left( a_1 \wedge a_2 \wedge \cdots \wedge a_r \right) + (-1)^r \left( a_1 \wedge a_2 \wedge \cdots \wedge a_r \right) a \right] = a \wedge a_1 \wedge a_2 \wedge \cdots \wedge a_r.$                                         | $\frac{1}{2} \left[ a \left( a_1 \wedge a_2 \wedge \cdots \wedge a_r \right) + (-1)^r \left( a_1 \wedge a_2 \wedge \cdots \wedge a_r \right) a \right] = a \wedge a_1 \wedge a_2 \wedge \cdots \wedge a_r.$          | $\frac{1}{2} \left[ a \left( a_1 \wedge a_2 \wedge \cdots \wedge a_r \right) + (-1)^r \left( a_1 \wedge a_2 \wedge \cdots \wedge a_r \right) a \right] = a \wedge a_1 \wedge a_2 \wedge \cdots \wedge a_r.$          |
| 1285.5935v1_<br>EQ0281 (273)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 057  | <div><span>■</span> 0.996</div> <div><span>●</span> N +0</div> | $\begin{aligned} \left( a^{i_r} \wedge \cdots \wedge a^{i_2} \wedge a^{i_1} \right) \downharpoonleft A_r &= \left( a^{i_r} \wedge \cdots \wedge a^{i_2} \right) \downharpoonleft \left( a^{i_1} \downharpoonleft A_r \right) \\ &= 0 \end{aligned}$ | $\left( a^{i_r} \wedge \cdots \wedge a^{i_2} \wedge a^{i_1} \right) \downharpoonleft A_r = \left( a^{i_r} \wedge \cdots \wedge a^{i_2} \right) \downharpoonleft \left( a^{i_1} \downharpoonleft A_r \right)$ $= 0$   | $\left( a^{i_r} \wedge \cdots \wedge a^{i_2} \wedge a^{i_1} \right) \downharpoonleft A_r = \left( a^{i_r} \wedge \cdots \wedge a^{i_2} \right) \downharpoonleft \left( a^{i_1} \downharpoonleft A_r \right)$ $= 0$   |
| 1285.5935v1_<br>EQ0178 (171)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 038  | <div><span>■</span> 0.996</div> <div><span>●</span> W +0</div> | $\frac{\boldsymbol{A}_{-1}}{\boldsymbol{A}_r} = (-1)^{r(r-1)/2} \frac{\boldsymbol{A}_r}{ \boldsymbol{A}_r ^2} = \frac{\boldsymbol{A}_r}{\boldsymbol{A}_r^2}.$                                                                                       | $\boldsymbol{A}_r^{-1} = (-1)^{r(r-1)/2} \frac{\boldsymbol{A}_r}{ \boldsymbol{A}_r ^2} = \frac{\boldsymbol{A}_r}{\boldsymbol{A}_r^2}.$                                                                               | $\boldsymbol{A}_r^{-1} = (-1)^{r(r-1)/2} \frac{\boldsymbol{A}_r}{ \boldsymbol{A}_r ^2} = \frac{\boldsymbol{A}_r}{\boldsymbol{A}_r^2}.$                                                                               |
| 1285.5935v1_<br>EQ0190 (183)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 041  | <div><span>■</span> 0.996</div> <div><span>●</span> X +1</div> | $\left( A^{\perp} \right)^{-1} = \left( A \boldsymbol{I}^{-1} \right)^{-1} = \boldsymbol{I} A^{-1}.$                                                                                                                                                | $\left( A^{\perp} \right)^{-1} = \left( A \boldsymbol{I}^{-1} \right)^{-1}$ $= \boldsymbol{I} A^{-1}.$                                                                                                               | $\left( A^{\perp} \right)^{-1} = \left( A \boldsymbol{I}^{-1} \right)^{-1}$ $= \boldsymbol{I} A^{-1}.$                                                                                                               |
| 1285.5935v1_<br>EQ0404 (388)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 081  | <div><span>■</span> 0.996</div> <div><span>●</span> N +0</div> | $\left( L \boldsymbol{v} \right) \downharpoonleft \hat{\boldsymbol{r}} = \frac{\left\langle L \boldsymbol{v} \boldsymbol{r} \right\rangle}{r} = \frac{l^2}{mr},$                                                                                    | $\left( L \boldsymbol{v} \right) \downharpoonleft \hat{\boldsymbol{r}} = \frac{\left\langle L \boldsymbol{v} \right\rangle}{r} = \frac{l^2}{mr},$                                                                    | $\left( L \boldsymbol{v} \right) \downharpoonleft \hat{\boldsymbol{r}} = \frac{\left\langle L \boldsymbol{v} \right\rangle}{r} = \frac{l^2}{mr},$                                                                    |
| 1285.5935v1_<br>EQ0116 (110)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 030  | <div><span>■</span> 0.997</div> <div><span>●</span></div>      | $\boldsymbol{A}_{-1} \boldsymbol{B}_s = a_1 a_2 \cdots a_r b_1 b_2 \cdots b_s$ $= a_1 \wedge a_2 \wedge \cdots \wedge a_r \wedge b_1 \wedge b_2 \wedge \cdots \wedge b_s$ $= \boldsymbol{A}_r \wedge \boldsymbol{B}_s$                              | $\boldsymbol{A}_r \boldsymbol{B}_s = a_1 a_2 \cdots a_r b_1 b_2 \cdots b_s$ $= a_1 \wedge a_2 \wedge \cdots \wedge a_r \wedge b_1 \wedge b_2 \wedge \cdots \wedge b_s$ $= \boldsymbol{A}_r \wedge \boldsymbol{B}_s.$ | $\boldsymbol{A}_r \boldsymbol{B}_s = a_1 a_2 \cdots a_r b_1 b_2 \cdots b_s$ $= a_1 \wedge a_2 \wedge \cdots \wedge a_r \wedge b_1 \wedge b_2 \wedge \cdots \wedge b_s$ $= \boldsymbol{A}_r \wedge \boldsymbol{B}_s.$ |
| 1285.5935v1_<br>EQ0119 (113)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 031  | <div><span>■</span> 0.997</div> <div><span>●</span> K +0</div> | $\left( a_1 a_2 \cdots a_r \right)^* = (-1)^r a_1 a_2 \cdots a_r,$                                                                                                                                                                                  | $\left( a_1 a_2 \cdots a_r \right)^* = (-1)^r a_1 a_2 \cdots a_r,$                                                                                                                                                   | $\left( a_1 a_2 \cdots a_r \right)^* = (-1)^r a_1 a_2 \cdots a_r,$                                                                                                                                                   |
| 1285.5935v1_<br>EQ0183 (176)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 040  | <div><span>■</span> 0.997</div> <div><span>●</span> K +0</div> | $A^{\perp} := A \downharpoonleft \boldsymbol{I}^{-1} = A \boldsymbol{I}^{-1}$                                                                                                                                                                       | $A^{\perp} := A \downharpoonleft \boldsymbol{I}^{-1} = A \boldsymbol{I}^{-1}$                                                                                                                                        | $A^{\perp} := A \downharpoonleft \boldsymbol{I}^{-1} = A \boldsymbol{I}^{-1}$                                                                                                                                        |
| <p><b>Conf.</b> MathPix confidence: <span>■</span> <math>\geq 0.9</math> <span>■</span> 0.5–0.9 <span>■</span> <math>&lt; 0.5</math> — no value</p> <p><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</p> <p><b>Residual</b> render vs scan (inkdrill): <span>●</span> C component <span>●</span> W weak <span>●</span> S stable <span>●</span> N noise <span>●</span> K clean <span>●</span> X crop overran its region <span>●</span> not measured</p> <p><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</p> |      |                                                                |                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                      |                                                                                                                                                                                                                      |



| Identifier                   | Page | Conf.                                         | LaTeX source                                                                                                                                                                                                                                                                               | Rendered                                                                                                                                                                                                                                                             | Scan image                                                                                                                                                                                                                                                           |
|------------------------------|------|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1205.5935v1_<br>E00127 (120) | 032  | <div><div></div>0.998</div> <div>● N +0</div> | $\begin{aligned} & \left( A \lfloor B \right)^* = A^* \lfloor B^* \\ & \left( A \wedge B \right)^* = A^* \wedge B^* . \end{aligned}$                                                                                                                                                       | $\begin{aligned} (A \lfloor B)^* &= A^* \lfloor B^* \\ (A \wedge B)^* &= A^* \wedge B^* . \end{aligned}$                                                                                                                                                             | $\begin{aligned} (A \lfloor B)^* &= A^* \lfloor B^* \\ (A \wedge B)^* &= A^* \wedge B^* . \end{aligned}$                                                                                                                                                             |
| 1205.5935v1_<br>E00444 (89)  | 088  | <div><div></div>0.998</div> <div>● N +0</div> | $\begin{aligned} & A^{\perp} = A \rfloor \boldsymbol{I}^{-1} \\ & \quad = A \boldsymbol{I}^{-1} \end{aligned}$                                                                                                                                                                             | $\begin{aligned} A^{\perp} &= A \rfloor \boldsymbol{I}^{-1} \\ &= A \boldsymbol{I}^{-1} \end{aligned}$                                                                                                                                                               | $\begin{aligned} A^{\perp} &= A \rfloor \boldsymbol{I}^{-1} \\ &= A \boldsymbol{I}^{-1} \end{aligned}$                                                                                                                                                               |
| 1205.5935v1_<br>E00160 (161) | 036  | <div><div></div>0.998</div> <div>● N +0</div> | $\begin{aligned} & A * (B \lfloor C) = \left\langle A^{\dagger} \rfloor (B \lfloor C) \right\rangle \\ & \quad = \left\langle (A^{\dagger} \rfloor B) \lfloor C \right\rangle \\ & \quad = (A^{\dagger} \rfloor B)^{\dagger} * C \\ & \quad = (B^{\dagger} \lfloor A) * C . \end{aligned}$ | $\begin{aligned} A * (B \lfloor C) &= \left\langle A^{\dagger} \rfloor (B \lfloor C) \right\rangle \\ &= \left\langle (A^{\dagger} \rfloor B) \lfloor C \right\rangle \\ &= (A^{\dagger} \rfloor B)^{\dagger} * C \\ &= (B^{\dagger} \lfloor A) * C . \end{aligned}$ | $\begin{aligned} A * (B \lfloor C) &= \left\langle A^{\dagger} \rfloor (B \lfloor C) \right\rangle \\ &= \left\langle (A^{\dagger} \rfloor B) \lfloor C \right\rangle \\ &= (A^{\dagger} \rfloor B)^{\dagger} * C \\ &= (B^{\dagger} \lfloor A) * C . \end{aligned}$ |
| 1205.5935v1_<br>E00137 (130) | 033  | <div><div></div>0.999</div> <div>● N +0</div> | $A^{\dagger} = A^{\dagger}$                                                                                                                                                                                                                                                                | $A^{\dagger} = A^{\dagger}$                                                                                                                                                                                                                                          | $A^{\dagger} = A^{\dagger}$                                                                                                                                                                                                                                          |
| 1205.5935v1_<br>E00146 (139) | 034  | <div><div></div>0.999</div> <div>● K +0</div> | $\left\langle A B \right\rangle = \left\langle A^{\dagger} B^{\dagger} \right\rangle$                                                                                                                                                                                                      | $\left\langle A B \right\rangle = \left\langle A^{\dagger} B^{\dagger} \right\rangle$                                                                                                                                                                                | $\left\langle A B \right\rangle = \left\langle A^{\dagger} B^{\dagger} \right\rangle$                                                                                                                                                                                |
| 1205.5935v1_<br>E00151 (144) | 034  | <div><div></div>0.999</div> <div>● K +0</div> | $A_{-} \wedge B_{-} = -B_{-} \wedge A_{-}$                                                                                                                                                                                                                                                 | $A_{-} \wedge B_{-} = -B_{-} \wedge A_{-}$                                                                                                                                                                                                                           | $A_{-} \wedge B_{-} = -B_{-} \wedge A_{-}$                                                                                                                                                                                                                           |
| 1205.5935v1_<br>E00191 (184) | 041  | <div><div></div>0.999</div> <div>● X +2</div> | $\begin{aligned} & \left( A^{\perp} \right)^{\perp} = A \\ & \quad = (-1)^n A^{\perp} \\ & \quad = (-1)^n (A^{\perp})^{\perp} \\ & \quad = (-1)^n A \end{aligned}$                                                                                                                         | $\begin{aligned} (A^{\perp})^{\perp} &= A^{\perp} \\ &= (-1)^n A^{\perp} \\ &= (-1)^n (A^{\perp})^{\perp} \\ &= (-1)^n A \end{aligned}$                                                                                                                              | $\begin{aligned} (A^{\perp})^{\perp} &= A^{\perp} \\ &= (-1)^n A^{\perp} \\ &= (-1)^n (A^{\perp})^{\perp} \\ &= (-1)^n A \end{aligned}$                                                                                                                              |
| 1205.5935v1_<br>E00199 (192) | 042  | <div><div></div>0.999</div> <div>●</div>      | $\begin{aligned} & (A \times B)^* = A^* \times B^* \\ & (A \times B)^{\dagger} = B^{\dagger} \times A^{\dagger} \\ & (A \times B)^{\ddagger} = B^{\ddagger} \times A^{\ddagger} . \end{aligned}$                                                                                           | $\begin{aligned} (A \times B)^* &= A^* \times B^* \\ (A \times B)^{\dagger} &= B^{\dagger} \times A^{\dagger} \\ (A \times B)^{\ddagger} &= B^{\ddagger} \times A^{\ddagger} . \end{aligned}$                                                                        | $\begin{aligned} (A \times B)^* &= A^* \times B^* \\ (A \times B)^{\dagger} &= B^{\dagger} \times A^{\dagger} \\ (A \times B)^{\ddagger} &= B^{\ddagger} \times A^{\ddagger} . \end{aligned}$                                                                        |
| 1205.5935v1_<br>E00288 (280) | 059  | <div><div></div>0.999</div> <div>● N +0</div> | $a^I = (-1)^{\sum_{j=1}^r (i_j - 1)} a_{I^c} a_N^{-1}$                                                                                                                                                                                                                                     | $a^I = (-1)^{\sum_{j=1}^r (i_j - 1)} a_{I^c} a_N^{-1}$                                                                                                                                                                                                               | $a^I = (-1)^{\sum_{j=1}^r (i_j - 1)} a_{I^c} a_N^{-1}$                                                                                                                                                                                                               |
| 1205.5935v1_<br>E00320 (310) | 065  | <div><div></div>0.999</div> <div>● N +0</div> | $F(a) = \sum_i \lambda_i (a * a^i) a_i$                                                                                                                                                                                                                                                    | $F(a) = \sum_i \lambda_i (a * a^i) a_i$                                                                                                                                                                                                                              | $F(a) = \sum_i \lambda_i (a * a^i) a_i$                                                                                                                                                                                                                              |

**Conf.** MathPix confidence:

≥0.9

0.5–0.9

<0.5

— no value

**LaTeX source** MathPix’s reading, unchanged;
**Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

**Residual** render vs scan (inkdrill):

C component

W weak

S stable

N noise

K clean

X crop overran its region

not measured

**[refined: basis]** beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Page | Conf.             | LaTeX source                                                                                                                                                                                                                                                                                                              | Rendered                                                                                                                                                                                                                                                                                                                 | Scan image                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1205.5935v1_<br>E09376 (362)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 077  | ■ 0.999<br>● X +1 | $\begin{aligned} & \bar{F}(\boldsymbol{I}) * \boldsymbol{I} = \boldsymbol{I} * F(\boldsymbol{I}) \\ & \det(\bar{F}) \boldsymbol{I} * \boldsymbol{I} = \det(F) \boldsymbol{I} * \boldsymbol{I} \\ & \det(\bar{F}) = \det(F) \end{aligned}$                                                                                 | $\begin{aligned} & \bar{F}(\boldsymbol{I}) * \boldsymbol{I} = \boldsymbol{I} * F(\boldsymbol{I}) \\ & \det(\bar{F}) \boldsymbol{I} * \boldsymbol{I} = \det(F) \boldsymbol{I} * \boldsymbol{I} \\ & \det(\bar{F}) = \det(F) \end{aligned}$                                                                                | $\begin{aligned} & \bar{F}(\boldsymbol{I}) * \boldsymbol{I} = \boldsymbol{I} * F(\boldsymbol{I}) \\ & \det(\bar{F}) \boldsymbol{I} * \boldsymbol{I} = \det(F) \boldsymbol{I} * \boldsymbol{I} \\ & \det(\bar{F}) = \det(F). \end{aligned}$                                                                                |
| 1205.5935v1_<br>E09492 (386)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 081  | ■ 0.999<br>● N +0 | $\begin{aligned} & (L \boldsymbol{v} - k \hat{\boldsymbol{r}})^2 = k^2 e^2 \\ & (L \boldsymbol{v})^2 - 2k(L \boldsymbol{v}) \rfloor \hat{\boldsymbol{r}} + k^2 = k^2 e^2 \end{aligned}$                                                                                                                                   | $\begin{aligned} & (L \boldsymbol{v} - k \hat{\boldsymbol{r}})^2 = k^2 e^2 \\ & (L \boldsymbol{v})^2 - 2k(L \boldsymbol{v}) \rfloor \hat{\boldsymbol{r}} + k^2 = k^2 e^2 \end{aligned}$                                                                                                                                  | $\begin{aligned} & (L \boldsymbol{v} - k \hat{\boldsymbol{r}})^2 = k^2 e^2 \\ & (L \boldsymbol{v})^2 - 2k(L \boldsymbol{v}) \rfloor \hat{\boldsymbol{r}} + k^2 = k^2 e^2 \end{aligned}$                                                                                                                                   |
| 1205.5935v1_<br>E09470 (128)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 090  | ■ 0.999<br>● N +0 | $\sum_i a_i a^i = \sum_i a^i a_i = n$                                                                                                                                                                                                                                                                                     | $\sum_i a_i a^i = \sum_i a^i a_i = n.$                                                                                                                                                                                                                                                                                   | $\sum_i a_i a^i = \sum_i a^i a_i = n.$                                                                                                                                                                                                                                                                                    |
| 1205.5935v1_<br>E09432 (58)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 086  | ■ 0.999<br>● W +0 | $\begin{aligned} & (A \rfloor B)^\dagger = B^\dagger \rfloor A^\dagger \\ & (A \rfloor B)^\dagger = B^\dagger \rfloor A^\dagger \\ & (A \wedge B)^\dagger = B^\dagger \wedge A^\dagger \end{aligned}$                                                                                                                     | $\begin{aligned} & (A \rfloor B)^\dagger = B^\dagger \rfloor A^\dagger \\ & (A \rfloor B)^\dagger = B^\dagger \rfloor A^\dagger \\ & (A \wedge B)^\dagger = B^\dagger \wedge A^\dagger \end{aligned}$                                                                                                                    | $\begin{aligned} & (A \rfloor B)^\dagger = B^\dagger \rfloor A^\dagger \\ & (A \rfloor B)^\dagger = B^\dagger \rfloor A^\dagger \\ & (A \wedge B)^\dagger = B^\dagger \wedge A^\dagger \end{aligned}$                                                                                                                     |
| 1205.5935v1_<br>E09035 (35)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 014  | ■ 0.999<br>● X +1 | $\begin{aligned} & a_1 \wedge a_2 \wedge \cdots \wedge a_r = \sum_{i,j,\dots,m} R_{1i} e_i \wedge R_{2j} e_j \wedge \cdots \wedge R_{rm} e_m \\ & = \det(R) e_1 \wedge e_2 \wedge \cdots \wedge e_r \end{aligned}$                                                                                                        | $\begin{aligned} & a_1 \wedge a_2 \wedge \cdots \wedge a_r = \sum_{i,j,\dots,m} R_{1i} e_i \wedge R_{2j} e_j \wedge \cdots \wedge R_{rm} e_m \\ & = \det(R) e_1 \wedge e_2 \wedge \cdots \wedge e_r \end{aligned}$                                                                                                       | $\begin{aligned} & a_1 \wedge a_2 \wedge \cdots \wedge a_r = \sum_{i,j,\dots,m} R_{1i} e_i \wedge R_{2j} e_j \wedge \cdots \wedge R_{rm} e_m \\ & = \det(R) e_1 \wedge e_2 \wedge \cdots \wedge e_r. \end{aligned}$                                                                                                       |
| 1205.5935v1_<br>E09040 (40)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 016  | ■ 0.999<br>● K +0 | $u \, v = u \downharpoonleft v + u \wedge v$                                                                                                                                                                                                                                                                              | $uv = u \rfloor v + u \wedge v$                                                                                                                                                                                                                                                                                          | $uv = u \rfloor v + u \wedge v$                                                                                                                                                                                                                                                                                           |
| 1205.5935v1_<br>E09090 (87)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 024  | ■ 0.999<br>●      | $\begin{aligned} & A_r \wedge (B_s \wedge C_t) = A_r \wedge \langle B_s C_t \rangle_{s+t} \\ & = \langle A_r \langle B_s C_t \rangle_{s+t} \rangle_{r+s+t} \\ & = \langle A_r B_s C_t \rangle_{r+s+t} \\ & = \langle \langle A_r B_s \rangle_{r+s} C_t \rangle_{r+s+t} \\ & = (A_r \wedge B_s) \wedge C_t. \end{aligned}$ | $\begin{aligned} & A_r \wedge (B_s \wedge C_t) = A_r \wedge \langle B_s C_t \rangle_{s+t} \\ & = \langle A_r \langle B_s C_t \rangle_{s+t} \rangle_{r+s+t} \\ & = \langle A_r B_s C_t \rangle_{r+s+t} \\ & = \langle \langle A_r B_s \rangle_{r+s} C_t \rangle_{r+s+t} \\ & = (A_r \wedge B_s) \wedge C_t \end{aligned}$ | $\begin{aligned} & A_r \wedge (B_s \wedge C_t) = A_r \wedge \langle B_s C_t \rangle_{s+t} \\ & = \langle A_r \langle B_s C_t \rangle_{s+t} \rangle_{r+s+t} \\ & = \langle A_r B_s C_t \rangle_{r+s+t} \\ & = \langle \langle A_r B_s \rangle_{r+s} C_t \rangle_{r+s+t} \\ & = (A_r \wedge B_s) \wedge C_t. \end{aligned}$ |
| 1205.5935v1_<br>E09148 (141)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 034  | ■ 0.999<br>● N +0 | $\left. A_r \right\rfloor B_s = (-1)^{r(s-1)} B_s \rfloor A_r.$                                                                                                                                                                                                                                                           | $A_r \rfloor B_s = (-1)^{r(s-1)} B_s \rfloor A_r.$                                                                                                                                                                                                                                                                       | $A_r \rfloor B_s = (-1)^{r(s-1)} B_s \rfloor A_r.$                                                                                                                                                                                                                                                                        |
| 1205.5935v1_<br>E09185 (178)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 040  | ■ 0.999<br>● N +1 | $A \boldsymbol{I} = \boldsymbol{I} A^{*(n-1)}$                                                                                                                                                                                                                                                                            | $A \boldsymbol{I} = \boldsymbol{I} A^{*(n-1)}.$                                                                                                                                                                                                                                                                          | $A \boldsymbol{I} = \boldsymbol{I} A^{*(n-1)}.$                                                                                                                                                                                                                                                                           |
| <div>Conf. MathPix confidence: <span>■ ≥0.9</span> <span>■ 0.5–0.9</span> <span>■ &lt;0.5</span> — no value</div> <div>LaTeX source MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</div> <div>Residual render vs scan (inkdrill): <span>● C component</span> <span>● W weak</span> <span>● S stable</span> <span>● N noise</span> <span>● K clean</span> <span>● X crop overran its region</span> <span>● not measured</span></div> <div><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</div> |      |                   |                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                           |

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Page | Conf.                                             | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                            | Scan image                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1285.5935v1_<br>EQ0256 (249)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 053  | <div><span>■</span> 0.999</div> <div>● N +0</div> | $\boldsymbol{A}_{\boldsymbol{r}}^{\perp} a^{*(n-r)} \left( \boldsymbol{A}_{\boldsymbol{r}}^{\perp} \right)^{-1} = -\boldsymbol{A}_{\boldsymbol{r}} a^{*r} \boldsymbol{A}_{\boldsymbol{r}}^{-1}.$                                                                                                                                                                                                                                                                                    | $\boldsymbol{A}_{\boldsymbol{r}}^{\perp} a^{*(n-r)} \left( \boldsymbol{A}_{\boldsymbol{r}}^{\perp} \right)^{-1} = -\boldsymbol{A}_{\boldsymbol{r}} a^{*r} \boldsymbol{A}_{\boldsymbol{r}}^{-1}.$                                                                                                                                                                                                                                    | $\boldsymbol{A}_{\boldsymbol{r}}^{\perp} a^{*(n-r)} \left( \boldsymbol{A}_{\boldsymbol{r}}^{\perp} \right)^{-1} = -\boldsymbol{A}_{\boldsymbol{r}} a^{*r} \boldsymbol{A}_{\boldsymbol{r}}^{-1}.$                                                                                                                                                                                                                                    |
| 1285.5935v1_<br>EQ0264 (257)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 055  | <div><span>■</span> 0.999</div> <div>● K +0</div> | $a^i := (-1)^{i-1} (a_1 \wedge a_2 \wedge \cdots \wedge \check{a}_i \wedge \cdots \wedge a_n) a_N^{-1}.$                                                                                                                                                                                                                                                                                                                                                                            | $a^i := (-1)^{i-1} (a_1 \wedge a_2 \wedge \cdots \wedge \check{a}_i \wedge \cdots \wedge a_n) a_N^{-1}.$                                                                                                                                                                                                                                                                                                                            | $a^i := (-1)^{i-1} (a_1 \wedge a_2 \wedge \cdots \wedge \check{a}_i \wedge \cdots \wedge a_n) a_N^{-1}.$                                                                                                                                                                                                                                                                                                                            |
| 1285.5935v1_<br>EQ0286 (278)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 058  | <div><span>■</span> 0.999</div> <div>● N +0</div> | $a^{\{N\}} = \frac{a_N}{ \boldsymbol{a}_N ^2}.$                                                                                                                                                                                                                                                                                                                                                                                                                                     | $a^N = \frac{a_N}{ a_N ^2}.$                                                                                                                                                                                                                                                                                                                                                                                                        | $a^N = \frac{a_N}{ a_N ^2}.$                                                                                                                                                                                                                                                                                                                                                                                                        |
| 1285.5935v1_<br>EQ0397 (381)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 080  | <div><span>■</span> 0.999</div> <div>● K +0</div> | $L \boldsymbol{v} \boldsymbol{r} = k(\hat{\boldsymbol{r}} \boldsymbol{r} + \boldsymbol{e} \boldsymbol{r}).$                                                                                                                                                                                                                                                                                                                                                                         | $L \boldsymbol{v} \boldsymbol{r} = k(\hat{\boldsymbol{r}} \boldsymbol{r} + \boldsymbol{e} \boldsymbol{r}).$                                                                                                                                                                                                                                                                                                                         | $L \boldsymbol{v} \boldsymbol{r} = k(\hat{\boldsymbol{r}} \boldsymbol{r} + \boldsymbol{e} \boldsymbol{r}).$                                                                                                                                                                                                                                                                                                                         |
| 1285.5935v1_<br>EQ0439 (72)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 087  | <div><span>■</span> 0.999</div> <div>● N +0</div> | $\begin{aligned} A * B &= \left\langle A^{\dagger} B \right\rangle \\ &= \left\langle A^{\dagger} \right\rfloor B \rangle \\ &= \left\langle A^{\dagger} \right\rfloor B \rangle \end{aligned}$                                                                                                                                                                                                                                                                                     | $\begin{aligned} A * B &= \left\langle A^{\dagger} B \right\rangle \\ &= \left\langle A^{\dagger} \right\rfloor B \rangle \\ &= \left\langle A^{\dagger} \right\rfloor B \rangle \end{aligned}$                                                                                                                                                                                                                                     | $\begin{aligned} A * B &= \left\langle A^{\dagger} B \right\rangle \\ &= \left\langle A^{\dagger} \right\rfloor B \rangle \\ &= \left\langle A^{\dagger} \right\rfloor B \rangle \end{aligned}$                                                                                                                                                                                                                                     |
| 1285.5935v1_<br>EQ0233 (226)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 049  | <div><span>■</span> 0.999</div> <div>● K +0</div> | $\left. \left. R_{\boldsymbol{A}_{\boldsymbol{r}}}(\boldsymbol{a}) \right\rfloor \boldsymbol{A}_{\boldsymbol{r}} = (a \wedge \boldsymbol{A}_{\boldsymbol{r}} \boldsymbol{A}_{\boldsymbol{r}}^{-1}) \right\rfloor \boldsymbol{A}_{\boldsymbol{r}} = \left\langle a \wedge \boldsymbol{A}_{\boldsymbol{r}} \boldsymbol{A}_{\boldsymbol{r}}^{-1} \boldsymbol{A}_{\boldsymbol{r}} \right\rangle_{r-1} = \left\langle a \wedge \boldsymbol{A}_{\boldsymbol{r}} \right\rangle_{r-1} = 0.$ | $R_{\boldsymbol{A}_{\boldsymbol{r}}}(\boldsymbol{a}) \rfloor \boldsymbol{A}_{\boldsymbol{r}} = (a \wedge \boldsymbol{A}_{\boldsymbol{r}} \boldsymbol{A}_{\boldsymbol{r}}^{-1}) \rfloor \boldsymbol{A}_{\boldsymbol{r}} = \langle a \wedge \boldsymbol{A}_{\boldsymbol{r}} \boldsymbol{A}_{\boldsymbol{r}}^{-1} \boldsymbol{A}_{\boldsymbol{r}} \rangle_{r-1} = \langle a \wedge \boldsymbol{A}_{\boldsymbol{r}} \rangle_{r-1} = 0.$ | $R_{\boldsymbol{A}_{\boldsymbol{r}}}(\boldsymbol{a}) \rfloor \boldsymbol{A}_{\boldsymbol{r}} = (a \wedge \boldsymbol{A}_{\boldsymbol{r}} \boldsymbol{A}_{\boldsymbol{r}}^{-1}) \rfloor \boldsymbol{A}_{\boldsymbol{r}} = \langle a \wedge \boldsymbol{A}_{\boldsymbol{r}} \boldsymbol{A}_{\boldsymbol{r}}^{-1} \boldsymbol{A}_{\boldsymbol{r}} \rangle_{r-1} = \langle a \wedge \boldsymbol{A}_{\boldsymbol{r}} \rangle_{r-1} = 0.$ |
| 1285.5935v1_<br>EQ0043 (43)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 017  | <div><span>■</span> 1.000</div> <div>● X +1</div> | $\begin{aligned} A_{\boldsymbol{r}} \rfloor a &:= \frac{1}{2} [A_{\boldsymbol{r}} a - (-1)^r a A_{\boldsymbol{r}}] \\ &= (-1)^{r-1} a \rfloor A_{\boldsymbol{r}} \end{aligned}$                                                                                                                                                                                                                                                                                                     | $\begin{aligned} A_{\boldsymbol{r}} \rfloor a &:= \frac{1}{2} [A_{\boldsymbol{r}} a - (-1)^r a A_{\boldsymbol{r}}] \\ &= (-1)^{r-1} a \rfloor A_{\boldsymbol{r}} \end{aligned}$                                                                                                                                                                                                                                                     | $\begin{aligned} A_{\boldsymbol{r}} \rfloor a &:= \frac{1}{2} [A_{\boldsymbol{r}} a - (-1)^r a A_{\boldsymbol{r}}] \\ &= (-1)^{r-1} a \rfloor A_{\boldsymbol{r}}. \end{aligned}$                                                                                                                                                                                                                                                    |
| 1285.5935v1_<br>EQ0113 (107)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 028  | <div><span>■</span> 1.000</div> <div>● K +0</div> | $a = a_1 - \left( \frac{a_1 \rfloor b_1}{a_2 \rfloor b_1} \right) a_2.$                                                                                                                                                                                                                                                                                                                                                                                                             | $a = a_1 - \left( \frac{a_1 \rfloor b_1}{a_2 \rfloor b_1} \right) a_2.$                                                                                                                                                                                                                                                                                                                                                             | $a = a_1 - \left( \frac{a_1 \rfloor b_1}{a_2 \rfloor b_1} \right) a_2.$                                                                                                                                                                                                                                                                                                                                                             |
| 1285.5935v1_<br>EQ0123 (117)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 031  | <div><span>■</span> 1.000</div> <div>●</div>      | $\langle A \rangle_r^* = \langle A^* \rangle_r,$                                                                                                                                                                                                                                                                                                                                                                                                                                    | $\langle A \rangle_r^* = \langle A^* \rangle_r,$                                                                                                                                                                                                                                                                                                                                                                                    | $\langle A \rangle_r^* = \langle A^* \rangle_r,$                                                                                                                                                                                                                                                                                                                                                                                    |
| 1285.5935v1_<br>EQ0128 (121)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 032  | <div><span>■</span> 1.000</div> <div>● K +0</div> | $\begin{aligned} a \downharpoonleft A &= \frac{1}{2} (aA - A^* a) \\ a \wedge A &= \frac{1}{2} (aA + A^* a), \end{aligned}$                                                                                                                                                                                                                                                                                                                                                         | $\begin{aligned} a \downharpoonleft A &= \frac{1}{2} (aA - A^* a) \\ a \wedge A &= \frac{1}{2} (aA + A^* a), \end{aligned}$                                                                                                                                                                                                                                                                                                         | $\begin{aligned} a \rfloor A &= \frac{1}{2} (aA - A^* a) \\ a \wedge A &= \frac{1}{2} (aA + A^* a), \end{aligned}$                                                                                                                                                                                                                                                                                                                  |
| 1285.5935v1_<br>EQ0163 (156)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 035  | <div><span>■</span> 1.000</div> <div>● N +0</div> | $\begin{aligned} A * B &= \sum_r A_r * B_r \\ &= \sum_r A_r^{\dagger} \rfloor B_r = \sum_r A_r^{\dagger} \rfloor B_r. \end{aligned}$                                                                                                                                                                                                                                                                                                                                                | $\begin{aligned} A * B &= \sum_r A_r * B_r \\ &= \sum_r A_r^{\dagger} \rfloor B_r = \sum_r A_r^{\dagger} \rfloor B_r. \end{aligned}$                                                                                                                                                                                                                                                                                                | $\begin{aligned} A * B &= \sum_r A_r * B_r \\ &= \sum_r A_r^{\dagger} \rfloor B_r = \sum_r A_r^{\dagger} \rfloor B_r. \end{aligned}$                                                                                                                                                                                                                                                                                                |
| <p><b>Conf.</b> MathPix confidence: <span>■</span> <math>\geq 0.9</math> <span>■</span> 0.5–0.9 <span>■</span> <math>&lt; 0.5</math> — no value</p> <p><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</p> <p><b>Residual</b> render vs scan (inkdrill): <span>●</span> C component <span>●</span> W weak <span>●</span> S stable <span>●</span> N noise <span>●</span> K clean <span>●</span> X crop overran its region <span>●</span> not measured</p> <p><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</p> |      |                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                     |

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Page | Conf.                                         | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Rendered                                                                                                                                                                                                                    | Scan image                                                                                                                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1205.5935v1_<br>EQ0172 (165)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 037  | <div><div></div>1.000</div> <div>● N +0</div> | $(A\ B)\ *(A\ C)=(B\ A)\ *(C\ A)= A ^{\wedge\{2\}}\ B\ *\ C$                                                                                                                                                                                                                                                                                                                                                                                                                  | $(AB)\ *(AC) = (BA)\ *(CA) =  A ^2 B\ * C.$                                                                                                                                                                                 | $(AB)\ *(AC) = (BA)\ *(CA) =  A ^2 B\ * C.$                                                                                                                                                                                 |
| 1205.5935v1_<br>EQ0184 (177)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 040  | <div><div></div>1.000</div> <div>● K +0</div> | $\begin{aligned} A_{\{r\}}\ \boldsymbol{\mathrm{I}}\ \&= (-1)^{\wedge\{r(n-1)\}}\ \boldsymbol{\mathrm{I}}\ A_{\{r\}}\ \backslash\&= \boldsymbol{\mathrm{I}}\ A_{\{r\}}^{\wedge\{*(n-1)\}},\end{aligned}$                                                                                                                                                                                                                                                                      | $A_r\boldsymbol{I} = (-1)^{r(n-1)}\boldsymbol{I}A_r$<br>$= \boldsymbol{I}A_r^{*(n-1)},$                                                                                                                                     | $A_r\boldsymbol{I} = (-1)^{r(n-1)}\boldsymbol{I}A_r$<br>$= \boldsymbol{I}A_r^{*(n-1)},$                                                                                                                                     |
| 1205.5935v1_<br>EQ0187 (180)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 040  | <div><div></div>1.000</div> <div>● K +0</div> | $(A\ B)^{\wedge\{\perp\}}=A\ B^{\wedge\{\perp\}}$                                                                                                                                                                                                                                                                                                                                                                                                                             | $(AB)^{\perp} = AB^{\perp}.$                                                                                                                                                                                                | $(AB)^{\perp} = AB^{\perp}.$                                                                                                                                                                                                |
| 1205.5935v1_<br>EQ0209 (202)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 044  | <div><div></div>1.000</div> <div>● K +0</div> | $\begin{aligned} \&\ \boldsymbol{\mathrm{e}}_{\{1\}}=-\boldsymbol{\mathrm{e}}_{\{2\}}\ \backslash\&\ \boldsymbol{\mathrm{e}}_{\{2\}}=\boldsymbol{\mathrm{e}}_{\{1\}},\end{aligned}$                                                                                                                                                                                                                                                                                           | $\boldsymbol{I}e_1 = -e_2$<br>$\boldsymbol{I}e_2 = e_1,$                                                                                                                                                                    | $\boldsymbol{I}e_1 = -e_2$<br>$\boldsymbol{I}e_2 = e_1,$                                                                                                                                                                    |
| 1205.5935v1_<br>EQ0262 (255)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 054  | <div><div></div>1.000</div> <div>● K +0</div> | $R\ \boldsymbol{\mathrm{R}}^{\{-1\}}=\boldsymbol{\mathrm{I}}$                                                                                                                                                                                                                                                                                                                                                                                                                 | $R\boldsymbol{I}R^{-1} = \boldsymbol{I},$                                                                                                                                                                                   | $R\boldsymbol{I}R^{-1} = \boldsymbol{I},$                                                                                                                                                                                   |
| 1205.5935v1_<br>EQ0268 (261)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 055  | <div><div></div>1.000</div> <div>● X +1</div> | $b\ \downharpoonleft c=\sum_{\{i,\ j\}}b_{\{i\}}\ c^{\{j\}}\backslashleft(a^{\{i\}}\ \downharpoonleft a_{\{j\}}\right)=\sum_{\{i\}}b_{\{i\}}\ c^{\{i\}}$                                                                                                                                                                                                                                                                                                                      | $b\downarrow c = \sum_{i,j}b_ic^j\left(a^i\downarrow a_j\right) = \sum_ib_ic^i,$                                                                                                                                            | $b\downharpoonleft c = \sum_{i,j}b_ic^j\left(a^i\downharpoonleft a_j\right) = \sum_ib_ic^i,$                                                                                                                                |
| 1205.5935v1_<br>EQ0299 (291)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 060  | <div><div></div>1.000</div> <div>● N +0</div> | $P_{\{a_{\{1\}}\}\ \wedge\cdots\ \wedge a_{\{r\}}\}(B)=\sum_{\{I\}}\backslashleft(B\ *\ a_{\{I\}}\right)\ a^{\{I\}}$                                                                                                                                                                                                                                                                                                                                                          | $P_{a_1\wedge\cdots\wedge a_r}(B) = \sum_I (B\ * a_I)\ a^I,$                                                                                                                                                                | $P_{a_1\wedge\cdots\wedge a_r}(B) = \sum_I (B\ * a_I)a^I,$                                                                                                                                                                  |
| 1205.5935v1_<br>EQ0317 (307)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 064  | <div><div></div>1.000</div> <div>● K +0</div> | $F(u)\rfloor F(v)=\frac{1}{2}\backslashleft[F(u+v)^{\wedge\{2\}}-F(u)^{\wedge\{2\}}-F(v)^{\wedge\{2\}}\right]$                                                                                                                                                                                                                                                                                                                                                                | $F(u)\rfloor F(v) = \frac{1}{2}\left[F(u+v)^2 - F(u)^2 - F(v)^2\right],$                                                                                                                                                    | $F(u)\rfloor F(v) = \frac{1}{2}\left[F(u+v)^2 - F(u)^2 - F(v)^2\right],$                                                                                                                                                    |
| 1205.5935v1_<br>EQ0348 (337)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 071  | <div><div></div>1.000</div> <div>● K +0</div> | $\begin{aligned} \mathcal{F}(A)\ \&= \mathcal{F}(1\ A)\ \backslash\&= \mathcal{F}(1\ \wedge A)\ \backslash\&= \mathcal{F}(1)\ \wedge\mathcal{F}(A)\ \backslash\&= 0,\end{aligned}$                                                                                                                                                                                                                                                                                            | $\mathcal{F}(A) = \mathcal{F}(1A)$<br>$= \mathcal{F}(1\wedge A)$<br>$= \mathcal{F}(1)\wedge \mathcal{F}(A)$<br>$= 0,$                                                                                                       | $\mathcal{F}(A) = \mathcal{F}(1A)$<br>$= \mathcal{F}(1\wedge A)$<br>$= \mathcal{F}(1)\wedge \mathcal{F}(A)$<br>$= 0,$                                                                                                       |
| 1205.5935v1_<br>EQ0370 (357)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 076  | <div><div></div>1.000</div> <div>● N +0</div> | $\operatorname{det}(F)=F(\boldsymbol{\mathrm{I}})^{\wedge\{\perp\}}$                                                                                                                                                                                                                                                                                                                                                                                                          | $\det(F) = F(\boldsymbol{I})^{\perp}.$                                                                                                                                                                                      | $\det(F) = F(\boldsymbol{I})^{\perp}.$                                                                                                                                                                                      |
| 1205.5935v1_<br>EQ0425 (43)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 085  | <div><div></div>1.000</div> <div>●</div>      | $\begin{gathered} A_{\{r\}}^{\wedge\{*\}}=(-1)^{\wedge\{r\}}\ A_{\{r\}}\ \backslash\&A^{\{*\}}=\ \angle A\angle_{\{+\}}\backslash\angle A\angle_{\{-}}\ \backslash\&A^{\{*\}}=\ \boldsymbol{\mathrm{I}}\ A^{\{*\ n\}}\ \boldsymbol{\mathrm{I}}^{\{-1\}}\ \backslash\&\angle A\ \angle_{\{\pm\}}=\frac{1}{2}\backslashleft(A\ \pm A^{\{*\}}\right)\ \backslash\&A^{\{*\}}=A\ \backslash\left(A^{\{-1\}}\right)^{\{*\}}=\backslashleft(A^{\{*\}}\right)^{\{-1\}}\end{gathered}$ | $A_r^* = (-1)^r A_r$<br>$A^* = \langle A\rangle_+ - \langle A\rangle_-$<br>$A^* = \boldsymbol{I}A^{*n}\boldsymbol{I}^{-1}$<br>$\langle A\rangle_{\pm} = \frac{1}{2}(A\pm A^*)$<br>$A^{**} = A$<br>$(A^{-1})^* = (A^*)^{-1}$ | $A_r^* = (-1)^r A_r$<br>$A^* = \langle A\rangle_+ - \langle A\rangle_-$<br>$A^* = \boldsymbol{I}A^{*n}\boldsymbol{I}^{-1}$<br>$\langle A\rangle_{\pm} = \frac{1}{2}(A\pm A^*)$<br>$A^{**} = A$<br>$(A^{-1})^* = (A^*)^{-1}$ |
| <div> <b>Conf.</b> MathPix confidence: <div><div></div>≥0.9</div> <div><div></div>0.5–0.9</div> <div><div></div>&lt;0.5</div> — no value         </div> <div> <b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ         </div> <div> <b>Residual</b> render vs scan (inkdrill): <div><div></div>C component</div> <div><div></div>W weak</div> <div><div></div>S stable</div> <div><div></div>N noise</div> <div><div></div>K clean</div> <div><div></div>X crop overran its region</div> <div><div></div>not measured</div> </div> <div> <b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.         </div> |      |                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                             |                                                                                                                                                                                                                             |

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Page | Conf.                                         | LaTeX source                                                                                                                                 | Rendered                                                                                                | Scan image                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| 1205.5935v1_<br>EQ0445 (92)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 088  | <div><div></div>1.000</div> <div>● N +0</div> | $\begin{aligned} A^{\{-\perp\}} \&= \mathbb{A} \rfloor \boldsymbol{I} \\ \&= \boldsymbol{A} \\ \&= \boldsymbol{I}^2 A^{\perp} \end{aligned}$ | $A^{-\perp} = A \mathbb{I}$ $= \boldsymbol{A} \boldsymbol{I}$ $= \boldsymbol{I}^2 A^{\perp}$            | $A^{-\perp} = A \mathbb{I}$ $= \boldsymbol{A} \boldsymbol{I}$ $= \boldsymbol{I}^2 A^{\perp}$            |
| 1205.5935v1_<br>EQ0451 (101)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 088  | <div><div></div>1.000</div> <div>● N +0</div> | $A^{\{\perp\}} * B^{\{\perp\}} =  \boldsymbol{I} ^{-2} A * B$                                                                                | $A^{\perp} * B^{\perp} =  \boldsymbol{I} ^{-2} A * B$                                                   | $A^{\perp} * B^{\perp} =  \boldsymbol{I} ^{-2} A * B$                                                   |
| 1205.5935v1_<br>EQ0463 (118)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 089  | <div><div></div>1.000</div> <div>● K +0</div> | $a^{\{i\}} = (-1)^{i-1} (a_1 \wedge a_2 \wedge \cdots \wedge \check{a}_i \wedge \cdots \wedge a_n) a_N^{-1}$                                 | $a^i = (-1)^{i-1} (a_1 \wedge a_2 \wedge \cdots \wedge \check{a}_i \wedge \cdots \wedge a_n) a_N^{-1}.$ | $a^i = (-1)^{i-1} (a_1 \wedge a_2 \wedge \cdots \wedge \check{a}_i \wedge \cdots \wedge a_n) a_N^{-1}.$ |
| 1205.5935v1_<br>EQ0471 (129)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 090  | <div><div></div>1.000</div> <div>● N +0</div> | $a^{\{N\}} = \frac{a_N}{ a_N ^2}$                                                                                                            | $a^N = \frac{a_N}{ a_N ^2}.$                                                                            | $a^N = \frac{a_N}{ a_N ^2}.$                                                                            |
| 1205.5935v1_<br>EQ0001 (1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 003  | <div><div></div>1.000</div> <div>● N +0</div> | $P_{\{u\}}(v) = \frac{u \cdot v}{ u ^2} u$                                                                                                   | $P_u(v) = \frac{u \cdot v}{ u ^2} u$                                                                    | $P_u(v) = \frac{u \cdot v}{ u ^2} u$                                                                    |
| 1205.5935v1_<br>EQ0002 (2)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 003  | <div><div></div>1.000</div> <div>● K +0</div> | $u \, v = \frac{1}{2} (u \, v + v \, u) + \frac{1}{2} (u \, v - v \, u)$                                                                     | $uv = \frac{1}{2}(uv + vu) + \frac{1}{2}(uv - vu).$                                                     | $uv = \frac{1}{2}(uv + vu) + \frac{1}{2}(uv - vu).$                                                     |
| 1205.5935v1_<br>EQ0003 (3)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 004  | <div><div></div>1.000</div> <div>● K +0</div> | $\frac{1}{2}(u \, v + v \, u) := u \cdot v$                                                                                                  | $\frac{1}{2}(uv + vu) := u \cdot v.$                                                                    | $\frac{1}{2}(uv + vu) := u \cdot v.$                                                                    |
| 1205.5935v1_<br>EQ0004 (4)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 004  | <div><div></div>1.000</div> <div>● N +0</div> | $u^{\{-1\}} := \frac{u}{u^2}$                                                                                                                | $u^{-1} := \frac{u}{u^2}$                                                                               | $u^{-1} := \frac{u}{u^2}$                                                                               |
| 1205.5935v1_<br>EQ0005 (5)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 004  | <div><div></div>1.000</div> <div>● K +0</div> | $P_{\{u\}}(v) = (v \cdot u) u^{\{-1\}}$                                                                                                      | $P_u(v) = (v \cdot u) u^{-1}.$                                                                          | $P_u(v) = (v \cdot u) u^{-1}.$                                                                          |
| 1205.5935v1_<br>EQ0006 (6)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 004  | <div><div></div>1.000</div> <div>● K +0</div> | $u \, v = u \cdot v + u \wedge v$                                                                                                            | $uv = u \cdot v + u \wedge v.$                                                                          | $uv = u \cdot v + u \wedge v.$                                                                          |
| 1205.5935v1_<br>EQ0007 (7)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 004  | <div><div></div>1.000</div> <div>● K +0</div> | $v \, u = u \cdot v - u \wedge v$                                                                                                            | $vu = u \cdot v - u \wedge v.$                                                                          | $vu = u \cdot v - u \wedge v.$                                                                          |
| 1205.5935v1_<br>EQ0008 (8)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 004  | <div><div></div>1.000</div> <div>● K +0</div> | $u \, v \, v \, u = (u \cdot v)^2 - (u \wedge v)^2$                                                                                          | $uvvu = (u \cdot v)^2 - (u \wedge v)^2.$                                                                | $uvvu = (u \cdot v)^2 - (u \wedge v)^2.$                                                                |
| 1205.5935v1_<br>EQ0009 (9)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 004  | <div><div></div>1.000</div> <div>● K +0</div> | $(u \wedge v)^2 = - u ^2  v ^2 \sin^2 \theta$                                                                                                | $(u \wedge v)^2 = - u ^2  v ^2 \sin^2 \theta.$                                                          | $(u \wedge v)^2 = - u ^2  v ^2 \sin^2 \theta.$                                                          |
| 1205.5935v1_<br>EQ0010 (10)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 004  | <div><div></div>1.000</div> <div>●</div>      | $u \wedge (v + \lambda u) = u \wedge v$                                                                                                      | $u \wedge (v + \lambda u) = u \wedge v.$                                                                | $u \wedge (v + \lambda u) = u \wedge v.$                                                                |
| 1205.5935v1_<br>EQ0011 (11)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 004  | <div><div></div>1.000</div> <div>● K +0</div> | $(u \wedge v) \wedge w = u \wedge (v \wedge w) = u \wedge v \wedge w$                                                                        | $(u \wedge v) \wedge w = u \wedge (v \wedge w) = u \wedge v \wedge w.$                                  | $(u \wedge v) \wedge w = u \wedge (v \wedge w) = u \wedge v \wedge w.$                                  |
| 1205.5935v1_<br>EQ0012 (12)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 005  | <div><div></div>1.000</div> <div>● K +0</div> | $x = \alpha a + \beta b$                                                                                                                     | $x = \alpha a + \beta b.$                                                                               | $x = \alpha a + \beta b.$                                                                               |
| 1205.5935v1_<br>EQ0013 (13)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 005  | <div><div></div>1.000</div> <div>● N +0</div> | $a \wedge x = \beta a \wedge b$                                                                                                              | $a \wedge x = \beta a \wedge b.$                                                                        | $a \wedge x = \beta a \wedge b.$                                                                        |
| 1205.5935v1_<br>EQ0014 (14)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 005  | <div><div></div>1.000</div> <div>● N +0</div> | $\beta = (a \wedge x)(a \wedge b)^{-1}$                                                                                                      | $\beta = (a \wedge x)(a \wedge b)^{-1}.$                                                                | $\beta = (a \wedge x)(a \wedge b)^{-1}.$                                                                |
| 1205.5935v1_<br>EQ0015 (15)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 005  | <div><div></div>1.000</div> <div>● N +0</div> | $\beta = \frac{a \wedge x}{a \wedge b}$                                                                                                      | $\beta = \frac{a \wedge x}{a \wedge b}.$                                                                | $\beta = \frac{a \wedge x}{a \wedge b}.$                                                                |
| <p><b>Conf.</b> MathPix confidence: <div><div></div>≥0.9</div> <div><div></div>0.5–0.9</div> <div><div></div>&lt;0.5</div> — no value</p> <p><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</p> <p><b>Residual</b> render vs scan (inkdrill): <div><div></div>C component</div> <div><div></div>W weak</div> <div><div></div>S stable</div> <div><div></div>N noise</div> <div><div></div>K clean</div> <div><div></div>X crop overran its region</div> <div><div></div>not measured</div></p> <p><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</p> |      |                                               |                                                                                                                                              |                                                                                                         |                                                                                                         |

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Page | Conf.                                         | LaTeX source                                                                                                                                                                      | Rendered                                                                                                                                                                          | Scan image                                                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1205.5935v1_<br>EQ0016 (16)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 005  | <div><div></div>1.000</div> <div>● N +0</div> | $x=\left(\frac{x\wedge b}{a\wedge b}\right)a+\left(\frac{a\wedge x}{a\wedge b}\right)b$                                                                                           | $x=\left(\frac{x\wedge b}{a\wedge b}\right)a+\left(\frac{a\wedge x}{a\wedge b}\right)b.$                                                                                          | $x=\left(\frac{x\wedge b}{a\wedge b}\right)a+\left(\frac{a\wedge x}{a\wedge b}\right)b.$                                                                                          |
| 1205.5935v1_<br>EQ0017 (17)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 005  | <div><div></div>1.000</div> <div>● N +0</div> | $x=\left(\frac{x\wedge b\wedge c}{a\wedge b\wedge c}\right)a+\left(\frac{a\wedge x\wedge c}{a\wedge b\wedge c}\right)b+\left(\frac{a\wedge b\wedge x}{a\wedge b\wedge c}\right)c$ | $x=\left(\frac{x\wedge b\wedge c}{a\wedge b\wedge c}\right)a+\left(\frac{a\wedge x\wedge c}{a\wedge b\wedge c}\right)b+\left(\frac{a\wedge b\wedge x}{a\wedge b\wedge c}\right)c$ | $x=\left(\frac{x\wedge b\wedge c}{a\wedge b\wedge c}\right)a+\left(\frac{a\wedge x\wedge c}{a\wedge b\wedge c}\right)b+\left(\frac{a\wedge b\wedge x}{a\wedge b\wedge c}\right)c$ |
| 1205.5935v1_<br>EQ0018 (18)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 005  | <div><div></div>1.000</div> <div>● K +0</div> | $x=p+\lambda a$                                                                                                                                                                   | $x=p+\lambda a$                                                                                                                                                                   | $x=p+\lambda a$                                                                                                                                                                   |
| 1205.5935v1_<br>EQ0019 (19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 006  | <div><div></div>1.000</div> <div>● X +1</div> | $x=\left(\frac{q\wedge b}{a\wedge b}\right)a+\left(\frac{a\wedge p}{a\wedge b}\right)b$                                                                                           | $x=\left(\frac{q\wedge b}{a\wedge b}\right)a+\left(\frac{a\wedge p}{a\wedge b}\right)b$                                                                                           | $x=\left(\frac{q\wedge b}{a\wedge b}\right)a+\left(\frac{a\wedge p}{a\wedge b}\right)b,$                                                                                          |
| 1205.5935v1_<br>EQ0021 (21)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 006  | <div><div></div>1.000</div> <div>● K +0</div> | $v^{\prime}=-\left(v\cdot n\right)n^{-1}+\left(v\wedge n\right)n^{-1}$                                                                                                            | $v'=-\left(v\cdot n\right)n^{-1}+\left(v\wedge n\right)n^{-1}.$                                                                                                                   | $v'=-\left(v\cdot n\right)n^{-1}+\left(v\wedge n\right)n^{-1}.$                                                                                                                   |
| 1205.5935v1_<br>EQ0023 (23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 006  | <div><div></div>1.000</div> <div>● N +1</div> | $v^{\prime}=v-2\frac{n\cdot v}{ n ^2}n$                                                                                                                                           | $v'=v-2\frac{n\cdot v}{ n ^2}n$                                                                                                                                                   | $v'=v-2\frac{n\cdot v}{ n ^2}n,$                                                                                                                                                  |
| 1205.5935v1_<br>EQ0024 (24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 007  | <div><div></div>1.000</div> <div>● N +0</div> | $\begin{aligned}v^{\prime}&=-m\left(-nv n^{-1}\right)\\&=\left(mn\right)v\left(mn\right)^{-1}\\&=RvR^{-1}\end{aligned}$                                                           | $\begin{aligned}v'&=-m\left(-nv n^{-1}\right)m^{-1}\\&=\left(mn\right)v\left(mn\right)^{-1}\\&=RvR^{-1}\end{aligned}$                                                             | $\begin{aligned}v'&=-m\left(-nv n^{-1}\right)m^{-1}\\&=\left(mn\right)v\left(mn\right)^{-1}\\&=RvR^{-1}\end{aligned}$                                                             |
| 1205.5935v1_<br>EQ0025 (25)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 007  | <div><div></div>1.000</div> <div>● K +0</div> | $\begin{aligned}R&=m\cdot n+m\wedge n\\&=m\cdot n-n\wedge m.\end{aligned}$                                                                                                        | $\begin{aligned}R&=mn\\&=m\cdot n+m\wedge n\\&=m\cdot n-n\wedge m.\end{aligned}$                                                                                                  | $\begin{aligned}R&=mn\\&=m\cdot n+m\wedge n\\&=m\cdot n-n\wedge m.\end{aligned}$                                                                                                  |
| 1205.5935v1_<br>EQ0026 (26)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 007  | <div><div></div>1.000</div> <div>● N +0</div> | $\begin{aligned}R&=\cos\left(\theta/2\right)-\sin\left(\theta/2\right)B\\&=\exp\left(-B\theta/2\right)\end{aligned}$                                                              | $\begin{aligned}R&=\cos\left(\theta/2\right)-\sin\left(\theta/2\right)B\\&=\exp\left(-B\theta/2\right)\end{aligned}$                                                              | $\begin{aligned}R&=\cos\left(\theta/2\right)-\sin\left(\theta/2\right)B\\&=\exp\left(-B\theta/2\right)\end{aligned}$                                                              |
| 1205.5935v1_<br>EQ0027 (27)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 008  | <div><div></div>1.000</div> <div>● K +0</div> | $\left[\left(v\wedge n\right)n^{-1}\right]\cdot n=0$                                                                                                                              | $\left[\left(v\wedge n\right)n^{-1}\right]\cdot n=0,$                                                                                                                             | $\left[\left(v\wedge n\right)n^{-1}\right]\cdot n=0,$                                                                                                                             |
| 1205.5935v1_<br>EQ0028 (28)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 011  | <div><div></div>1.000</div> <div>● K +0</div> | $\frac{1}{2}\left(uv+vu\right)=\frac{1}{2}\left[\left(u+v\right)^2-u^2-v^2\right]$                                                                                                | $\frac{1}{2}\left(uv+vu\right)=\frac{1}{2}\left[\left(u+v\right)^2-u^2-v^2\right]$                                                                                                | $\frac{1}{2}\left(uv+vu\right)=\frac{1}{2}\left[\left(u+v\right)^2-u^2-v^2\right]$                                                                                                |
| 1205.5935v1_<br>EQ0029 (29)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 012  | <div><div></div>1.000</div> <div>● N +0</div> | $\lambda A=A\lambda\in\mathcal{G}_r.$                                                                                                                                             | $\lambda A=A\lambda\in\mathcal{G}_r.$                                                                                                                                             | $\lambda A=A\lambda\in\mathcal{G}_r.$                                                                                                                                             |
| 1205.5935v1_<br>EQ0030 (30)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 013  | <div><div></div>1.000</div> <div>●</div>      | $a_{\{1\}}\wedge a_{\{2\}}\wedge\cdots\wedge a_{\{r\}}:=\frac{1}{r!}\sum_{\sigma}(\operatorname{sgn}\sigma)a_{\sigma(1)}a_{\sigma(2)}\cdots a_{\sigma(r)}$                        | $a_1\wedge a_2\wedge\cdots\wedge a_r:=\frac{1}{r!}\sum_{\sigma}(\operatorname{sgn}\sigma)a_{\sigma(1)}a_{\sigma(2)}\cdots a_{\sigma(r)}$                                          | $a_1\wedge a_2\wedge\cdots\wedge a_r:=\frac{1}{r!}\sum_{\sigma}(\operatorname{sgn}\sigma)a_{\sigma(1)}a_{\sigma(2)}\cdots a_{\sigma(r)}$                                          |
| 1205.5935v1_<br>EQ0031 (31)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 013  | <div><div></div>1.000</div> <div>● K +0</div> | $e_{\{1\}}\wedge e_{\{2\}}\wedge\cdots\wedge e_{\{r\}}=\frac{1}{r!}\sum_{\sigma}(\operatorname{sgn}\sigma)e_{\sigma(1)}e_{\sigma(2)}\cdots e_{\sigma(r)}.$                        | $e_1\wedge e_2\wedge\cdots\wedge e_r=\frac{1}{r!}\sum_{\sigma}(\operatorname{sgn}\sigma)e_{\sigma(1)}e_{\sigma(2)}\cdots e_{\sigma(r)}.$                                          | $e_1\wedge e_2\wedge\cdots\wedge e_r=\frac{1}{r!}\sum_{\sigma}(\operatorname{sgn}\sigma)e_{\sigma(1)}e_{\sigma(2)}\cdots e_{\sigma(r)}.$                                          |
| <p><b>Conf.</b> MathPix confidence: <div><div></div>≥0.9</div> <div><div></div>0.5–0.9</div> <div><div></div>&lt;0.5</div> — no value</p> <p><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</p> <p><b>Residual</b> render vs scan (inkdrill): <div><div></div>C component</div> <div><div></div>W weak</div> <div><div></div>S stable</div> <div><div></div>N noise</div> <div><div></div>K clean</div> <div><div></div>X crop overran its region</div> <div><div></div>not measured</div></p> <p><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</p> |      |                                               |                                                                                                                                                                                   |                                                                                                                                                                                   |                                                                                                                                                                                   |

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Page | Conf.                                             | LaTeX source                                                                                                                                                                                            | Rendered                                                                                                                                                                                  | Scan image                                                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1205.5935v1_ E00032 (32)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 013  | <div> <div>■ 1.000</div> <div>● K +0</div> </div> | $\begin{aligned} e_{\{1\}} \wedge e_{\{2\}} \wedge \cdots \wedge e_{\{r\}} &= \frac{1}{r!} \sum_{\sigma} e_{\{1\}} e_{\{2\}} \cdots e_{\{r\}} \\ &= e_{\{1\}} e_{\{2\}} \cdots e_{\{r\}} \end{aligned}$ | $e_1 \wedge e_2 \wedge \cdots \wedge e_r = \frac{1}{r!} \sum_{\sigma} e_1 e_2 \cdots e_r = e_1 e_2 \cdots e_r$                                                                            | $e_1 \wedge e_2 \wedge \cdots \wedge e_r = \frac{1}{r!} \sum_{\sigma} e_1 e_2 \cdots e_r = e_1 e_2 \cdots e_r$                                                                            |
| 1205.5935v1_ E00033 (33)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 014  | <div> <div>■ 1.000</div> <div>● X +1</div> </div> | $e_{\{1\}} \wedge e_{\{2\}} = e_{\{1\}} e_{\{2\}}$                                                                                                                                                      | $e_1 \wedge e_2 = e_1 e_2$                                                                                                                                                                | $e_1 \wedge e_2 = e_1 e_2,$                                                                                                                                                               |
| 1205.5935v1_ E00036 (36)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 014  | <div> <div>■ 1.000</div> <div>● K +0</div> </div> | $\begin{aligned} a_{\{1\}} \wedge a_{\{2\}} \wedge \cdots \wedge a_{\{r\}} &= e_{\{1\}} \wedge e_{\{2\}} \wedge \cdots \wedge e_{\{r\}} \\ &= e_{\{1\}} e_{\{2\}} \cdots e_{\{r\}} \end{aligned}$       | $a_1 \wedge a_2 \wedge \cdots \wedge a_r = e_1 \wedge e_2 \wedge \cdots \wedge e_r = e_1 e_2 \cdots e_r$                                                                                  | $a_1 \wedge a_2 \wedge \cdots \wedge a_r = e_1 \wedge e_2 \wedge \cdots \wedge e_r = e_1 e_2 \cdots e_r$                                                                                  |
| 1205.5935v1_ E00052 (51)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 018  | <div> <div>■ 1.000</div> <div>● K +0</div> </div> | $a \wedge A_{\{r\}} := \frac{1}{2} \left[ a A_{\{r\}} + (-1)^r A_{\{r\}} a \right]$                                                                                                                     | $a \wedge A_r := \frac{1}{2} [a A_r + (-1)^r A_r a],$                                                                                                                                     | $a \wedge A_r := \frac{1}{2} [a A_r + (-1)^r A_r a],$                                                                                                                                     |
| 1205.5935v1_ E00054 (53)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 018  | <div> <div>■ 1.000</div> <div>● K +0</div> </div> | $a \wedge A_{\{+\}} = A_{\{+\}} \wedge a \quad \text{while} \quad a \wedge A_{\{-}} = -A_{\{-}} \wedge a.$                                                                                              | $a \wedge A_+ = A_+ \wedge a \quad \text{while} \quad a \wedge A_- = -A_- \wedge a.$                                                                                                      | $a \wedge A_+ = A_+ \wedge a \quad \text{while} \quad a \wedge A_- = -A_- \wedge a.$                                                                                                      |
| 1205.5935v1_ E00056 (55)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 018  | <div> <div>■ 1.000</div> <div>● K +0</div> </div> | $a \wedge a_{\{1\}} \wedge a_{\{2\}} \wedge \cdots \wedge a_{\{r\}} = a \wedge e_{\{1\}} \wedge e_{\{2\}} \wedge \cdots \wedge e_{\{r\}}$                                                               | $a \wedge a_1 \wedge a_2 \wedge \cdots \wedge a_r = a \wedge e_1 \wedge e_2 \wedge \cdots \wedge e_r$                                                                                     | $a \wedge a_1 \wedge a_2 \wedge \cdots \wedge a_r = a \wedge e_1 \wedge e_2 \wedge \cdots \wedge e_r$                                                                                     |
| 1205.5935v1_ E00057 (56)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 018  | <div> <div>■ 1.000</div> <div>● X +1</div> </div> | $\frac{1}{2} \left[ a (e_1 e_2 \cdots e_r) + (-1)^r (e_1 e_2 \cdots e_r) a \right] = a \wedge e_1 \wedge e_2 \wedge \cdots \wedge e_r.$                                                                 | $\frac{1}{2} [a (e_1 e_2 \cdots e_r) + (-1)^r (e_1 e_2 \cdots e_r) a] = a \wedge e_1 \wedge e_2 \wedge \cdots \wedge e_r$                                                                 | $\frac{1}{2} [a (e_1 e_2 \cdots e_r) + (-1)^r (e_1 e_2 \cdots e_r) a] = a \wedge e_1 \wedge e_2 \wedge \cdots \wedge e_r.$                                                                |
| 1205.5935v1_ E00059 (58)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 019  | <div> <div>■ 1.000</div> <div>● K +0</div> </div> | $\frac{1}{r+1} \left[ \frac{1}{r!} \sum_{\pi} (\text{sgn } \pi) a e_{\pi(1)} e_{\pi(2)} \cdots e_{\pi(r)} \right] = \frac{1}{r+1} a e_1 e_2 \cdots e_r.$                                                | $\frac{1}{r+1} \left[ \frac{1}{r!} \sum_{\pi} (\text{sgn } \pi) a e_{\pi(1)} e_{\pi(2)} \cdots e_{\pi(r)} \right] = \frac{1}{r+1} a e_1 e_2 \cdots e_r.$                                  | $\frac{1}{r+1} \left[ \frac{1}{r!} \sum_{\pi} (\text{sgn } \pi) a e_{\pi(1)} e_{\pi(2)} \cdots e_{\pi(r)} \right] = \frac{1}{r+1} a e_1 e_2 \cdots e_r.$                                  |
| 1205.5935v1_ E00064 (63)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 020  | <div> <div>■ 1.000</div> <div>● N +0</div> </div> | $\begin{aligned} & \frac{1}{2r+2} [a e_1 e_2 \cdots e_r - (-1)^r e_1 e_2 \cdots e_r a] \\ &= \frac{1}{2} [a e_1 e_2 \cdots e_r + (-1)^r e_1 e_2 \cdots e_r a], \end{aligned}$                           | $\frac{1}{2r+2} [a e_1 e_2 \cdots e_r - (-1)^r e_1 e_2 \cdots e_r a] = \frac{1}{2} [a e_1 e_2 \cdots e_r + (-1)^r e_1 e_2 \cdots e_r a],$                                                 | $\frac{1}{2r+2} [a e_1 e_2 \cdots e_r - (-1)^r e_1 e_2 \cdots e_r a] = \frac{1}{2} [a e_1 e_2 \cdots e_r + (-1)^r e_1 e_2 \cdots e_r a],$                                                 |
| 1205.5935v1_ E00065 (64)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 020  | <div> <div>■ 1.000</div> <div>● N +0</div> </div> | $a \wedge A_{\{r\}} = a \wedge \downharpoonleft A_{\{r\}} + a \wedge A_{\{r\}}$                                                                                                                         | $a A_r = a \wedge A_r + a \wedge A_r,$                                                                                                                                                    | $a A_r = a \wedge A_r + a \wedge A_r,$                                                                                                                                                    |
| 1205.5935v1_ E00067 (66)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 020  | <div> <div>■ 1.000</div> <div>● K +0</div> </div> | $A_{\{r\}} a = A_{\{r\}} \lfloor a + A_{\{r\}} \wedge a, \rfloor$                                                                                                                                       | $A_r a = A_r \lfloor a + A_r \wedge a,$                                                                                                                                                   | $A_r a = A_r \lfloor a + A_r \wedge a,$                                                                                                                                                   |
| 1205.5935v1_ E00069 (68)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 020  | <div> <div>■ 1.000</div> <div>● X +1</div> </div> | $A_{\{r\}} B_{\{s\}} = \sum_{j=0}^{\min\{r,s\}} \langle A_r B_s \rangle_{ r-s +2j}$                                                                                                                     | $A_r B_s = \sum_{j=0}^{\min\{r,s\}} \langle A_r B_s \rangle_{ r-s +2j}$                                                                                                                   | $A_r B_s = \sum_{j=0}^{\min\{r,s\}} \langle A_r B_s \rangle_{ r-s +2j}.$                                                                                                                  |
| 1205.5935v1_ E00071 (69)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 021  | <div> <div>■ 1.000</div> <div>●</div> </div>      | $\begin{aligned} & \sum_{j=0}^{\min\{r-1,s\}} a \langle \mathbf{A}_{r-1} B_s \rangle_{ r-1-s +2j} \\ &= \sum_{j=0}^{r-1} a \langle \mathbf{A}_{r-1} B_s \rangle_{s-r+2j+1} \end{aligned}$               | $\begin{aligned} & \sum_{j=0}^{\min\{r-1,s\}} a \langle \mathbf{A}_{r-1} B_s \rangle_{ r-1-s +2j} \\ &= \sum_{j=0}^{r-1} a \langle \mathbf{A}_{r-1} B_s \rangle_{s-r+2j+1} \end{aligned}$ | $\begin{aligned} & \sum_{j=0}^{\min\{r-1,s\}} a \langle \mathbf{A}_{r-1} B_s \rangle_{ r-1-s +2j} \\ &= \sum_{j=0}^{r-1} a \langle \mathbf{A}_{r-1} B_s \rangle_{s-r+2j+1} \end{aligned}$ |
| <p><b>Conf.</b> MathPix confidence: ■ ≥0.9   ■ 0.5–0.9   ■ &lt;0.5   — no value</p> <p><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</p> <p><b>Residual</b> render vs scan (inkdrill): ● C component   ● W weak   ● S stable   ● N noise   ● K clean   ● X crop overran its region   ● not measured</p> <p><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</p> |      |                                                   |                                                                                                                                                                                                         |                                                                                                                                                                                           |                                                                                                                                                                                           |



| Identifier                   | Page | Conf.                                                          | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Scan image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------|------|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1285.5935v1_<br>EQ0139 (132) | 033  | <div><span>■</span> 1.000</div> <div><span>●</span> K +0</div> | $\left(A^{-1}\right)^{\dagger}=\left(A^{\dagger}\right)^{-1}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | $\left(A^{-1}\right)^{\dagger}=\left(A^{\dagger}\right)^{-1}.$                                                                                                                                                                                                                                                                                                                                                                                                                              | $\left(A^{-1}\right)^{\dagger}=\left(A^{\dagger}\right)^{-1}.$                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 1285.5935v1_<br>EQ0141 (134) | 033  | <div><span>■</span> 1.000</div> <div><span>●</span> N +0</div> | $\begin{aligned} \langle A\,B\rangle_r&=\left(-1\right)^{r\left(r-1\right) / 2}\left\langle\left(A\,B\right)^{\dagger}\right\rangle_r \\ &=\left(-1\right)^{r\left(r-1\right) / 2}\left\langle B^{\dagger} A^{\dagger}\right\rangle_r \end{aligned}$                                                                                                                                                                                                                                                                                  | $\begin{aligned} \langle A B\rangle_r &= (-1)^{r(r-1) / 2}\left\langle (A B)^{\dagger}\right\rangle_r \\ &= (-1)^{r(r-1) / 2}\left\langle B^{\dagger} A^{\dagger}\right\rangle_r \end{aligned}$                                                                                                                                                                                                                                                                                             | $\begin{aligned} \langle A B\rangle_r &= (-1)^{r(r-1) / 2}\left\langle (A B)^{\dagger}\right\rangle_r \\ &= (-1)^{r(r-1) / 2}\left\langle B^{\dagger} A^{\dagger}\right\rangle_r \end{aligned}$                                                                                                                                                                                                                                                                                               |
| 1285.5935v1_<br>EQ0142 (135) | 033  | <div><span>■</span> 1.000</div> <div><span>●</span> K +0</div> | $\langle A\,B\rangle=\left\langle B^{\dagger} A^{\dagger}\right\rangle$                                                                                                                                                                                                                                                                                                                                                                                                                                                               | $\langle A B\rangle=\left\langle B^{\dagger} A^{\dagger}\right\rangle.$                                                                                                                                                                                                                                                                                                                                                                                                                     | $\langle A B\rangle=\left\langle B^{\dagger} A^{\dagger}\right\rangle .$                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 1285.5935v1_<br>EQ0144 (137) | 033  | <div><span>■</span> 1.000</div> <div><span>●</span> K +0</div> | $\langle A\,B\rangle=\langle B\,A\rangle$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | $\langle A B\rangle=\langle B A\rangle$                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $\langle A B\rangle=\langle B A\rangle$                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 1285.5935v1_<br>EQ0145 (138) | 033  | <div><span>■</span> 1.000</div> <div><span>●</span> K +0</div> | $\langle A\,B\,\cdots C\,D\rangle=\langle D\,A\,B\,\cdots C\,\rangle$                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $\langle A B \cdots C D\rangle=\langle D A B \cdots C\rangle,$                                                                                                                                                                                                                                                                                                                                                                                                                              | $\langle A B \cdots C D\rangle=\langle D A B \cdots C\rangle ,$                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 1285.5935v1_<br>EQ0147 (140) | 034  | <div><span>■</span> 1.000</div> <div><span>●</span> X +1</div> | $\begin{aligned} \left\langle A_r B_s\right\rangle_{r+s-2 j} &=\left(-1\right)^{(r+s-2 j)(r+s-2 j-1) / 2}\left\langle\left(A_r B_s\right)^{\dagger}\right\rangle_{r+s-2 j} \\ &=\left(-1\right)^{(r+s-2 j)(r+s-2 j-1) / 2}\left\langle B_s^{\dagger} A_r^{\dagger}\right\rangle_{r+s-2 j} \\ &=\left(-1\right)^{(r+s-2 j)(r+s-2 j-1) / 2}\left(-1\right)^{r(r-1) / 2}\left(-1\right)^{s(s-1) / 2}\left\langle B_s A_r\right\rangle_{r+s-2 j} \\ &=\left(-1\right)^{r s-j}\left\langle B_s A_r\right\rangle_{r+s-2 j} . \end{aligned}$ | $\begin{aligned} \left\langle A_r B_s\right\rangle_{r+s-2 j} &= (-1)^{(r+s-2 j)(r+s-2 j-1) / 2}\left\langle\left(A_r B_s\right)^{\dagger}\right\rangle_{r+s-2 j} \\ &= (-1)^{(r+s-2 j)(r+s-2 j-1) / 2}\left\langle B_s^{\dagger} A_r^{\dagger}\right\rangle_{r+s-2 j} \\ &= (-1)^{(r+s-2 j)(r+s-2 j-1) / 2}\left(-1\right)^{r(r-1) / 2}\left(-1\right)^{s(s-1) / 2}\left\langle B_s A_r\right\rangle_{r+s-2 j} \\ &= (-1)^{r s-j}\left\langle B_s A_r\right\rangle_{r+s-2 j} \end{aligned}$ | $\begin{aligned} \left\langle A_r B_s\right\rangle_{r+s-2 j} &= (-1)^{(r+s-2 j)(r+s-2 j-1) / 2}\left\langle\left(A_r B_s\right)^{\dagger}\right\rangle_{r+s-2 j} \\ &= (-1)^{(r+s-2 j)(r+s-2 j-1) / 2}\left\langle B_s^{\dagger} A_r^{\dagger}\right\rangle_{r+s-2 j} \\ &= (-1)^{(r+s-2 j)(r+s-2 j-1) / 2}\left(-1\right)^{r(r-1) / 2}\left(-1\right)^{s(s-1) / 2}\left\langle B_s A_r\right\rangle_{r+s-2 j} \\ &= (-1)^{r s-j}\left\langle B_s A_r\right\rangle_{r+s-2 j} . \end{aligned}$ |
| 1285.5935v1_<br>EQ0150 (143) | 034  | <div><span>■</span> 1.000</div> <div><span>●</span> N +0</div> | $A_{\{r\}} \wedge B_{\{s\}}=(-1)^{r\,s} B_{\{s\}} \wedge A_{\{r\}}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $A_r \wedge B_s=\left(-1\right)^{r s} B_s \wedge A_r,$                                                                                                                                                                                                                                                                                                                                                                                                                                      | $A_r \wedge B_s=\left(-1\right)^{r s} B_s \wedge A_r,$                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 1285.5935v1_<br>EQ0158 (151) | 035  | <div><span>■</span> 1.000</div> <div><span>●</span> X +1</div> | $\begin{aligned} A_{\{r\}}^{\dagger} &=A_{\{r\}}^{*\dagger} \\ &=\left(-1\right)^r\left(-1\right)^{r\left(r-1\right) / 2} A_r \\ &=\left(-1\right)^{r(r+1) / 2} A_r . \end{aligned}$                                                                                                                                                                                                                                                                                                                                                  | $\begin{aligned} A_r^{\dagger} &=A_r^{*\dagger} \\ &=\left(-1\right)^r\left(-1\right)^{r\left(r-1\right) / 2} A_r \\ &=\left(-1\right)^{r(r+1) / 2} A_r . \end{aligned}$                                                                                                                                                                                                                                                                                                                    | $\begin{aligned} A_r^{\dagger} &=A_r^{*\dagger} \\ &=\left(-1\right)^r\left(-1\right)^{r\left(r-1\right) / 2} A_r \\ &=\left(-1\right)^{r(r+1) / 2} A_r . \end{aligned}$                                                                                                                                                                                                                                                                                                                      |
| 1285.5935v1_<br>EQ0159 (152) | 035  | <div><span>■</span> 1.000</div> <div><span>●</span> N +0</div> | $A * B:=\langle A^{\dagger}\rangle B$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $A * B:=\left\langle A^{\dagger} B\right\rangle .$                                                                                                                                                                                                                                                                                                                                                                                                                                          | $A * B:=\left\langle A^{\dagger} B\right\rangle .$                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1285.5935v1_<br>EQ0160 (153) | 035  | <div><span>■</span> 1.000</div> <div><span>●</span> N +0</div> | $\left.A_{\{r\}} * B_{\{s\}}=\left\langle A_{\{r\}}^{\dagger}\right\rangle \right\rfloor B_{\{s\}}\right\rangle=\left\langle A_{\{r\}}^{\dagger}\right\rangle \left\lfloor B_{\{s\}}\right\rangle$                                                                                                                                                                                                                                                                                                                                    | $A_r * B_s=\left\langle A_r^{\dagger}\right\rangle B_s=\left\langle A_r^{\dagger}\right\rangle\left[B_s\right] .$                                                                                                                                                                                                                                                                                                                                                                           | $A_r * B_s=\left\langle A_r^{\dagger}\right\rangle B_s=\left\langle A_r^{\dagger}\right\rangle\left[B_s\right] .$                                                                                                                                                                                                                                                                                                                                                                             |
| 1285.5935v1_<br>EQ0162 (155) | 035  | <div><span>■</span> 1.000</div> <div><span>●</span> N +0</div> | $\left.A_{\{r\}} * B_{\{s\}}=\left\langle A_{\{r\}}^{\dagger}\right\rangle \right\rfloor B_{\{s\}}\right\rangle=\left\langle A_{\{r\}}^{\dagger}\right\rangle \left\lfloor B_{\{s\}}\right\rangle$                                                                                                                                                                                                                                                                                                                                    | $A_r * B_s=\left(A_r^{\dagger}\right] B_s) \delta_{r s}=\left(A_r^{\dagger}\left[B_s\right) \delta_{r s} .$                                                                                                                                                                                                                                                                                                                                                                                 | $A_r * B_s=\left(A_r^{\dagger}\right] B_s) \delta_{r s}=\left(A_r^{\dagger}\left[B_s\right) \delta_{r s} .$                                                                                                                                                                                                                                                                                                                                                                                   |
| 1285.5935v1_<br>EQ0164 (157) | 035  | <div><span>■</span> 1.000</div> <div><span>●</span> N +0</div> | $A * B=A^* * B^* .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $A * B=A^* * B^* .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | $A * B=A^* * B^* .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 1285.5935v1_<br>EQ0165 (158) | 036  | <div><span>■</span> 1.000</div> <div><span>●</span> N +0</div> | $A * B=B * A=A^{\dagger} * B^{\dagger}=A^{\ddagger} * B^{\ddagger}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $A * B=B * A=A^{\dagger} * B^{\dagger}=A^{\ddagger} * B^{\ddagger}$                                                                                                                                                                                                                                                                                                                                                                                                                         | $A * B=B * A=A^{\dagger} * B^{\dagger}=A^{\ddagger} * B^{\ddagger}$                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 1285.5935v1_<br>EQ0169 (162) | 036  | <div><span>■</span> 1.000</div> <div><span>●</span></div>      | $ A ^2:=A * A$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | $ A ^2:=A * A .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $ A ^2:=A * A .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

**Conf.** MathPix confidence: ■  $\geq 0.9$  ■ 0.5–0.9 ■  $< 0.5$  — no value

**LaTeX source** MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

**Residual** render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● X crop overran its region ● not measured

**[refined: *basis*]** beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Page | Conf.                                       | LaTeX source                                                                                                                                                                                                                                    | Rendered                                                                                                                                                                                     | Scan image                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1205.5935v1_<br>EQ0173 (166)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 037  | <div><div></div>■1.000<div>K +0</div></div> | $ A\ B ^{\wedge 2}= A ^{\wedge 2} B ^{\wedge 2}$                                                                                                                                                                                                | $ AB ^2= A ^2 B ^2.$                                                                                                                                                                         | $ AB ^2= A ^2 B ^2.$                                                                                                                                                                         |
| 1205.5935v1_<br>EQ0174 (167)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 037  | <div><div></div>■1.000<div>W +1</div></div> | $\begin{aligned} (A\ B)\ *(A\ C) \&=\left\langle A\ B\right\rangle^{\dagger}A\ C\right\rangle\\ &=\left\langle B^{\dagger}A^{\dagger}AC\right\rangle\\ &=\left\langle  A ^2\ B^{\dagger}C\right\rangle\\ &= A ^2\ B\ *C\end{aligned}$           | $(AB)*(AC)=\langle(AB)^{\dagger}AC\rangle$<br>$=\langle B^{\dagger}A^{\dagger}AC\rangle$<br>$=\langle  A ^2B^{\dagger}C\rangle$<br>$= A ^2B*C$                                               | $(AB)*(AC)=\langle(AB)^{\dagger}AC\rangle$<br>$=\langle B^{\dagger}A^{\dagger}AC\rangle$<br>$=\langle  A ^2B^{\dagger}C\rangle$<br>$= A ^2B*C$                                               |
| 1205.5935v1_<br>EQ0175 (168)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 037  | <div><div></div>■1.000<div>K +0</div></div> | $A^{-1}=\frac{A^{\dag}}{ A ^2}$                                                                                                                                                                                                                 | $A^{-1}=\frac{A^{\dagger}}{ A ^2},$                                                                                                                                                          | $A^{-1}=\frac{A^{\dagger}}{ A ^2},$                                                                                                                                                          |
| 1205.5935v1_<br>EQ0176 (169)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 037  | <div><div></div>■1.000<div>K +0</div></div> | $\left A^{-1}\right ^{\wedge 2}= A ^{\wedge -2}$                                                                                                                                                                                                | $\left A^{-1}\right ^2= A ^{-2}.$                                                                                                                                                            | $ A^{-1} ^2= A ^{-2}.$                                                                                                                                                                       |
| 1205.5935v1_<br>EQ0179 (172)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 038  | <div><div></div>■1.000<div>X +2</div></div> | $(a\wedge b)^{-1}=\frac{b\wedge a}{a^2b^2\sin^{\wedge 2}\theta}$                                                                                                                                                                                | $(a\wedge b)^{-1}=\frac{b\wedge a}{a^2b^2\sin^2\theta}$                                                                                                                                      | $(a\wedge b)^{-1}=\frac{b\wedge a}{a^2b^2\sin^2\theta}.$                                                                                                                                     |
| 1205.5935v1_<br>EQ0181 (174)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 039  | <div><div></div>■1.000<div>K +0</div></div> | $ \boldsymbol{I} ^{\wedge 2}={e_1}^{\wedge 2}\ {e_2}^{\wedge 2}\ \cdots\ {e_n}^{\wedge 2}\\=(-1)^{\wedge p}$                                                                                                                                    | $ \boldsymbol{I} ^2=e_1^2e_2^2\cdots e_n^2=(-1)^p.$                                                                                                                                          | $ \boldsymbol{I} ^2=e_1^2e_2^2\cdots e_n^2=(-1)^p.$                                                                                                                                          |
| 1205.5935v1_<br>EQ0182 (175)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 039  | <div><div></div>■1.000<div>K +0</div></div> | $ \boldsymbol{I} ^{\wedge 2}=(-1)^{n(n-1)/2+p}$                                                                                                                                                                                                 | $\boldsymbol{I}^2=(-1)^{n(n-1)/2+p}.$                                                                                                                                                        | $\boldsymbol{I}^2=(-1)^{n(n-1)/2+p}.$                                                                                                                                                        |
| 1205.5935v1_<br>EQ0196 (189)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 042  | <div><div></div>■1.000<div>K +0</div></div> | $A\times(B\ C)=(A\times B)\ C+B(A\times C)$                                                                                                                                                                                                     | $A\times(BC)=(A\times B)C+B(A\times C),$                                                                                                                                                     | $A\times(BC)=(A\times B)C+B(A\times C),$                                                                                                                                                     |
| 1205.5935v1_<br>EQ0203 (196)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 043  | <div><div></div>■1.000<div>N +1</div></div> | $A_{\wedge 2}\times A_r=\frac{1}{2}(A_{\wedge 2}\ A_r-A_r\ A_{\wedge 2})\right\rangle$                                                                                                                                                          | $A_2\times A_r=\frac{1}{2}(A_2A_r-A_rA_2)=\langle A_2A_r\rangle_r$                                                                                                                           | $A_2\times A_r=\frac{1}{2}(A_2A_r-A_rA_2)=\langle A_2A_r\rangle_r.$                                                                                                                          |
| 1205.5935v1_<br>EQ0208 (201)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 044  | <div><div></div>■1.000<div>K +0</div></div> | $\left e_{\wedge 1}\ e_{\wedge 2}\right ^{\wedge 2}=\left(e_{\wedge 1}\ e_{\wedge 2}\right)^{\dagger}e_{\wedge 1}\ e_{\wedge 2}=e_2e_1e_1e_2=1$                                                                                                 | $ e_1e_2 ^2=(e_1e_2)^{\dagger}e_1e_2=e_2e_1e_1e_2=1.$                                                                                                                                        | $ e_1e_2 ^2=(e_1e_2)^{\dagger}e_1e_2=e_2e_1e_1e_2=1.$                                                                                                                                        |
| 1205.5935v1_<br>EQ0210 (203)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 044  | <div><div></div>■1.000<div>K +0</div></div> | $Z\&=e_{\wedge 1}\ z\ \ \ z\&=e_{\wedge 1}\ Z.\end{aligned}$                                                                                                                                                                                    | $Z=e_1z$<br>$z=e_1Z.$                                                                                                                                                                        | $Z=e_1z$<br>$z=e_1Z.$                                                                                                                                                                        |
| 1205.5935v1_<br>EQ0213 (206)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 045  | <div><div></div>■1.000<div>N +0</div></div> | $\begin{aligned} z^{\prime}\&=W\ z\ \ \&=\exp (-\boldsymbol{\theta})\ z\ \ \&=\exp (-\boldsymbol{\theta} / 2)\ \exp (-\boldsymbol{\theta} / 2)\ z\ \ \&=\\ &=\exp (-\boldsymbol{\theta} / 2)\ z\ \exp (\boldsymbol{\theta} / 2)\ \end{aligned}$ | $z'=Wz$<br>$=\exp(-\boldsymbol{I}\theta)z$<br>$=\exp(-\boldsymbol{I}\theta/2)\exp(-\boldsymbol{I}\theta/2)z$<br>$=\exp(-\boldsymbol{I}\theta/2)z\exp(\boldsymbol{I}\theta/2)$<br>$=RzR^{-1}$ | $z'=Wz$<br>$=\exp(-\boldsymbol{I}\theta)z$<br>$=\exp(-\boldsymbol{I}\theta/2)\exp(-\boldsymbol{I}\theta/2)z$<br>$=\exp(-\boldsymbol{I}\theta/2)z\exp(\boldsymbol{I}\theta/2)$<br>$=RzR^{-1}$ |
| <div><div>Conf.</div>MathPix confidence:<div><div>■≥0.9</div><div>■0.5–0.9</div><div>■&lt;0.5</div><div>—no value</div></div><div>LaTeX source</div>MathPix’s reading, unchanged;<div>Rendered</div>the same reading, with a display environment (align/algnat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ<div>Residual</div>render vs scan (inkdrill):<div><div>●C component</div><div>●W weak</div><div>●S stable</div><div>●N noise</div><div>●K clean</div><div>●X crop overran its region</div><div>●not measured</div></div><div>[refined: basis]</div>beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</div> |      |                                             |                                                                                                                                                                                                                                                 |                                                                                                                                                                                              |                                                                                                                                                                                              |

| Identifier                   | Page | Conf.                                          | LaTeX source                                                                                                                                                                                                           | Rendered                                                                                                                                                                                                               | Scan image                                                                                                                                                                                                             |
|------------------------------|------|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1205.5935v1_<br>EQ0214 (207) | 045  | <div><div>■ 1.000</div><div>● X +2</div></div> | $\begin{aligned} Z^{\prime} &= e_1 z^{\prime} \\ &= e_1 W z \\ &= w z \\ &= W^{\dagger} Z \\ &= \exp(\boldsymbol{I} \theta) Z \end{aligned}$                                                                           | $\begin{aligned} Z' &= e_1 z' \\ &= e_1 W z \\ &= w z \\ &= W^{\dagger} Z \\ &= \exp(\boldsymbol{I} \theta) Z \end{aligned}$                                                                                           | $\begin{aligned} Z' &= e_1 z' \\ &= e_1 W z \\ &= w z \\ &= W^{\dagger} Z \\ &= \exp(\boldsymbol{I} \theta) Z. \end{aligned}$                                                                                          |
| 1205.5935v1_<br>EQ0215 (208) | 046  | <div><div>■ 1.000</div><div>●</div></div>      | $a \times b = (a \wedge b)^{\perp}.$                                                                                                                                                                                   | $a \times b = (a \wedge b)^{\perp}.$                                                                                                                                                                                   | $a \times b = (a \wedge b)^{\perp}.$                                                                                                                                                                                   |
| 1205.5935v1_<br>EQ0217 (210) | 046  | <div><div>■ 1.000</div><div>● N +0</div></div> | $a \cdot b \times c = (a \wedge b \wedge c)^{\perp}.$                                                                                                                                                                  | $a \cdot b \times c = (a \wedge b \wedge c)^{\perp}.$                                                                                                                                                                  | $a \cdot b \times c = (a \wedge b \wedge c)^{\perp}.$                                                                                                                                                                  |
| 1205.5935v1_<br>EQ0219 (212) | 046  | <div><div>■ 1.000</div><div>● N +0</div></div> | $\boldsymbol{A}_2 + \boldsymbol{B}_2 = a \wedge b + a \wedge c = a \wedge (b + c)$                                                                                                                                     | $\boldsymbol{A}_2 + \boldsymbol{B}_2 = a \wedge b + a \wedge c = a \wedge (b + c)$                                                                                                                                     | $\boldsymbol{A}_2 + \boldsymbol{B}_2 = a \wedge b + a \wedge c = a \wedge (b + c)$                                                                                                                                     |
| 1205.5935v1_<br>EQ0220 (213) | 046  | <div><div>■ 1.000</div><div>● N +0</div></div> | $\begin{aligned} a \boldsymbol{B} &= (a_{\parallel} + a_{\perp}) a_{\parallel} b \\ &= a_{\parallel}^2 b + a_{\perp} a_{\parallel} b \\ &= a_{\parallel}^2 b + a_{\perp} \wedge a_{\parallel} \wedge b. \end{aligned}$ | $\begin{aligned} a \boldsymbol{B} &= (a_{\parallel} + a_{\perp}) a_{\parallel} b \\ &= a_{\parallel}^2 b + a_{\perp} a_{\parallel} b \\ &= a_{\parallel}^2 b + a_{\perp} \wedge a_{\parallel} \wedge b. \end{aligned}$ | $\begin{aligned} a \boldsymbol{B} &= (a_{\parallel} + a_{\perp}) a_{\parallel} b \\ &= a_{\parallel}^2 b + a_{\perp} a_{\parallel} b \\ &= a_{\parallel}^2 b + a_{\perp} \wedge a_{\parallel} \wedge b. \end{aligned}$ |
| 1205.5935v1_<br>EQ0222 (215) | 047  | <div><div>■ 1.000</div><div>● K +0</div></div> | $a \boldsymbol{T} = a^2 b c = a^2 b \wedge c.$                                                                                                                                                                         | $a \boldsymbol{T} = a^2 b c = a^2 b \wedge c.$                                                                                                                                                                         | $a \boldsymbol{T} = a^2 b c = a^2 b \wedge c.$                                                                                                                                                                         |
| 1205.5935v1_<br>EQ0224 (217) | 047  | <div><div>■ 1.000</div><div>● N +0</div></div> | $\boldsymbol{B} \boldsymbol{T} = baabc = a^2 b^2 c,$                                                                                                                                                                   | $\boldsymbol{B} \boldsymbol{T} = baabc = a^2 b^2 c,$                                                                                                                                                                   | $\boldsymbol{B} \boldsymbol{T} = baabc = a^2 b^2 c,$                                                                                                                                                                   |
| 1205.5935v1_<br>EQ0227 (220) | 047  | <div><div>■ 1.000</div><div>● N +0</div></div> | $\boldsymbol{B}_1^2 = \boldsymbol{B}_2^2 = \boldsymbol{B}_3^2 = -1$                                                                                                                                                    | $\boldsymbol{B}_1^2 = \boldsymbol{B}_2^2 = \boldsymbol{B}_3^2 = -1$                                                                                                                                                    | $\boldsymbol{B}_1^2 = \boldsymbol{B}_2^2 = \boldsymbol{B}_3^2 = -1$                                                                                                                                                    |
| 1205.5935v1_<br>EQ0230 (223) | 048  | <div><div>■ 1.000</div><div>● K +0</div></div> | $R_u(v) = v \wedge u u^{-1} = v \wedge u \lfloor u^{-1}.$                                                                                                                                                              | $R_u(v) = v \wedge u u^{-1} = v \wedge u \lfloor u^{-1}.$                                                                                                                                                              | $R_u(v) = v \wedge u u^{-1} = v \wedge u \lfloor u^{-1}.$                                                                                                                                                              |
| 1205.5935v1_<br>EQ0244 (237) | 051  | <div><div>■ 1.000</div><div>● K +0</div></div> | $v^{\prime} = -n v n^{-1}.$                                                                                                                                                                                            | $v' = -n v n^{-1}.$                                                                                                                                                                                                    | $v' = -n v n^{-1}.$                                                                                                                                                                                                    |
| 1205.5935v1_<br>EQ0253 (246) | 053  | <div><div>■ 1.000</div><div>● K +0</div></div> | $\boldsymbol{B}^{\prime} := \boldsymbol{A}_r \boldsymbol{B}^{*r} \boldsymbol{A}_r^{-1}.$                                                                                                                               | $\boldsymbol{B}' := \boldsymbol{A}_r \boldsymbol{B}^{*r} \boldsymbol{A}_r^{-1}.$                                                                                                                                       | $\boldsymbol{B}' := \boldsymbol{A}_r \boldsymbol{B}^{*r} \boldsymbol{A}_r^{-1}.$                                                                                                                                       |
| 1205.5935v1_<br>EQ0255 (248) | 053  | <div><div>■ 1.000</div><div>● K +0</div></div> | $\boldsymbol{A}^* = \boldsymbol{I} \boldsymbol{A}^{*n} \boldsymbol{I}^{-1}.$                                                                                                                                           | $\boldsymbol{A}^* = \boldsymbol{I} \boldsymbol{A}^{*n} \boldsymbol{I}^{-1}.$                                                                                                                                           | $\boldsymbol{A}^* = \boldsymbol{I} \boldsymbol{A}^{*n} \boldsymbol{I}^{-1}.$                                                                                                                                           |
| 1205.5935v1_<br>EQ0257 (250) | 053  | <div><div>■ 1.000</div><div>● N +0</div></div> | $\boldsymbol{A}_r^{\perp} \boldsymbol{B}^{*(n-r)} (\boldsymbol{A}_r^{\perp})^{-1} = (\boldsymbol{A}_r \boldsymbol{B}^{*r} \boldsymbol{A}_r^{-1})^*.$                                                                   | $\boldsymbol{A}_r^{\perp} \boldsymbol{B}^{*(n-r)} (\boldsymbol{A}_r^{\perp})^{-1} = (\boldsymbol{A}_r \boldsymbol{B}^{*r} \boldsymbol{A}_r^{-1})^*.$                                                                   | $\boldsymbol{A}_r^{\perp} \boldsymbol{B}^{*(n-r)} (\boldsymbol{A}_r^{\perp})^{-1} = (\boldsymbol{A}_r \boldsymbol{B}^{*r} \boldsymbol{A}_r^{-1})^*.$                                                                   |
| 1205.5935v1_<br>EQ0259 (252) | 053  | <div><div>■ 1.000</div><div>● K +0</div></div> | $v^{\prime} = R v R^{-1}$                                                                                                                                                                                              | $v' = R v R^{-1}$                                                                                                                                                                                                      | $v' = R v R^{-1}$                                                                                                                                                                                                      |
| 1205.5935v1_<br>EQ0261 (254) | 054  | <div><div>■ 1.000</div><div>● K +0</div></div> | $\boldsymbol{A}^{\prime} := \boldsymbol{R} \boldsymbol{A} \boldsymbol{R}^{-1},$                                                                                                                                        | $\boldsymbol{A}' := \boldsymbol{R} \boldsymbol{A} \boldsymbol{R}^{-1},$                                                                                                                                                | $\boldsymbol{A}' := \boldsymbol{R} \boldsymbol{A} \boldsymbol{R}^{-1},$                                                                                                                                                |
| 1205.5935v1_<br>EQ0269 (262) | 055  | <div><div>■ 1.000</div><div>● K +0</div></div> | $\sum_i a_i a^i = \sum_i a^i a_i = n.$                                                                                                                                                                                 | $\sum_i a_i a^i = \sum_i a^i a_i = n.$                                                                                                                                                                                 | $\sum_i a_i a^i = \sum_i a^i a_i = n.$                                                                                                                                                                                 |
| 1205.5935v1_<br>EQ0272 (265) | 056  | <div><div>■ 1.000</div><div>● K +0</div></div> | $a_I := a_{i_1} \wedge a_{i_2} \wedge \cdots \wedge a_{i_r},$                                                                                                                                                          | $a_I := a_{i_1} \wedge a_{i_2} \wedge \cdots \wedge a_{i_r},$                                                                                                                                                          | $a_I := a_{i_1} \wedge a_{i_2} \wedge \cdots \wedge a_{i_r},$                                                                                                                                                          |

**Conf.** MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

**LaTeX source** MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

**Residual** render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● X crop overran its region ● not measured

**[refined: *basis*]** beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Page | Conf.                                             | LaTeX source                                                                                                                                                        | Rendered                                                                                                                                    | Scan image                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 1205.5935v1_E00273 (266)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 056  | <div><span>■</span> 1.000</div> <div>● N +0</div> | $a^{\{I\}} * a_{\{J\}} = \delta_{IJ}^I$                                                                                                                             | $a^I * a_J = \delta_J^I$                                                                                                                    | $a^I * a_J = \delta_J^I$                                                                                                                    |
| 1205.5935v1_E00275                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 056  | <div><span>■</span> 1.000</div> <div>● N +0</div> | $\left(a^{\{i_r\}} \wedge \cdots \wedge a^{\{i_2\}} \wedge a^{\{i_1\}} \right) * \left(a_{\{i_1\}} \wedge a_{\{i_2\}} \wedge \cdots \wedge a_{\{i_r\}} \right)$     | $\left(a^{i_r} \wedge \cdots \wedge a^{i_2} \wedge a^{i_1} \right) * \left(a_{i_1} \wedge a_{i_2} \wedge \cdots \wedge a_{i_r} \right)$     | $\left(a^{i_r} \wedge \cdots \wedge a^{i_2} \wedge a^{i_1} \right) * \left(a_{i_1} \wedge a_{i_2} \wedge \cdots \wedge a_{i_r} \right)$     |
| 1205.5935v1_E00277 (269)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 057  | <div><span>■</span> 1.000</div> <div>● N +0</div> | $\left(a^{\{i_r\}} \wedge \cdots \wedge a^{\{i_2\}} \wedge a^{\{i_1\}} \right) * \left(a_{\{i_1\}} \wedge a_{\{i_2\}} \wedge \cdots \wedge a_{\{i_r\}} \right) = 1$ | $\left(a^{i_r} \wedge \cdots \wedge a^{i_2} \wedge a^{i_1} \right) * \left(a_{i_1} \wedge a_{i_2} \wedge \cdots \wedge a_{i_r} \right) = 1$ | $\left(a^{i_r} \wedge \cdots \wedge a^{i_2} \wedge a^{i_1} \right) * \left(a_{i_1} \wedge a_{i_2} \wedge \cdots \wedge a_{i_r} \right) = 1$ |
| 1205.5935v1_E00278 (270)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 057  | <div><span>■</span> 1.000</div> <div>● K +0</div> | $A = \sum_{\{I\}} A^{\{I\}} a_{\{I\}} \quad \text{where} \quad A^{\{I\}} = A * a^{\{I\}}$                                                                           | $A = \sum_I A^I a_I \quad \text{where} \quad A^I = A * a^I$                                                                                 | $A = \sum_I A^I a_I \quad \text{where} \quad A^I = A * a^I$                                                                                 |
| 1205.5935v1_E00279 (271)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 057  | <div><span>■</span> 1.000</div> <div>●</div>      | $\operatorname{dim} \mathcal{G}^n = \sum_{r=0}^n \operatorname{dim} \mathcal{G}_r^n = \sum_{r=0}^n \binom{n}{r} = 2^n$                                              | $\dim \mathcal{G}^n = \sum_{r=0}^n \dim \mathcal{G}_r^n = \sum_{r=0}^n \binom{n}{r} = 2^n$                                                  | $\dim \mathcal{G}^n = \sum_{r=0}^n \dim \mathcal{G}_r^n = \sum_{r=0}^n \binom{n}{r} = 2^n$                                                  |
| 1205.5935v1_E00280 (272)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 057  | <div><span>■</span> 1.000</div> <div>● N +0</div> | $B * C = \sum_{\{I, J\}} B_{\{I\}} C^{\{J\}} \left(a^{\{I\}} * a_{\{J\}} \right) = \sum_{\{I\}} B_{\{I\}} C^{\{I\}}$                                                | $B * C = \sum_{I, J} B_I C^J \left(a^I * a_J \right) = \sum_I B_I C^I$                                                                      | $B * C = \sum_{I, J} B_I C^J \left(a^I * a_J \right) = \sum_I B_I C^I$                                                                      |
| 1205.5935v1_E00292 (204)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 059  | <div><span>■</span> 1.000</div> <div>● N +0</div> | $a^{\{I\}} a_{\{N\}} = (-1)^{\sum_{j=1}^r \left(i_j - 1\right)} a_{I^c}$                                                                                            | $a^I a_N = (-1)^{\sum_{j=1}^r (i_j - 1)} a_{I^c}$                                                                                           | $a^I a_N = (-1)^{\sum_{j=1}^r (i_j - 1)} a_{I^c}$                                                                                           |
| 1205.5935v1_E00294 (206)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 060  | <div><span>■</span> 1.000</div> <div>● K +0</div> | $B * C = \sum_{\{I\}} (-1)^{I_{\{m\}}} B_{\{I\}} C_{\{I\}} = \sum_{\{I\}} (-1)^{I_{\{m\}}} B^{\{I\}} C^{\{I\}}$                                                     | $B * C = \sum_I (-1)^{I_m} B_I C_I = \sum_I (-1)^{I_m} B^I C^I$                                                                             | $B * C = \sum_I (-1)^{I_m} B_I C_I = \sum_I (-1)^{I_m} B^I C^I$                                                                             |
| 1205.5935v1_E00300 (292)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 061  | <div><span>■</span> 1.000</div> <div>● N +0</div> | $\operatorname{rank}(F) + \operatorname{null}(F) = \operatorname{dim} U$                                                                                            | $\operatorname{rank}(F) + \operatorname{null}(F) = \dim U$                                                                                  | $\operatorname{rank}(F) + \operatorname{null}(F) = \dim U$                                                                                  |
| 1205.5935v1_E00303 (295)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 062  | <div><span>■</span> 1.000</div> <div>● N +0</div> | $\begin{aligned} \bar{F}(v) &= \sum_i \left[ a^i * \bar{F}(v) \right] a_i \\ &= \sum_i \left[ F(a^i) * v \right] a_i. \end{aligned}$                                | $\begin{aligned} \bar{F}(v) &= \sum_i \left[ a^i * \bar{F}(v) \right] a_i \\ &= \sum_i \left[ F(a^i) * v \right] a_i. \end{aligned}$        | $\begin{aligned} \bar{F}(v) &= \sum_i \left[ a^i * \bar{F}(v) \right] a_i \\ &= \sum_i \left[ F(a^i) * v \right] a_i. \end{aligned}$        |
| 1205.5935v1_E00304 (296)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 063  | <div><span>■</span> 1.000</div> <div>● N +0</div> | $\overline{\alpha F + \beta G} = \alpha \bar{F} + \beta \bar{G}$                                                                                                    | $\overline{\alpha F + \beta G} = \alpha \bar{F} + \beta \bar{G}$                                                                            | $\overline{\alpha F + \beta G} = \alpha \bar{F} + \beta \bar{G}$                                                                            |
| 1205.5935v1_E00305 (297)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 063  | <div><span>■</span> 1.000</div> <div>● N +0</div> | $\overline{\bar{F}} = F$                                                                                                                                            | $\overline{\bar{F}} = F$                                                                                                                    | $\overline{\bar{F}} = F$                                                                                                                    |
| 1205.5935v1_E00306 (298)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 063  | <div><span>■</span> 1.000</div> <div>● N +0</div> | $w * G F(u) = \bar{G}(w) * F(u) \\ = \bar{F} \bar{G}(w) * u,$                                                                                                       | $w * GF(u) = \bar{G}(w) * F(u) \\ = \bar{F} \bar{G}(w) * u,$                                                                                | $w * GF(u) = \bar{G}(w) * F(u) \\ = \bar{F} \bar{G}(w) * u,$                                                                                |
| 1205.5935v1_E00307 (299)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 063  | <div><span>■</span> 1.000</div> <div>● N +0</div> | $\overline{GF} = \bar{F} \bar{G}$                                                                                                                                   | $\overline{GF} = \bar{F} \bar{G}$                                                                                                           | $\overline{GF} = \bar{F} \bar{G}$                                                                                                           |
| 1205.5935v1_E00308 (300)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 063  | <div><span>■</span> 1.000</div> <div>● K +0</div> | $\overline{F^n} = (\bar{F})^n$                                                                                                                                      | $\overline{F^n} = (\bar{F})^n$                                                                                                              | $\overline{F^n} = (\bar{F})^n$                                                                                                              |
| 1205.5935v1_E00310 (302)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 063  | <div><span>■</span> 1.000</div> <div>● N +0</div> | $\operatorname{Ker}(F) = \operatorname{Range}(\bar{F})^\perp$                                                                                                       | $\operatorname{Ker}(F) = \operatorname{Range}(\bar{F})^\perp$                                                                               | $\operatorname{Ker}(F) = \operatorname{Range}(\bar{F})^\perp$                                                                               |
| 1205.5935v1_E00311 (303)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 063  | <div><span>■</span> 1.000</div> <div>● K +0</div> | $\operatorname{rank}(F) = \operatorname{rank}(\bar{F})$                                                                                                             | $\operatorname{rank}(F) = \operatorname{rank}(\bar{F})$                                                                                     | $\operatorname{rank}(F) = \operatorname{rank}(\bar{F})$                                                                                     |
| <p><b>Conf.</b> MathPix confidence: <span>■</span> <math>\geq 0.9</math> <span>■</span> 0.5–0.9 <span>■</span> <math>&lt; 0.5</math> — no value</p> <p><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</p> <p><b>Residual</b> render vs scan (inkdrill): <span>●</span> C component <span>●</span> W weak <span>●</span> S stable <span>●</span> N noise <span>●</span> K clean <span>●</span> X crop overran its region <span>●</span> not measured</p> <p><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</p> |      |                                                   |                                                                                                                                                                     |                                                                                                                                             |                                                                                                                                             |

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Page | Conf.                                          | LaTeX source                                                                                                                                                                                                                       | Rendered                                                                                                                                                                                                                           | Scan image                                                                                                                                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1205.5935v1_<br>EQ0314 (304)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 063  | <div><div></div>■1.000</div> <div>● X +1</div> | $\begin{aligned} \operatorname{rank}(F) &= \operatorname{dim} U - \operatorname{dim} \operatorname{Range}(\bar{F})^\perp \\ &= \operatorname{dim} \operatorname{Range}(\bar{F}) \\ &= \operatorname{rank}(\bar{F}). \end{aligned}$ | $\begin{aligned} \operatorname{rank}(F) &= \operatorname{dim} U - \operatorname{dim} \operatorname{Range}(\bar{F})^\perp \\ &= \operatorname{dim} \operatorname{Range}(\bar{F}) \\ &= \operatorname{rank}(\bar{F}). \end{aligned}$ | $\begin{aligned} \operatorname{rank}(F) &= \operatorname{dim} U - \operatorname{dim} \operatorname{Range}(\bar{F})^\perp \\ &= \operatorname{dim} \operatorname{Range}(\bar{F}) \\ &= \operatorname{rank}(\bar{F}). \end{aligned}$ |
| 1205.5935v1_<br>EQ0319 (309)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 065  | <div><div></div>■1.000</div> <div>● N -1</div> | $F = \frac{1}{2}(F + \bar{F}) + \frac{1}{2}(F - \bar{F})$                                                                                                                                                                          | $F = \frac{1}{2}(F + \bar{F}) + \frac{1}{2}(F - \bar{F}).$                                                                                                                                                                         | $F = \frac{1}{2}(F + \bar{F}) + \frac{1}{2}(F - \bar{F}).$                                                                                                                                                                         |
| 1205.5935v1_EQ0326                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 066  | <div><div></div>■1.000</div> <div>● N +0</div> | $\begin{aligned} F(u) &= (-1)^r (a_r \cdots a_2 a_1) u (a_1^{-1} a_2^{-1} \cdots a_r^{-1}) \\ &= (a_r \cdots a_2 a_1) u^{*r} (a_r \cdots a_2 a_1)^{-1} \end{aligned}$                                                              | $\begin{aligned} F(u) &= (-1)^r (a_r \cdots a_2 a_1) u (a_1^{-1} a_2^{-1} \cdots a_r^{-1}) \\ &= (a_r \cdots a_2 a_1) u^{*r} (a_r \cdots a_2 a_1)^{-1} \end{aligned}$                                                              | $\begin{aligned} F(u) &= (-1)^r (a_r \cdots a_2 a_1) u (a_1^{-1} a_2^{-1} \cdots a_r^{-1}) \\ &= (a_r \cdots a_2 a_1) u^{*r} (a_r \cdots a_2 a_1)^{-1} \end{aligned}$                                                              |
| 1205.5935v1_<br>EQ0327 (316)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 067  | <div><div></div>■1.000</div> <div>●</div>      | $A_r u^{*r} A_r^{-1}$                                                                                                                                                                                                              | $A_r u^{*r} A_r^{-1}$                                                                                                                                                                                                              | $A_r u^{*r} A_r^{-1}$                                                                                                                                                                                                              |
| 1205.5935v1_<br>EQ0328 (317)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 067  | <div><div></div>■1.000</div> <div>● X +1</div> | $F(B) = A_r B^{*r} A_r^{-1}$                                                                                                                                                                                                       | $F(B) = A_r B^{*r} A_r^{-1}$                                                                                                                                                                                                       | $F(B) = A_r B^{*r} A_r^{-1}$                                                                                                                                                                                                       |
| 1205.5935v1_<br>EQ0329 (318)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 067  | <div><div></div>■1.000</div> <div>● N +0</div> | $A_r \boldsymbol{I}^{*r} A_r^{-1} = (-1)^r \boldsymbol{I},$                                                                                                                                                                        | $A_r \boldsymbol{I}^{*r} A_r^{-1} = (-1)^r \boldsymbol{I},$                                                                                                                                                                        | $A_r \boldsymbol{I}^{*r} A_r^{-1} = (-1)^r \boldsymbol{I},$                                                                                                                                                                        |
| 1205.5935v1_<br>EQ0330 (319)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 067  | <div><div></div>■1.000</div> <div>● N +0</div> | $B_s^{-1} A_r \times u = 0 \quad \text{for all } u.$                                                                                                                                                                               | $B_s^{-1} A_r \times u = 0 \quad \text{for all } u.$                                                                                                                                                                               | $B_s^{-1} A_r \times u = 0 \quad \text{for all } u.$                                                                                                                                                                               |
| 1205.5935v1_<br>EQ0331 (320)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 067  | <div><div></div>■1.000</div> <div>● N +0</div> | $a \, b = \left( \frac{a-b}{\sqrt{2}} \right) \left( \frac{a+b}{\sqrt{2}} \right),$                                                                                                                                                | $a \, b = \left( \frac{a-b}{\sqrt{2}} \right) \left( \frac{a+b}{\sqrt{2}} \right),$                                                                                                                                                | $a \, b = \left( \frac{a-b}{\sqrt{2}} \right) \left( \frac{a+b}{\sqrt{2}} \right),$                                                                                                                                                |
| 1205.5935v1_<br>EQ0333 (322)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 068  | <div><div></div>■1.000</div> <div>● K +0</div> | $\boldsymbol{B} u \boldsymbol{B}^{-1} = -u \boldsymbol{B} \boldsymbol{B}^{-1} = -u.$                                                                                                                                               | $\boldsymbol{B} u \boldsymbol{B}^{-1} = -u \boldsymbol{B} \boldsymbol{B}^{-1} = -u.$                                                                                                                                               | $\boldsymbol{B} u \boldsymbol{B}^{-1} = -u \boldsymbol{B} \boldsymbol{B}^{-1} = -u.$                                                                                                                                               |
| 1205.5935v1_<br>EQ0334 (323)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 068  | <div><div></div>■1.000</div> <div>● K +0</div> | $1 = \mu^2 + \lambda^2.$                                                                                                                                                                                                           | $1 = \mu^2 + \lambda^2.$                                                                                                                                                                                                           | $1 = \mu^2 + \lambda^2.$                                                                                                                                                                                                           |
| 1205.5935v1_<br>EQ0336 (325)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 068  | <div><div></div>■1.000</div> <div>● K +0</div> | $1 = \mu^2 - \lambda^2.$                                                                                                                                                                                                           | $1 = \mu^2 - \lambda^2.$                                                                                                                                                                                                           | $1 = \mu^2 - \lambda^2.$                                                                                                                                                                                                           |
| 1205.5935v1_<br>EQ0338 (327)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 069  | <div><div></div>■1.000</div> <div>● K +0</div> | $-1 = \mu^2 - \lambda^2,$                                                                                                                                                                                                          | $-1 = \mu^2 - \lambda^2,$                                                                                                                                                                                                          | $-1 = \mu^2 - \lambda^2,$                                                                                                                                                                                                          |
| 1205.5935v1_<br>EQ0342 (331)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 070  | <div><div></div>■1.000</div> <div>● K +0</div> | $F(a_1 \wedge \cdots \wedge a_r) := F(a_1) \wedge \cdots \wedge F(a_r).$                                                                                                                                                           | $F(a_1 \wedge \cdots \wedge a_r) := F(a_1) \wedge \cdots \wedge F(a_r).$                                                                                                                                                           | $F(a_1 \wedge \cdots \wedge a_r) := F(a_1) \wedge \cdots \wedge F(a_r).$                                                                                                                                                           |
| 1205.5935v1_<br>EQ0345 (334)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 071  | <div><div></div>■1.000</div> <div>● K +0</div> | $A_0 = A_0^2,$                                                                                                                                                                                                                     | $A_0 = A_0^2,$                                                                                                                                                                                                                     | $A_0 = A_0^2,$                                                                                                                                                                                                                     |
| 1205.5935v1_<br>EQ0346 (335)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 071  | <div><div></div>■1.000</div> <div>● K +0</div> | $A_1 = 2 A_0 A_1.$                                                                                                                                                                                                                 | $A_1 = 2 A_0 A_1.$                                                                                                                                                                                                                 | $A_1 = 2 A_0 A_1.$                                                                                                                                                                                                                 |
| 1205.5935v1_<br>EQ0347 (336)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 071  | <div><div></div>■1.000</div> <div>● N +0</div> | $A_r = 2 A_0 A_r.$                                                                                                                                                                                                                 | $A_r = 2 A_0 A_r.$                                                                                                                                                                                                                 | $A_r = 2 A_0 A_r.$                                                                                                                                                                                                                 |
| 1205.5935v1_<br>EQ0349 (338)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 071  | <div><div></div>■1.000</div> <div>● N +0</div> | $\overline{\mathcal{F}}(B) * A := B * \mathcal{F}(A) \quad \text{for all } A \in \mathcal{G}^1 \text{ and } B \in \mathcal{G}^2.$                                                                                                  | $\overline{\mathcal{F}}(B) * A := B * \mathcal{F}(A) \quad \text{for all } A \in \mathcal{G}^1 \text{ and } B \in \mathcal{G}^2.$                                                                                                  | $\overline{\mathcal{F}}(B) * A := B * \mathcal{F}(A) \quad \text{for all } A \in \mathcal{G}^1 \text{ and } B \in \mathcal{G}^2.$                                                                                                  |
| 1205.5935v1_<br>EQ0350 (339)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 072  | <div><div></div>■1.000</div> <div>● K +0</div> | $\langle [F](A) \rangle_r = [F](\langle A \rangle_r).$                                                                                                                                                                             | $\langle [F](A) \rangle_r = [F](\langle A \rangle_r).$                                                                                                                                                                             | $\langle [F](A) \rangle_r = [F](\langle A \rangle_r).$                                                                                                                                                                             |
| <p><b>Conf.</b> MathPix confidence: <div><div></div>■≥0.9 <div></div>■0.5–0.9 <div></div>■&lt;0.5 — no value</div></p> <p><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</p> <p><b>Residual</b> render vs scan (inkdrill): <div><div></div>●C component <div></div>●W weak <div></div>●S stable <div></div>●N noise <div></div>●K clean <div></div>●X crop overran its region <div></div>●not measured</div></p> <p><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</p> |      |                                                |                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                    |

| Identifier               | Page | Conf.                                          | LaTeX source                                                                                                                                                                                                                                                                                                            | Rendered                                                                                                                                                                                                                                         | Scan image                                                                                                                                                                                                                                       |
|--------------------------|------|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1205.5935v1_E00353 (342) | 072  | <div><div>■ 1.000</div><div>● N +0</div></div> | $\begin{aligned} & \left[ [G] \left( \left[ F \right] \left( u_1 \wedge \cdots \wedge u_r \right) \right) \right] = \left[ G \right] \left( \left[ F \right] \left( u_1 \wedge \cdots \wedge u_r \right) \right) \\ & = GF(u_1) \wedge \cdots \wedge GF(u_r) \\ & = [GF] (u_1 \wedge \cdots \wedge u_r). \end{aligned}$ | $\begin{aligned} [G] [[F] (u_1 \wedge \cdots \wedge u_r)] &= [G] [F(u_1) \wedge \cdots \wedge F(u_r)] \\ &= GF(u_1) \wedge \cdots \wedge GF(u_r) \\ &= [GF] (u_1 \wedge \cdots \wedge u_r). \end{aligned}$                                       | $\begin{aligned} [G] [[F] (u_1 \wedge \cdots \wedge u_r)] &= [G] [F(u_1) \wedge \cdots \wedge F(u_r)] \\ &= GF(u_1) \wedge \cdots \wedge GF(u_r) \\ &= [GF] (u_1 \wedge \cdots \wedge u_r). \end{aligned}$                                       |
| 1205.5935v1_E00365       | 075  | <div><div>■ 1.000</div><div>● K +0</div></div> | $F(\boldsymbol{B})=\exp \left( -\boldsymbol{B}\theta /2\right) \boldsymbol{B}\exp \left( \boldsymbol{B}\theta /2\right)$                                                                                                                                                                                                | $F(\boldsymbol{B})=\exp(-\boldsymbol{B}\theta/2)\boldsymbol{B}\exp(\boldsymbol{B}\theta/2)$                                                                                                                                                      | $F(\boldsymbol{B})=\exp(-\boldsymbol{B}\theta/2)\boldsymbol{B}\exp(\boldsymbol{B}\theta/2)$                                                                                                                                                      |
| 1205.5935v1_E00366 (353) | 076  | <div><div>■ 1.000</div><div>● K +0</div></div> | $\begin{aligned} & \exp(-\boldsymbol{B}\theta/2)\exp(\boldsymbol{B}\theta/2)\boldsymbol{B} \\ &= \boldsymbol{B}, \end{aligned}$                                                                                                                                                                                         | $\begin{aligned} &= \exp(-\boldsymbol{B}\theta/2)\exp(\boldsymbol{B}\theta/2)\boldsymbol{B} \\ &= \boldsymbol{B}, \end{aligned}$                                                                                                                 | $\begin{aligned} &= \exp(-\boldsymbol{B}\theta/2)\exp(\boldsymbol{B}\theta/2)\boldsymbol{B} \\ &= \boldsymbol{B}, \end{aligned}$                                                                                                                 |
| 1205.5935v1_E00367 (354) | 076  | <div><div>■ 1.000</div><div>●</div></div>      | $\begin{aligned} & \boldsymbol{A} \wedge F(a) = \lambda^{-1} \lambda \boldsymbol{A} \wedge F(a) \\ &= \lambda^{-1} F(\boldsymbol{A}) \wedge F(a) \\ &= \lambda^{-1} F(\boldsymbol{A} \wedge a) \\ &= 0, \end{aligned}$                                                                                                  | $\begin{aligned} \boldsymbol{A} \wedge F(a) &= \lambda^{-1} \lambda \boldsymbol{A} \wedge F(a) \\ &= \lambda^{-1} F(\boldsymbol{A}) \wedge F(a) \\ &= \lambda^{-1} F(\boldsymbol{A} \wedge a) \\ &= 0, \end{aligned}$                            | $\begin{aligned} \boldsymbol{A} \wedge F(a) &= \lambda^{-1} \lambda \boldsymbol{A} \wedge F(a) \\ &= \lambda^{-1} F(\boldsymbol{A}) \wedge F(a) \\ &= \lambda^{-1} F(\boldsymbol{A} \wedge a) \\ &= 0, \end{aligned}$                            |
| 1205.5935v1_E00369 (356) | 076  | <div><div>■ 1.000</div><div>● K +0</div></div> | $F(\boldsymbol{I})=: \operatorname{det}(F) \boldsymbol{I}$                                                                                                                                                                                                                                                              | $F(\boldsymbol{I})=: \det(F)\boldsymbol{I}.$                                                                                                                                                                                                     | $F(\boldsymbol{I})=: \det(F)\boldsymbol{I}.$                                                                                                                                                                                                     |
| 1205.5935v1_E00371       | 076  | <div><div>■ 1.000</div><div>● N +0</div></div> | $\begin{aligned} & \det(FG) \boldsymbol{I} = FG(\boldsymbol{I}) \\ &= F(\det(G) \boldsymbol{I}) \end{aligned}$                                                                                                                                                                                                          | $\begin{aligned} \det(FG) \boldsymbol{I} &= FG(\boldsymbol{I}) \\ &= F(\det(G) \boldsymbol{I}) \end{aligned}$                                                                                                                                    | $\begin{aligned} \det(FG) \boldsymbol{I} &= FG(\boldsymbol{I}) \\ &= F(\det(G) \boldsymbol{I}) \end{aligned}$                                                                                                                                    |
| 1205.5935v1_E00372 (358) | 077  | <div><div>■ 1.000</div><div>● N +0</div></div> | $\begin{aligned} & \det(G) F(\boldsymbol{I}) \\ &= \det(G) \det(F) \boldsymbol{I}, \end{aligned}$                                                                                                                                                                                                                       | $\begin{aligned} &= \det(G) F(\boldsymbol{I}) \\ &= \det(G) \det(F) \boldsymbol{I}, \end{aligned}$                                                                                                                                               | $\begin{aligned} &= \det(G) F(\boldsymbol{I}) \\ &= \det(G) \det(F) \boldsymbol{I}, \end{aligned}$                                                                                                                                               |
| 1205.5935v1_E00373 (359) | 077  | <div><div>■ 1.000</div><div>● K +0</div></div> | $\operatorname{det}(FG)=\operatorname{det}(F)\operatorname{det}(G)$                                                                                                                                                                                                                                                     | $\det(FG)=\det(F)\det(G).$                                                                                                                                                                                                                       | $\det(FG)=\det(F)\det(G).$                                                                                                                                                                                                                       |
| 1205.5935v1_E00374 (360) | 077  | <div><div>■ 1.000</div><div>● N +0</div></div> | $\begin{aligned} & \operatorname{det}\left(F^{-1}\right) \operatorname{det}(F)=\operatorname{det}\left(F^{-1} F\right) \\ &= \operatorname{det}(\mathrm{Id}) \\ &= 1 \end{aligned}$                                                                                                                                     | $\begin{aligned} \det \left( F^{-1} \right) \det (F) &= \det \left( F^{-1} F \right) \\ &= \det (\mathrm{Id}) \\ &= 1 \end{aligned}$                                                                                                             | $\begin{aligned} \det (F^{-1}) \det (F) &= \det (F^{-1} F) \\ &= \det (\mathrm{Id}) \\ &= 1 \end{aligned}$                                                                                                                                       |
| 1205.5935v1_E00375 (361) | 077  | <div><div>■ 1.000</div><div>● K +0</div></div> | $\operatorname{det}\left(F^{-1}\right)=\operatorname{det}(F)^{-1}$                                                                                                                                                                                                                                                      | $\det \left( F^{-1} \right) = \det (F)^{-1}.$                                                                                                                                                                                                    | $\det (F^{-1}) = \det (F)^{-1}.$                                                                                                                                                                                                                 |
| 1205.5935v1_E00378 (364) | 077  | <div><div>■ 1.000</div><div>● N +0</div></div> | $\begin{aligned} & \operatorname{det}(F) a_1 \wedge \cdots \wedge a_n = F\left(a_1 \wedge \cdots \wedge a_n\right) \\ &= F\left(a_1\right) \wedge \cdots \wedge F\left(a_n\right) \\ &= \lambda_1 \cdots \lambda_n a_1 \wedge \cdots \wedge a_n \end{aligned}$                                                          | $\begin{aligned} \det (F) a_1 \wedge \cdots \wedge a_n &= F\left(a_1 \wedge \cdots \wedge a_n\right) \\ &= F\left(a_1\right) \wedge \cdots \wedge F\left(a_n\right) \\ &= \lambda_1 \cdots \lambda_n a_1 \wedge \cdots \wedge a_n \end{aligned}$ | $\begin{aligned} \det (F) a_1 \wedge \cdots \wedge a_n &= F\left(a_1 \wedge \cdots \wedge a_n\right) \\ &= F\left(a_1\right) \wedge \cdots \wedge F\left(a_n\right) \\ &= \lambda_1 \cdots \lambda_n a_1 \wedge \cdots \wedge a_n \end{aligned}$ |

**Conf.** MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value

**LaTeX source** MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

**Residual** render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● X crop overran its region ● not measured

**[refined: *basis*]** beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

| Identifier                   | Page | Conf.                                         | LaTeX source                                                                                                                                                                                               | Rendered                                                                                                                                                                                                                                                                                               | Scan image                                                                                                                                                                                                                                                                                             |
|------------------------------|------|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1205.5935v1_<br>EQ0379 (365) | 077  | <div><div></div>1.000</div> <div>● K +0</div> | $\operatorname{det}(F)=\lambda_1\cdots\lambda_n$                                                                                                                                                           | $\det(F)=\lambda_1\cdots\lambda_n.$                                                                                                                                                                                                                                                                    | $\det(F)=\lambda_1\cdots\lambda_n.$                                                                                                                                                                                                                                                                    |
| 1205.5935v1_<br>EQ0380 (366) | 077  | <div><div></div>1.000</div> <div>● X +1</div> | $\left[\operatorname{det}(F)=(-1)^r\right]\text{.}\end{itemize}$                                                                                                                                           | $\det(F)=(-1)^r.$                                                                                                                                                                                                                                                                                      | $\det(F)=(-1)^r.$                                                                                                                                                                                                                                                                                      |
| 1205.5935v1_<br>EQ0381 (367) | 077  | <div><div></div>1.000</div> <div>● N +0</div> | $F^{-1}(A)=\frac{\bar{F}\left(A^{-\perp}\right)^{\perp}}{\operatorname{det}(F)}$                                                                                                                           | $F^{-1}(A)=\frac{\bar{F}\left(A^{-\perp}\right)^{\perp}}{\det(F)}.$                                                                                                                                                                                                                                    | $F^{-1}(A)=\frac{\bar{F}(A^{-\perp})^{\perp}}{\det(F)}.$                                                                                                                                                                                                                                               |
| 1205.5935v1_<br>EQ0383 (369) | 078  | <div><div></div>1.000</div> <div>● K +0</div> | $\dot{A}=\frac{1}{2}\boldsymbol{r}\wedge\boldsymbol{v}$                                                                                                                                                    | $\dot{A}=\frac{1}{2}\boldsymbol{r}\wedge\boldsymbol{v}$                                                                                                                                                                                                                                                | $\dot{A}=\frac{1}{2}\boldsymbol{r}\wedge\boldsymbol{v}$                                                                                                                                                                                                                                                |
| 1205.5935v1_<br>EQ0384 (370) | 078  | <div><div></div>1.000</div> <div>● K +0</div> | $L:=\boldsymbol{r}\wedge\boldsymbol{p}=m\boldsymbol{r}\wedge\dot{\boldsymbol{A}}\wedge\boldsymbol{v}=2\,m\,\dot{\boldsymbol{A}}$                                                                           | $L:=\boldsymbol{r}\wedge\boldsymbol{p}=m\boldsymbol{r}\wedge\boldsymbol{v}=2m\dot{A},$                                                                                                                                                                                                                 | $L:=\boldsymbol{r}\wedge\boldsymbol{p}=m\boldsymbol{r}\wedge\boldsymbol{v}=2m\dot{A},$                                                                                                                                                                                                                 |
| 1205.5935v1_<br>EQ0385 (371) | 078  | <div><div></div>1.000</div> <div>● X +1</div> | $\begin{aligned}\dot{L}&= \boldsymbol{v}\wedge\boldsymbol{v}+m\boldsymbol{r}\wedge\dot{\boldsymbol{v}}\\&= m\boldsymbol{r}\wedge\dot{\boldsymbol{v}}\\&= \boldsymbol{r}\wedge\boldsymbol{F},\end{aligned}$ | $\begin{aligned}\dot{L}&= m\boldsymbol{v}\wedge\boldsymbol{v}+m\boldsymbol{r}\wedge\dot{\boldsymbol{v}}\\&= m\boldsymbol{r}\wedge\dot{\boldsymbol{v}}\\&= \boldsymbol{r}\wedge\boldsymbol{F},\end{aligned}$                                                                                            | $\begin{aligned}\dot{L}&= m\boldsymbol{v}\wedge\boldsymbol{v}+m\boldsymbol{r}\wedge\dot{\boldsymbol{v}}\\&= m\boldsymbol{r}\wedge\dot{\boldsymbol{v}}\\&= \boldsymbol{r}\wedge\boldsymbol{F},\end{aligned}$                                                                                            |
| 1205.5935v1_<br>EQ0386 (372) | 079  | <div><div></div>1.000</div> <div>●</div>      | $\begin{aligned}L&= \boldsymbol{r}\wedge\boldsymbol{p}=m\boldsymbol{r}\wedge\boldsymbol{v}\wedge\boldsymbol{r}\wedge\dot{\boldsymbol{r}}+r\dot{\boldsymbol{r}}\wedge\boldsymbol{r}\end{aligned}$           | $\begin{aligned}L&= \boldsymbol{r}\wedge\boldsymbol{p}=m\boldsymbol{r}\wedge\boldsymbol{v}\\&= m\boldsymbol{r}\hat{\boldsymbol{r}}\wedge(\dot{\boldsymbol{r}}\hat{\boldsymbol{r}}+r\dot{\hat{\boldsymbol{r}}})\\&= m\boldsymbol{r}^2\hat{\boldsymbol{r}}\wedge\dot{\hat{\boldsymbol{r}}}\end{aligned}$ | $\begin{aligned}L&= \boldsymbol{r}\wedge\boldsymbol{p}=m\boldsymbol{r}\wedge\boldsymbol{v}\\&= m\boldsymbol{r}\hat{\boldsymbol{r}}\wedge(\dot{\boldsymbol{r}}\hat{\boldsymbol{r}}+r\dot{\hat{\boldsymbol{r}}})\\&= m\boldsymbol{r}^2\hat{\boldsymbol{r}}\wedge\dot{\hat{\boldsymbol{r}}}\end{aligned}$ |
| 1205.5935v1_<br>EQ0391 (377) | 079  | <div><div></div>1.000</div> <div>● N +0</div> | $\dot{\boldsymbol{v}}=-\frac{k}{mr^2}\hat{\boldsymbol{r}}$                                                                                                                                                 | $\dot{\boldsymbol{v}}=-\frac{k}{mr^2}\hat{\boldsymbol{r}}.$                                                                                                                                                                                                                                            | $\dot{\boldsymbol{v}}=-\frac{k}{mr^2}\hat{\boldsymbol{r}}.$                                                                                                                                                                                                                                            |
| 1205.5935v1_<br>EQ0393 (378) | 079  | <div><div></div>1.000</div> <div>● N +1</div> | $\frac{d}{dt}(L\boldsymbol{v}-k\hat{\boldsymbol{r}})=0$                                                                                                                                                    | $\frac{d}{dt}(L\boldsymbol{v}-k\hat{\boldsymbol{r}})=0$                                                                                                                                                                                                                                                | $\frac{d}{dt}(L\boldsymbol{v}-k\hat{\boldsymbol{r}})=0.$                                                                                                                                                                                                                                               |
| 1205.5935v1_EQ0394           | 079  | <div><div></div>1.000</div> <div>● K +0</div> | $L\boldsymbol{v}=L\lfloor\boldsymbol{v}+L\wedge\boldsymbol{v}$                                                                                                                                             | $L\boldsymbol{v}=L\lfloor\boldsymbol{v}+L\wedge\boldsymbol{v}$                                                                                                                                                                                                                                         | $L\boldsymbol{v}=L\lfloor\boldsymbol{v}+L\wedge\boldsymbol{v}$                                                                                                                                                                                                                                         |
| 1205.5935v1_<br>EQ0395 (379) | 080  | <div><div></div>1.000</div> <div>● K +0</div> | $\begin{aligned}&=L\lfloor\boldsymbol{v}+m\boldsymbol{r}\wedge\boldsymbol{v}\wedge\boldsymbol{v}\\&=L\lfloor\boldsymbol{v}.\end{aligned}$                                                                  | $\begin{aligned}&=L\lfloor\boldsymbol{v}+m\boldsymbol{r}\wedge\boldsymbol{v}\wedge\boldsymbol{v}\\&=L\lfloor\boldsymbol{v}.\end{aligned}$                                                                                                                                                              | $\begin{aligned}&=L\lfloor\boldsymbol{v}+m\boldsymbol{r}\wedge\boldsymbol{v}\wedge\boldsymbol{v}\\&=L\lfloor\boldsymbol{v}.\end{aligned}$                                                                                                                                                              |
| 1205.5935v1_<br>EQ0405 (389) | 081  | <div><div></div>1.000</div> <div>● K +0</div> | $\begin{aligned}l^2v^2-\frac{2kl^2}{mr}&=k^2\left(e^2-1\right)\left(e^2-1\right)\frac{2l^2}{m}\left(\frac{1}{2}mv^2-\frac{k}{r}\right)&=k^2\left(e^2-1\right)\left(e^2-1\right)\end{aligned}$              | $\begin{aligned}l^2v^2-\frac{2kl^2}{mr}&=k^2\left(e^2-1\right)\\ \frac{2l^2}{m}\left(\frac{1}{2}mv^2-\frac{k}{r}\right)&=k^2\left(e^2-1\right)\end{aligned}$                                                                                                                                           | $\begin{aligned}l^2v^2-\frac{2kl^2}{mr}&=k^2(e^2-1)\\ \frac{2l^2}{m}\left(\frac{1}{2}mv^2-\frac{k}{r}\right)&=k^2(e^2-1)\end{aligned}$                                                                                                                                                                 |
| 1205.5935v1_<br>EQ0406 (390) | 081  | <div><div></div>1.000</div> <div>● K +0</div> | $E=\frac{mk^2}{2l^2}\left(e^2-1\right)$                                                                                                                                                                    | $E=\frac{mk^2}{2l^2}\left(e^2-1\right).$                                                                                                                                                                                                                                                               | $E=\frac{mk^2}{2l^2}(e^2-1).$                                                                                                                                                                                                                                                                          |

**Conf.** MathPix confidence: 

≥0.9

0.5–0.9

<0.5

 — no value

**LaTeX source** MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

**Residual** render vs scan (inkdrill): 

C component

W weak

S stable

N noise

K clean

X crop overran its region

not measured

**[refined: *basis*]** beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

| Identifier                  | Page | Conf.                                             | LaTeX source                                                                                                                                                                                                                                                                 | Rendered                                                                                                                                                                                                                            | Scan image                                                                                                                                                                                                                          |
|-----------------------------|------|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1285.5935v1_<br>EQ0407 (1)  | 083  | <div> <div>■ 1.000</div> <div>● X +1</div> </div> | $\begin{gathered} A_{\{r\}} B_{\{s\}} = \sum_{j=0}^{\min\{r,s\}} \langle A_{\{r\}} B_{\{s\}} \rangle_{ r-s +2j} \\ \langle A_{\{r\}} B_{\{s\}} \rangle_{r+s-2j} = (-1)^{r-s-j} \langle B_{\{s\}} A_{\{r\}} \rangle_{r+s-2j} \end{gathered}$                                  | $A_r B_s = \sum_{j=0}^{\min\{r,s\}} \langle A_r B_s \rangle_{ r-s +2j}$ $\langle A_r B_s \rangle_{r+s-2j} = (-1)^{rs-j} \langle B_s A_r \rangle_{r+s-2j}$                                                                           | $A_r B_s = \sum_{j=0}^{\min\{r,s\}} \langle A_r B_s \rangle_{ r-s +2j}$ $\langle A_r B_s \rangle_{r+s-2j} = (-1)^{rs-j} \langle B_s A_r \rangle_{r+s-2j}$                                                                           |
| 1285.5935v1_<br>EQ0408 (5)  | 083  | <div> <div>■ 1.000</div> <div>● K +0</div> </div> | $\begin{aligned} & \langle A B \rangle = \langle B A \rangle \\ & \langle A^* B^* \rangle = \langle A^\dagger B^\dagger \rangle \\ & \langle A^{\ddagger} B^{\ddagger} \rangle = \langle A^\ddagger B^\ddagger \rangle \end{aligned}$                                        | $\langle AB \rangle = \langle BA \rangle$ $= \langle A^* B^* \rangle$ $= \langle A^\dagger B^\dagger \rangle$ $= \langle A^\ddagger B^\ddagger \rangle$                                                                             | $\langle AB \rangle = \langle BA \rangle$ $= \langle A^* B^* \rangle$ $= \langle A^\dagger B^\dagger \rangle$ $= \langle A^\ddagger B^\ddagger \rangle$                                                                             |
| 1285.5935v1_<br>EQ0411 (12) | 083  | <div> <div>■ 1.000</div> <div>● N +0</div> </div> | $a_1 \wedge a_2 \wedge \cdots \wedge a_r = \langle a_1 a_2 \cdots a_r \rangle_r$                                                                                                                                                                                             | $a_1 \wedge a_2 \wedge \cdots \wedge a_r = \langle a_1 a_2 \cdots a_r \rangle_r$                                                                                                                                                    | $a_1 \wedge a_2 \wedge \cdots \wedge a_r = \langle a_1 a_2 \cdots a_r \rangle_r$                                                                                                                                                    |
| 1285.5935v1_<br>EQ0412 (13) | 084  | <div> <div>■ 1.000</div> <div>●</div> </div>      | $\begin{aligned} & a \downarrow A = \frac{1}{2} (aA - A^* a) \\ & a \wedge A = \frac{1}{2} (aA + A^* a) \end{aligned}$                                                                                                                                                       | $a \downarrow A = \frac{1}{2} (aA - A^* a)$ $a \wedge A = \frac{1}{2} (aA + A^* a)$                                                                                                                                                 | $a \downarrow A = \frac{1}{2} (aA - A^* a)$ $a \wedge A = \frac{1}{2} (aA + A^* a)$                                                                                                                                                 |
| 1285.5935v1_<br>EQ0414 (17) | 084  | <div> <div>■ 1.000</div> <div>● N +0</div> </div> | $a \wedge A \wedge b = -b \wedge A \wedge a$                                                                                                                                                                                                                                 | $a \wedge A \wedge b = -b \wedge A \wedge a$                                                                                                                                                                                        | $a \wedge A \wedge b = -b \wedge A \wedge a$                                                                                                                                                                                        |
| 1285.5935v1_<br>EQ0420 (29) | 084  | <div> <div>■ 1.000</div> <div>● W +0</div> </div> | $\begin{aligned} & A \wedge (B \wedge C) = (A \wedge B) \wedge C \\ & A \downarrow (B \downarrow C) = (A \downarrow B) \downarrow C \\ & A \downarrow (B \uparrow C) = (A \wedge B) \uparrow C \\ & A \downarrow (B \wedge C) = (A \downarrow B) \downarrow C \end{aligned}$ | $A \wedge (B \wedge C) = (A \wedge B) \wedge C$ $A \downarrow (B \downarrow C) = (A \downarrow B) \downarrow C$ $A \downarrow (B \uparrow C) = (A \wedge B) \uparrow C$ $A \downarrow (B \wedge C) = (A \downarrow B) \downarrow C$ | $A \wedge (B \wedge C) = (A \wedge B) \wedge C$ $A \downarrow (B \downarrow C) = (A \downarrow B) \downarrow C$ $A \downarrow (B \uparrow C) = (A \wedge B) \uparrow C$ $A \downarrow (B \wedge C) = (A \downarrow B) \downarrow C$ |
| 1285.5935v1_<br>EQ0421 (32) | 084  | <div> <div>■ 1.000</div> <div>● N +0</div> </div> | $A B_s A^\dagger = \langle A B_s A^\dagger \rangle_s$                                                                                                                                                                                                                        | $A B_s A^\dagger = \langle A B_s A^\dagger \rangle_s$                                                                                                                                                                               | $A B_s A^\dagger = \langle A B_s A^\dagger \rangle_s$                                                                                                                                                                               |
| 1285.5935v1_<br>EQ0422 (33) | 084  | <div> <div>■ 1.000</div> <div>● N +0</div> </div> | $(A B A^\dagger) \wedge (A C A^\dagger) =  A ^2 A (B \wedge C) A^\dagger$                                                                                                                                                                                                    | $(A B A^\dagger) \wedge (A C A^\dagger) =  A ^2 A (B \wedge C) A^\dagger$                                                                                                                                                           | $(A B A^\dagger) \wedge (A C A^\dagger) =  A ^2 A (B \wedge C) A^\dagger$                                                                                                                                                           |
| 1285.5935v1_<br>EQ0427 (49) | 085  | <div> <div>■ 1.000</div> <div>● N +0</div> </div> | $A \boldsymbol{I} = \boldsymbol{I} A^{*(n-1)}$                                                                                                                                                                                                                               | $A \boldsymbol{I} = \boldsymbol{I} A^{*(n-1)}$                                                                                                                                                                                      | $A \boldsymbol{I} = \boldsymbol{I} A^{*(n-1)}$                                                                                                                                                                                      |
| 1285.5935v1_<br>EQ0429 (54) | 086  | <div> <div>■ 1.000</div> <div>● K +0</div> </div> | $A_r^\dagger = (-1)^{r(r-1)/2} A_r$                                                                                                                                                                                                                                          | $A_r^\dagger = (-1)^{r(r-1)/2} A_r$                                                                                                                                                                                                 | $A_r^\dagger = (-1)^{r(r-1)/2} A_r$                                                                                                                                                                                                 |
| 1285.5935v1_<br>EQ0430 (55) | 086  | <div> <div>■ 1.000</div> <div>● K +0</div> </div> | $A^{\dagger\dagger} = A$                                                                                                                                                                                                                                                     | $A^{\dagger\dagger} = A$                                                                                                                                                                                                            | $A^{\dagger\dagger} = A$                                                                                                                                                                                                            |
| 1285.5935v1_<br>EQ0431 (56) | 086  | <div> <div>■ 1.000</div> <div>● K +0</div> </div> | $(A^{-1})^\dagger = (A^\dagger)^{-1}$                                                                                                                                                                                                                                        | $(A^{-1})^\dagger = (A^\dagger)^{-1}$                                                                                                                                                                                               | $(A^{-1})^\dagger = (A^\dagger)^{-1}$                                                                                                                                                                                               |
| 1285.5935v1_<br>EQ0434 (64) | 086  | <div> <div>■ 1.000</div> <div>● N +0</div> </div> | $A^{\ddagger} = A^{*\dagger} = A^{\dagger*}$                                                                                                                                                                                                                                 | $A^{\ddagger} = A^{*\dagger} = A^{\dagger*}$                                                                                                                                                                                        | $A^{\ddagger} = A^{*\dagger} = A^{\dagger*}$                                                                                                                                                                                        |

**Conf.** MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

**LaTeX source** MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

**Residual** render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● X crop overran its region ● not measured

**[refined: *basis*]** beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.



| Identifier                                                                                                                                                                                                                                                                                                           | Page | Conf.                                                            | LaTeX source                                                                                                                                                                                                                                                             | Rendered                                                                                                                                                                                          | Scan image                                                                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1205.5935v1_<br>EQ0457 (108)                                                                                                                                                                                                                                                                                         | 088  | <div><div>1.000</div><div><div></div><div>N +0</div></div></div> | $\begin{aligned} & (A \times B)^{*}=A^{*} \times B^{*} \backslash \\ & \backslash \& (A \times B)^{\{\dagger\}}=B^{\{\dagger\}} \times A^{\{\dagger\}} \\ & \backslash \backslash \& (A \times B)^{\{\ddagger\}}=B^{\{\ddagger\}} \times A^{\{\ddagger\}} \end{aligned}$ | $\begin{aligned} (A \times B)^{*} &= A^{*} \times B^{*} \\ (A \times B)^{\dagger} &= B^{\dagger} \times A^{\dagger} \\ (A \times B)^{\ddagger} &= B^{\ddagger} \times A^{\ddagger} \end{aligned}$ | $\begin{aligned} (A \times B)^{*} &= A^{*} \times B^{*} \\ (A \times B)^{\dagger} &= B^{\dagger} \times A^{\dagger} \\ (A \times B)^{\ddagger} &= B^{\ddagger} \times A^{\ddagger} \end{aligned}$ |
| 1205.5935v1_<br>EQ0458 (110)                                                                                                                                                                                                                                                                                         | 089  | <div><div>1.000</div><div><div></div><div>K +0</div></div></div> | $\lambda \times A=0$                                                                                                                                                                                                                                                     | $\lambda \times A=0$                                                                                                                                                                              | $\lambda \times A=0$                                                                                                                                                                              |
| 1205.5935v1_<br>EQ0460 (113)                                                                                                                                                                                                                                                                                         | 089  | <div><div>1.000</div><div><div></div><div>K +0</div></div></div> | $A_{\{2\}} \times A_{\{r\}}=\left\langle A_{2} A_{r}\right\rangle _{r}$                                                                                                                                                                                                  | $A_2 \times A_r=\left\langle A_2 A_r\right\rangle _r$                                                                                                                                             | $A_2 \times A_r=\left\langle A_2 A_r\right\rangle _r$                                                                                                                                             |
| 1205.5935v1_<br>EQ0466 (122)                                                                                                                                                                                                                                                                                         | 089  | <div><div>1.000</div><div><div></div><div>K +0</div></div></div> | $a_{\{I\}} * a^{\{J\}}=\delta _{I}^{\{J\}}$                                                                                                                                                                                                                              | $a_I * a^J = \delta_I^J,$                                                                                                                                                                         | $a_I * a^J = \delta_I^J,$                                                                                                                                                                         |
| 1205.5935v1_<br>EQ0472 (131)                                                                                                                                                                                                                                                                                         | 090  | <div><div>1.000</div><div><div></div><div>N +0</div></div></div> | $\begin{aligned} & P_{\{a_{\{1\}} \wedge \cdots \wedge a_{\{r\}}\}}(B) \& \\ & =\sum_{\{I\}}\left(B * a_{\{I\}}\right) a^{\{I\}} \backslash \backslash \& =\sum_{\{I\}}\left(B * a^{\{I\}}\right) a_{\{I\}} \end{aligned}$                                               | $\begin{aligned} P_{a_1 \wedge \cdots \wedge a_r}(B) &= \sum_I(B * a_I) a^I \\ &= \sum_I(B * a^I) a_I \end{aligned}$                                                                              | $\begin{aligned} P_{a_1 \wedge \cdots \wedge a_r}(B) &= \sum_I(B * a_I) a^I \\ &= \sum_I(B * a^I) a_I \end{aligned}$                                                                              |
| 1205.5935v1_<br>EQ0473 (132)                                                                                                                                                                                                                                                                                         | 090  | <div><div>1.000</div><div><div></div><div>N +0</div></div></div> | $\bar{\mathsf{F}}(\mathsf{B}) * \mathsf{A}=\mathsf{B} * \mathsf{F}(\mathsf{A})$                                                                                                                                                                                          | $\bar{F}(B) * A = B * F(A)$                                                                                                                                                                       | $\overline{F}(B) * A = B * F(A)$                                                                                                                                                                  |
| 1205.5935v1_<br>EQ0475 (134)                                                                                                                                                                                                                                                                                         | 090  | <div><div>1.000</div><div><div></div><div>N +0</div></div></div> | $\overline{\bar{\mathsf{F}}}=\mathsf{F}$                                                                                                                                                                                                                                 | $\overline{\overline{F}} = F$                                                                                                                                                                     | $\overline{\overline{F}} = F$                                                                                                                                                                     |
| 1205.5935v1_<br>EQ0476 (135)                                                                                                                                                                                                                                                                                         | 090  | <div><div>1.000</div><div><div></div><div>W +0</div></div></div> | $\overline{\mathsf{G} \mathsf{F}}=\bar{\mathsf{F}} \bar{\mathsf{G}}$                                                                                                                                                                                                     | $\overline{GF} = \bar{F} \bar{G}$                                                                                                                                                                 | $\overline{GF} = \bar{F} \bar{G}$                                                                                                                                                                 |
| 1205.5935v1_<br>EQ0477 (136)                                                                                                                                                                                                                                                                                         | 090  | <div><div>1.000</div><div><div></div><div>N +0</div></div></div> | $\bar{\mathsf{F}}^{\{-1\}}=\overline{\mathsf{F}^{\{-1\}}}$                                                                                                                                                                                                               | $\bar{F}^{-1} = \overline{F^{-1}}$                                                                                                                                                                | $\overline{F}^{-1} = \overline{F^{-1}}$                                                                                                                                                           |
| 1205.5935v1_<br>EQ0478 (137)                                                                                                                                                                                                                                                                                         | 090  | <div><div>1.000</div><div><div></div><div>K +0</div></div></div> | $\operatorname{det}(\bar{\mathsf{F}})=\operatorname{det}(\mathsf{F})$                                                                                                                                                                                                    | $\det(\bar{F}) = \det(F)$                                                                                                                                                                         | $\det(\overline{F}) = \det(F)$                                                                                                                                                                    |
| <div><div>Conf.</div><div>MathPix confidence:</div><div><div><div>1.000</div></div><div><div>0.5–0.9</div></div><div><div>&lt;0.5</div></div><div>— no value</div></div></div>                                                                                                                                       |      |                                                                  |                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                   |                                                                                                                                                                                                   |
| <div><div>LaTeX source</div><div>MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</div></div>                           |      |                                                                  |                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                   |                                                                                                                                                                                                   |
| <div><div>Residual</div><div>render vs scan (inkdrill):</div><div><div><div>C component</div></div><div><div>W weak</div></div><div><div>S stable</div></div><div><div>N noise</div></div><div><div>K clean</div></div><div><div>X crop overran its region</div></div><div><div>not measured</div></div></div></div> |      |                                                                  |                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                   |                                                                                                                                                                                                   |
| <div><div>[refined: <i>basis</i>]</div><div>beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</div></div> |      |                                                                  |                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                   |                                                                                                                                                                                                   |

Tables

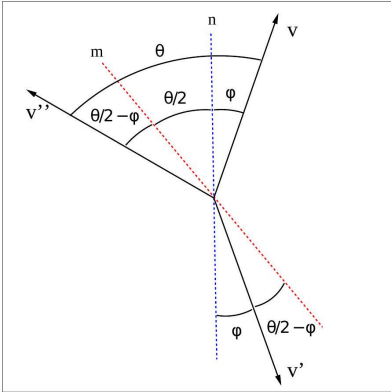
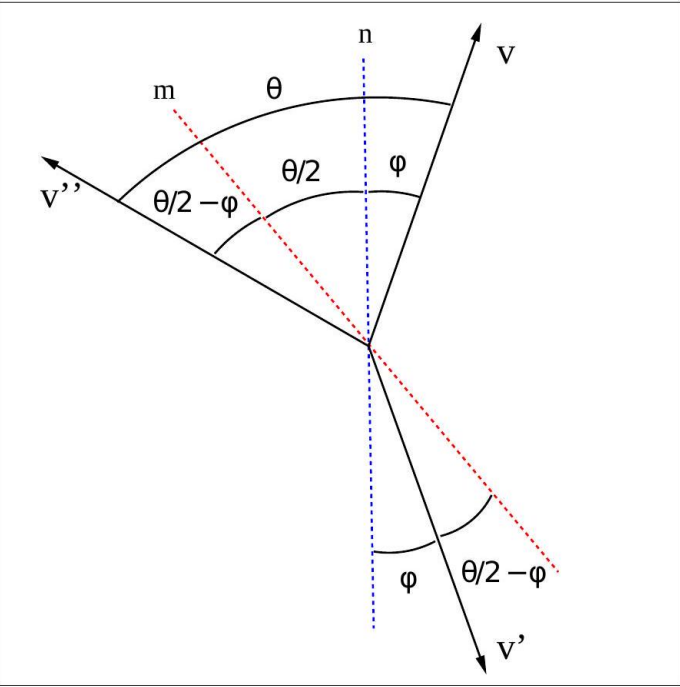
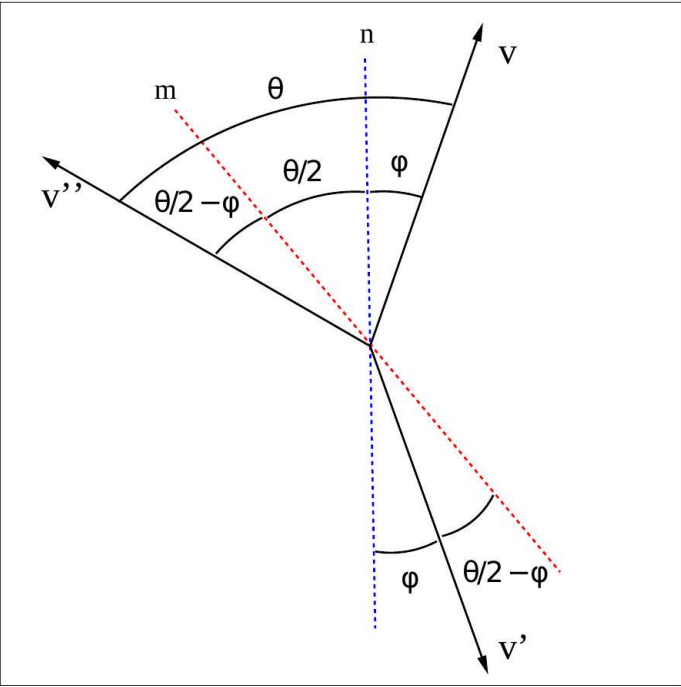
| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Page | Conf. | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                              | Rendered                                                                                                                                                                                                                                                                                                                                                                                                           | Scan image                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1205.5935v1_TAB_001                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 081  | —     | $\begin{tabular}{t l l } \hline \mathcal{G} & \text{Geometric algebra} \\ \hline \mathcal{G}_r & \text{Grade-}r \text{ subspace of } \mathcal{G} \text{ (space of } r\text{-vectors)} \\ \hline \mathcal{G}^n & \text{Geometric algebra of an } n\text{-dimensional vector space} \\ \hline A, B, \text{ etc.} & \text{General multivector} \\ \hline \lambda, \mu, \text{ etc.} & \text{Scalar} \\ \hline \end{tabular}$ | <div><math>\mathcal{G}</math>Geometric algebra</div> <div><math>\mathcal{G}_r</math>Grade- <math>r</math> subspace of <math>\mathcal{G}</math> (space of <math>r</math>-vectors)</div> <div><math>\mathcal{G}^n</math>Geometric algebra of an <math>n</math>-dimensional vector space</div> <div><math>A, B, \text{ etc.}</math>General multivector</div> <div><math>\lambda, \mu, \text{ etc.}</math>Scalar</div> | <div><math>\mathcal{G}</math>Geometric algebra</div> <div><math>\mathcal{G}_r</math>Grade-<math>r</math> subspace of <math>\mathcal{G}</math> (space of <math>r</math>-vectors)</div> <div><math>\mathcal{G}^n</math>Geometric algebra of an <math>n</math>-dimensional vector space</div> <div><math>A, B, \text{ etc.}</math>General multivector</div> <div><math>\lambda, \mu, \text{ etc.}</math>Scalar</div> |
| <div><b>Conf.</b> MathPix confidence: <span>■</span> <math>\geq 0.9</math> <span>■</span> 0.5–0.9 <span>■</span> <math>&lt; 0.5</math> — no value</div> <div><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</div> <div><b>Residual</b> render vs scan (inkdrill): <span>●</span> C component <span>●</span> W weak <span>●</span> S stable <span>●</span> N noise <span>●</span> K clean <span>●</span> X crop overran its region <span>●</span> not measured</div> <div><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</div> |      |       |                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                   |

| Identifier                 | Page                                                           | Conf. | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     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                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Scan image                 |        |       |                                                 |                    |            |       |                        |       |                       |     |                |                       |                        |                     |                              |                       |                        |                       |                       |          |                |       |                         |           |                               |             |                |              |                           |         |                     |           |             |              |                     |      |                                  |               |                                    |                 |                                   |              |                              |         |                               |              |                           |                           |                                                      |                           |                                                     |                         |              |                      |                                                             |       |                                    |                      |                                                                |    |                   |                 |               |                   |              |                  |                |                  |             |                |                |           |                    |               |               |
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| 1205.5935v1_TAB_002        | 082                                                            | —     | $\begin{tabular}[t]{t}[l l] \hline (a, b, u, v), \text{ etc.} \\ \& \text{Vector} \\\hline (A_r) \& (r)\text{-vector (sometimes grade- } (r) \text{ part of } (A) ) \\\hline (\boldsymbol{A}_r) \& (r)\text{-blade} \\\hline (A_+) \& \text{Even-grade multivector} \\\hline (A_-) \& \text{Odd-grade multivector} \\\hline I \& \text{Volume element} \\\hline \langle A \rangle_r \& \text{Grade- } r \text{ part of } A \\\hline \langle A \rangle \& \text{Scalar (grade-0) part of } A \\\hline \langle A \rangle_+ \& \text{Even-grade part of } A \\\hline \langle A \rangle_- \& \text{Odd-grade part of } A \\\hline A^{-1} \& \text{Inverse of } A \\\hline A^* \& \text{Grade involution of } A \\\hline A^{**r} \& r \text{ times grade involuted } A \\\hline A^\dagger \& \text{Reverse of } A \\\hline A^\ddagger \& \text{Clifford conjugate of } A \\\hline  A ^2 \& \text{Squared norm of } A \\\hline A^\perp \& \text{Dual of } A \\\hline A^{-\perp} \& \text{Inverse dual of } A \\\hline AB \& \text{Geometric product of } A \text{ and } B \\\hline A \rfloor B \& \text{Left inner product of } A \text{ into } B \\\hline A \llcorner B \& \text{Right inner product of } A \text{ by } B \\\hline A \wedge B \& \text{Outer product of } A \text{ and } B \\\hline A * B \& \text{Scalar product of } A \text{ and } B \\\hline A \times B \& \text{Commutator of } A \text{ and } B \\\hline P_{\boldsymbol{A}_r}(B) \& \text{Orthogonal projection of } B \text{ into } \boldsymbol{A}_r \\\hline R_{\boldsymbol{A}_r}(B) \& \text{Orthogonal rejection of } B \text{ from } \boldsymbol{A}_r \\\hline U, V, W, \text{ etc.} \& \text{Vector space} \\\hline F, G, \text{ etc.} \& \text{Linear function of vectors (or its outermorphism extension)} \\\hline F_U \& \text{Restriction of } F \text{ to subspace } U \\\hline F_{\boldsymbol{A}} \& \text{Restriction of } F \text{ to subspace represented by } \boldsymbol{A} \\\hline \text{Id} \& \text{Identity function} \\\hline \text{Ker}(F) \& \text{Kernel of } F \\\hline \text{Range}(F) \& \text{Range of } F \\\hline \text{null}(F) \& \text{Nullity of } F \\\hline \text{rank}(F) \& \text{Rank of } F \\\hline \overline{F} \& \text{Adjoint of } F \\\hline \det(F) \& \text{Determinant of } F \\\hline \mathcal{F} \& \text{Outermorphism} \end{tabular}$ | <table><tr><td><math>a, b, u, v, \text{ etc.}</math></td><td>Vector</td></tr><tr><td><math>A_r</math></td><td><math>r</math>-vector (sometimes grade- <math>r</math> part of <math>A</math> )</td></tr><tr><td><math>\boldsymbol{A}_r</math></td><td><math>r</math>-blade</td></tr><tr><td><math>A_+</math></td><td>Even-grade multivector</td></tr><tr><td><math>A_-</math></td><td>Odd-grade multivector</td></tr><tr><td><math>I</math></td><td>Volume element</td></tr><tr><td><math>\langle A \rangle_r</math></td><td>Grade- <math>r</math> part of <math>A</math></td></tr><tr><td><math>\langle A \rangle</math></td><td>Scalar (grade-0) part of <math>A</math></td></tr><tr><td><math>\langle A \rangle_+</math></td><td>Even-grade part of <math>A</math></td></tr><tr><td><math>\langle A \rangle_-</math></td><td>Odd-grade part of <math>A</math></td></tr><tr><td><math>A^{-1}</math></td><td>Inverse of <math>A</math></td></tr><tr><td><math>A^*</math></td><td>Grade involution of <math>A</math></td></tr><tr><td><math>A^{**r}</math></td><td><math>r</math> times grade involuted <math>A</math></td></tr><tr><td><math>A^\dagger</math></td><td>Reverse of <math>A</math></td></tr><tr><td><math>A^\ddagger</math></td><td>Clifford conjugate of <math>A</math></td></tr><tr><td><math> A ^2</math></td><td>Squared norm of <math>A</math></td></tr><tr><td><math>A^\perp</math></td><td>Dual of <math>A</math></td></tr><tr><td><math>A^{-\perp}</math></td><td>Inverse dual of <math>A</math></td></tr><tr><td><math>AB</math></td><td>Geometric product of <math>A</math> and <math>B</math></td></tr><tr><td><math>A \rfloor B</math></td><td>Left inner product of <math>A</math> into <math>B</math></td></tr><tr><td><math>A \llcorner B</math></td><td>Right inner product of <math>A</math> by <math>B</math></td></tr><tr><td><math>A \wedge B</math></td><td>Outer product of <math>A</math> and <math>B</math></td></tr><tr><td><math>A * B</math></td><td>Scalar product of <math>A</math> and <math>B</math></td></tr><tr><td><math>A \times B</math></td><td>Commutator of <math>A</math> and <math>B</math></td></tr><tr><td><math>P_{\boldsymbol{A}_r}(B)</math></td><td>Orthogonal projection of <math>B</math> into <math>\boldsymbol{A}_r</math></td></tr><tr><td><math>R_{\boldsymbol{A}_r}(B)</math></td><td>Orthogonal rejection of <math>B</math> from <math>\boldsymbol{A}_r</math></td></tr><tr><td><math>U, V, W, \text{ etc.}</math></td><td>Vector space</td></tr><tr><td><math>F, G, \text{ etc.}</math></td><td>Linear function of vectors (or its outermorphism extension)</td></tr><tr><td><math>F_U</math></td><td>Restriction of <math>F</math> to subspace <math>U</math></td></tr><tr><td><math>F_{\boldsymbol{A}}</math></td><td>Restriction of <math>F</math> to subspace represented by <math>\boldsymbol{A}</math></td></tr><tr><td>Id</td><td>Identity function</td></tr><tr><td><math>\text{Ker}(F)</math></td><td>Kernel of <math>F</math></td></tr><tr><td><math>\text{Range}(F)</math></td><td>Range of <math>F</math></td></tr><tr><td><math>\text{null}(F)</math></td><td>Nullity of <math>F</math></td></tr><tr><td><math>\text{rank}(F)</math></td><td>Rank of <math>F</math></td></tr><tr><td><math>\overline{F}</math></td><td>Adjoint of <math>F</math></td></tr><tr><td><math>\det(F)</math></td><td>Determinant of <math>F</math></td></tr><tr><td><math>\mathcal{F}</math></td><td>Outermorphism</td></tr></table> | $a, b, u, v, \text{ etc.}$ | Vector | $A_r$ | $r$ -vector (sometimes grade- $r$ part of $A$ ) | $\boldsymbol{A}_r$ | $r$ -blade | $A_+$ | Even-grade multivector | $A_-$ | Odd-grade multivector | $I$ | Volume element | $\langle A \rangle_r$ | Grade- $r$ part of $A$ | $\langle A \rangle$ | Scalar (grade-0) part of $A$ | $\langle A \rangle_+$ | Even-grade part of $A$ | $\langle A \rangle_-$ | Odd-grade part of $A$ | $A^{-1}$ | Inverse of $A$ | $A^*$ | Grade involution of $A$ | $A^{**r}$ | $r$ times grade involuted $A$ | $A^\dagger$ | Reverse of $A$ | $A^\ddagger$ | Clifford conjugate of $A$ | $ A ^2$ | Squared norm of $A$ | $A^\perp$ | Dual of $A$ | $A^{-\perp}$ | Inverse dual of $A$ | $AB$ | Geometric product of $A$ and $B$ | $A \rfloor B$ | Left inner product of $A$ into $B$ | $A \llcorner B$ | Right inner product of $A$ by $B$ | $A \wedge B$ | Outer product of $A$ and $B$ | $A * B$ | Scalar product of $A$ and $B$ | $A \times B$ | Commutator of $A$ and $B$ | $P_{\boldsymbol{A}_r}(B)$ | Orthogonal projection of $B$ into $\boldsymbol{A}_r$ | $R_{\boldsymbol{A}_r}(B)$ | Orthogonal rejection of $B$ from $\boldsymbol{A}_r$ | $U, V, W, \text{ etc.}$ | Vector space | $F, G, \text{ etc.}$ | Linear function of vectors (or its outermorphism extension) | $F_U$ | Restriction of $F$ to subspace $U$ | $F_{\boldsymbol{A}}$ | Restriction of $F$ to subspace represented by $\boldsymbol{A}$ | Id | Identity function | $\text{Ker}(F)$ | Kernel of $F$ | $\text{Range}(F)$ | Range of $F$ | $\text{null}(F)$ | Nullity of $F$ | $\text{rank}(F)$ | Rank of $F$ | $\overline{F}$ | Adjoint of $F$ | $\det(F)$ | Determinant of $F$ | $\mathcal{F}$ | Outermorphism |
| $a, b, u, v, \text{ etc.}$ | Vector                                                         |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| $A_r$                      | $r$ -vector (sometimes grade- $r$ part of $A$ )                |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| $\boldsymbol{A}_r$         | $r$ -blade                                                     |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| $A_+$                      | Even-grade multivector                                         |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| $A_-$                      | Odd-grade multivector                                          |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| $I$                        | Volume element                                                 |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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   |                                    |                      |                                                                |    |                   |                 |               |                   |              |                  |                |                  |             |                |                |           |                    |               |               |
| $\langle A \rangle_r$      | Grade- $r$ part of $A$                                         |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| $\langle A \rangle$        | Scalar (grade-0) part of $A$                                   |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| $\langle A \rangle_+$      | Even-grade part of $A$                                         |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| $\langle A \rangle_-$      | Odd-grade part of $A$                                          |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| $A^{-1}$                   | Inverse of $A$                                                 |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| $A^*$                      | Grade involution of $A$                                        |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| $A^{**r}$                  | $r$ times grade involuted $A$                                  |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| $A^\dagger$                | Reverse of $A$                                                 |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| $A^\ddagger$               | Clifford conjugate of $A$                                      |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| $ A ^2$                    | Squared norm of $A$                                            |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| $A^\perp$                  | Dual of $A$                                                    |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| $A^{-\perp}$               | Inverse dual of $A$                                            |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| $AB$                       | Geometric product of $A$ and $B$                               |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| $A \rfloor B$              | Left inner product of $A$ into $B$                             |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| $A \llcorner B$            | Right inner product of $A$ by $B$                              |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| $A \wedge B$               | Outer product of $A$ and $B$                                   |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| $A * B$                    | Scalar product of $A$ and $B$                                  |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| $A \times B$               | Commutator of $A$ and $B$                                      |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| $P_{\boldsymbol{A}_r}(B)$  | Orthogonal projection of $B$ into $\boldsymbol{A}_r$           |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| $R_{\boldsymbol{A}_r}(B)$  | Orthogonal rejection of $B$ from $\boldsymbol{A}_r$            |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| $U, V, W, \text{ etc.}$    | Vector space                                                   |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| $F, G, \text{ etc.}$       | Linear function of vectors (or its outermorphism extension)    |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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   |                                    |                      |                                                                |    |                   |                 |               |                   |              |                  |                |                  |             |                |                |           |                    |               |               |
| $F_U$                      | Restriction of $F$ to subspace $U$                             |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| $F_{\boldsymbol{A}}$       | Restriction of $F$ to subspace represented by $\boldsymbol{A}$ |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| Id                         | Identity function                                              |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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   |                                    |                      |                                                                |    |                   |                 |               |                   |              |                  |                |                  |             |                |                |           |                    |               |               |
| $\text{Ker}(F)$            | Kernel of $F$                                                  |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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|                     |                              |                       |                        |                       |                       |          |                |       |                         |           |                               |             |                |              |                           |         |                     |           |             |              |                     |      |                                  |               |                                    |                 |                                   |              |                              |         |                               |              |                           |                           |                                                      |                           |                                                     |                         |              |                      |                                                             |       |                                    |                      |                                                                |    |                   |                 |               |                   |              |                  |                |                  |             |                |                |           |                    |               |               |
| $\text{Range}(F)$          | Range of $F$                                                   |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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   |                                    |                      |                                                                |    |                   |                 |               |                   |              |                  |                |                  |             |                |                |           |                    |               |               |
| $\text{null}(F)$           | Nullity of $F$                                                 |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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   |                                    |                      |                                                                |    |                   |                 |               |                   |              |                  |                |                  |             |                |                |           |                    |               |               |
| $\text{rank}(F)$           | Rank of $F$                                                    |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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   |                                    |                      |                                                                |    |                   |                 |               |                   |              |                  |                |                  |             |                |                |           |                    |               |               |
| $\overline{F}$             | Adjoint of $F$                                                 |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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   |                                    |                      |                                                                |    |                   |                 |               |                   |              |                  |                |                  |             |                |                |           |                    |               |               |
| $\det(F)$                  | Determinant of $F$                                             |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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   |                                    |                      |                                                                |    |                   |                 |               |                   |              |                  |                |                  |             |                |                |           |                    |               |               |
| $\mathcal{F}$              | Outermorphism                                                  |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Page | Conf. | LaTeX source | Rendered | Scan image |
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| <div><b>Conf.</b> MathPix confidence: <span>■</span> <math>\geq 0.9</math> <span>■</span> 0.5–0.9 <span>■</span> <math>&lt; 0.5</math> — no value</div> <div><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</div> <div><b>Residual</b> render vs scan (inkdrill): <span>●</span> C component <span>●</span> W weak <span>●</span> S stable <span>●</span> N noise <span>●</span> K clean <span>●</span> X crop overran its region <span>●</span> not measured</div> <div><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</div> |      |       |              |          |            |

Image regions — rendered against scan

Two comparable cells per row. **Rendered**: the region’s own LaTeX compiled as its OWN document, or — where no LaTeX exists — the scan again, so every row has two cells and the ink difference reads as a floor rather than as missing-versus-present. A duplicated row is marked *(dup)* and its distance is a SELF-comparison, not agreement between two sources. **Scan**: the tex.zip image whose filename region 5-tuple matches this row, else the CDN crop. The Class column carries the residual class beside MathPix’s confidence, as the equation rows do. Setting a region’s LaTeX does NOT say it renders to what is on the page — that is what the two columns are for, and 281 is the open question they exist to let a reader answer. A region with no LaTeX can be reconstructed with pdfdrill vision; verify any LLM result against the real ink with inkdrill.

| Identifier           | Page | Class   | Source                                                      | Author source                                                                        | Rendered                                                                             | Scan                                                                                 |
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| 1205.5935v1_DIA_0001 | 007  | — (dup) | fd36c8bf-285b-42a6-a95b-6c070a0b062c-07_898_898_249_640.jpg |  |  |  |